

# DIGITAL LOGIC

Handwritten Notes

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GateWallah Parakram C 2024

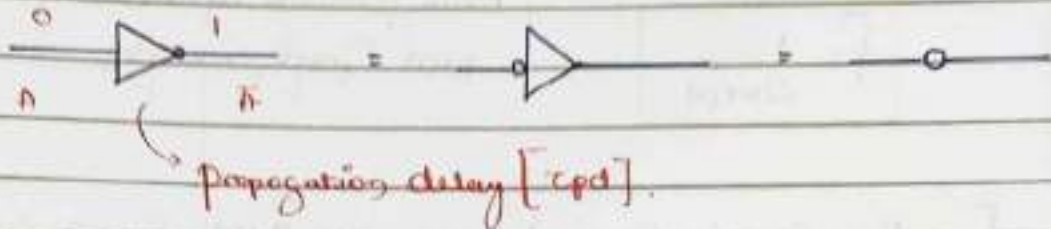
(Hinglish Weekday)

Other Notes Uploaded Below

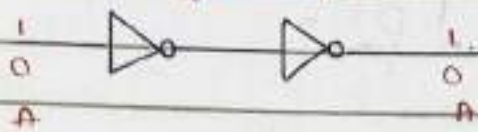
<https://t.me/pwgatenotes>

# LOGIC GATES

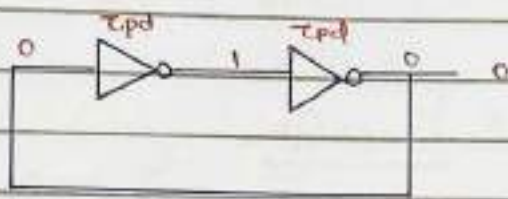
## 1. NOT Gate [Inverter, Negation, Complement Logic].



Buffer (for even)



No. of not gates are even - Buffer.  
No. of not gate odd - Inverter.  
[High.]

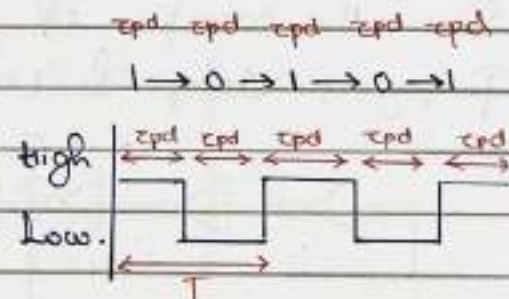
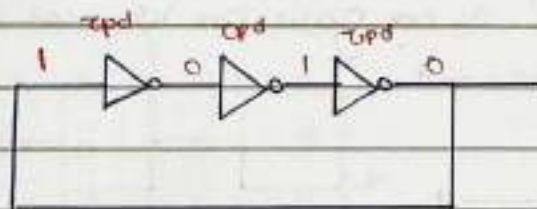


2cpd 2cpd 2cpd 2cpd  
1 → 1 → 1 → 1 →

2cpd 2cpd 2cpd  
0 → 0 → 0 → [Low]

Whenever even numbers of 'not' gate in circuit (loop).

1. Basic memory element
2. Bistable multivibrator
3. DC Generator ( $P = 0\%$ ).



When odd no. of NOT Gate:

1. Astable multivibrator
2. Square wave generator
3. Clock generator
4. Free running circuit
5. Ring oscillator.

$$T = 2cpd$$

$$f = \frac{1}{2cpd}$$

$$f = \frac{1}{T}$$

$$T = 2 \times N \times \tau_{pd} \quad N = \text{no. of not gate in loop.}$$

$$f = \frac{1}{2N\tau_{pd}}$$

\* Both formulae applicable when number of NOT Gate are odd.

Q1. For the circuit given below, all NOT Gate are identical to each other and having Propagation delay  $10 \text{ ps}$ . Find the frequency of generated wave form?  $10^{12}$

$$N = 5$$

$$\tau_{pd} = 10 \times 10^{-12}$$

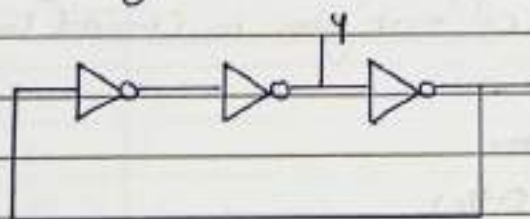
$$f = \frac{1}{2N\tau_{pd}} = \frac{1}{2 \times 5 \times 10 \times 10^{-12}} = \frac{10^{12}}{10 \times 10} \text{ Hz}$$

(odd)

$$= 10^9 \text{ Hz}$$

$$\therefore f = 10 \text{ GHz}$$

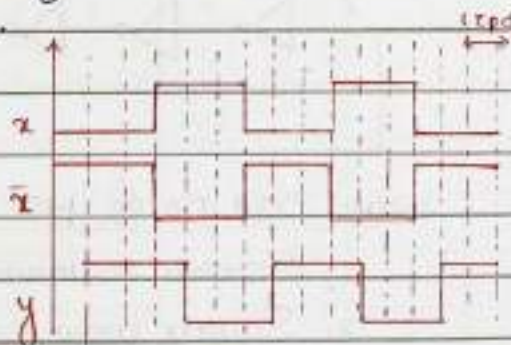
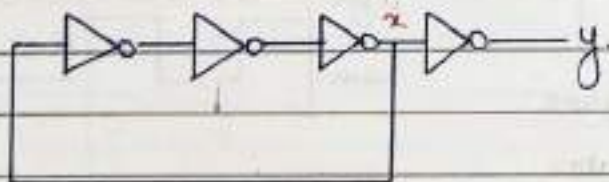
Q2. Circuit given below are called. Astable Multivibrator



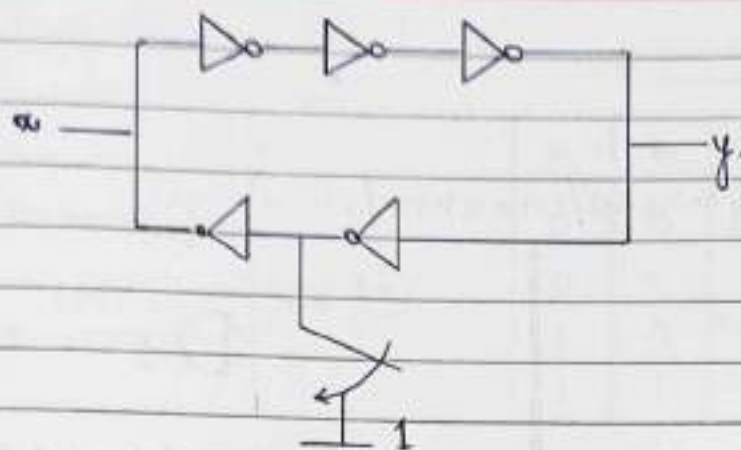
Note: always, For calculation of frequency consider the no. of NOT gate in loop.

HW

Q3. Sketch the waveform of  $y$  if all not gate are identical and having Propagation delay of 1 microsecond.



Q4. For the circuit given below  $x$  and  $y$  condition will be -  $x$  and  $y$  both toggle



Odd no. of not gate in the loop  
 $\therefore$  x and y will toggle

Basic Gate

NOT

AND

OR

Universal Gate

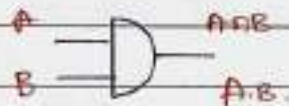
NAND

NOR

Exclusive Gate

X-OR

X-NOR

AND Gate

$$A = \{1, 2, 3, 4\} \quad B = \{2, 3, 4\}$$

$$A \cap B = \{2, 3, 4\}$$

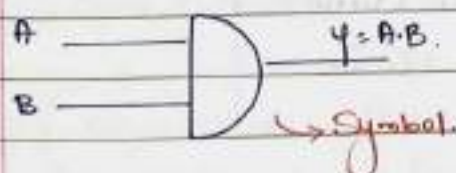
$$\begin{array}{l|l} 0 \cdot 0 = 0 & A \cdot A = A \\ 0 \cdot 1 = 0 & A \cdot 1 = A \\ 1 \cdot 0 = 0 & A \cdot 0 = 0 \\ 1 \cdot 1 = 1 & A \cdot \bar{A} = 0 \end{array}$$

OR Gate

$$A = \{1, 2\} \quad B = \{3, 4\}$$

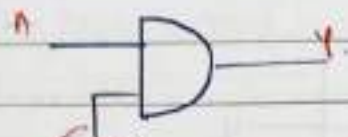
$$A \cup B = \{1, 2, 3, 4\}$$

$$\begin{array}{l|l} 0 + 0 = 0 & A + A = A \\ 0 + 1 = 1 & A + 0 = A \\ 1 + 0 = 1 & A + 1 = 1 \\ 1 + 1 = 1 & A + \bar{A} = 1 \end{array}$$

AND Gate [Intersection]

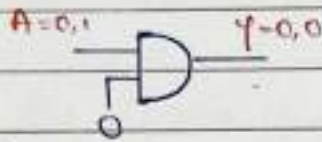
| A | B | $Y = A \cdot B$ |
|---|---|-----------------|
| 0 | 0 | 0               |
| 0 | 1 | 0               |
| 1 | 0 | 0               |
| 1 | 1 | 1               |

Truth Table



floating terminal / Control Input / Strobe.

④



Control '0' disabled.

⑤



Control '1' enabled.

\* Commutative Law :  $A \cdot B = B \cdot A$

\* Associative Law :  $A \cdot B \cdot C = (A \cdot B) \cdot C$

} AND is both commutative and associative.



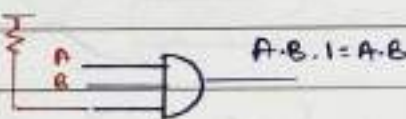
TTL = Transistor Transistor Logic.

①



(Favourite)

②



pull up arrangement.

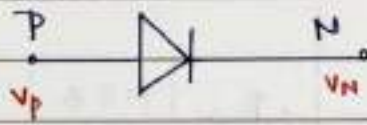
(Crazy)

③

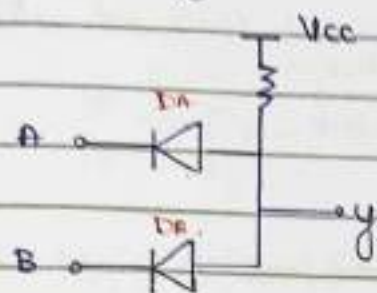


(Noise margin ↓)

"Diode"



$V_p > V_n \rightarrow$  Forward bias  $\rightarrow$  Shorted circuit.  
 $V_p < V_n \rightarrow$  Reverse bias  $\rightarrow$  Open circuit.

Circuit diagram: [AND]

| A | B | D <sub>A</sub> | D <sub>B</sub> | y |
|---|---|----------------|----------------|---|
| 0 | 0 | F <sub>B</sub> | F <sub>B</sub> | 0 |
| 0 | 1 | F <sub>B</sub> | R <sub>E</sub> | 0 |
| 1 | 0 | R <sub>S</sub> | F <sub>B</sub> | 0 |
| 1 | 1 | R <sub>E</sub> | R <sub>E</sub> | 1 |

Note:

- \* Whenever logic are designed by TTL the floating terminal always works as high.
- \* Whenever logic are designed by ECL (Emitter Coupled Logic) then floating terminal always works as low.
- \* Noise Margin: Maximum noise added to the input which will not affect the output are called Noise margin.
- \* Fan out: The number of logic driven by the logic are called fan out.
- \* Fan in: Number of input of a logic are called fan in.
- \* ECL is the fastest logic among all the logic family.

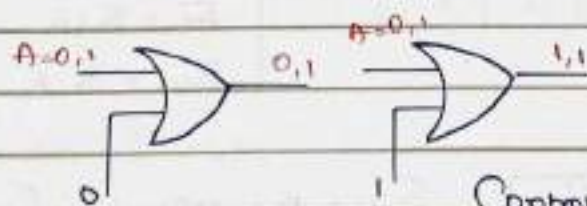
OR Gate.

1. Symbol. A+B.

## 2. Truth Table

| A | B | y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

## 3. Enable/Disable



Control '0' enabled.

Control '1' disabled.

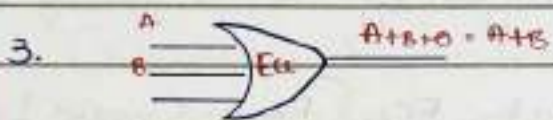
4. Commutative:  $A+B = B+A$
5. Associative:  $A+(B+C) = (A+B)+C$



ECL - Emitter Coupled  
Logic.

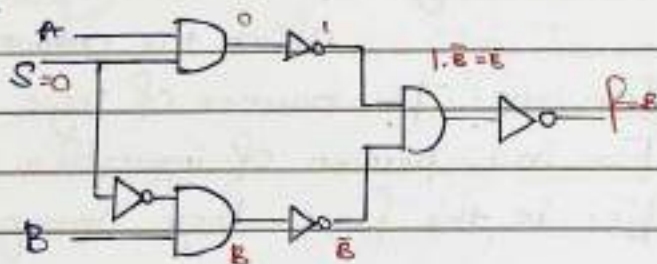


Pull down arrangement.

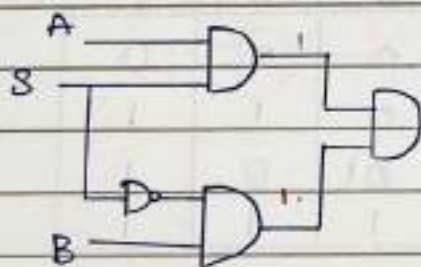


Q1. Output at  $S=0$  is B

$\rightarrow f = B$



Q2.  $f=1$  for combination of A, B, C



[None of the options are correct.]

De Morgan's Law

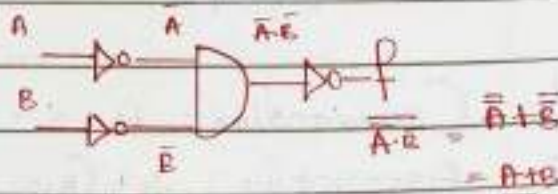
$$\overline{AB} = \overline{A} + \overline{B}$$

$$\overline{A+B} = \overline{A} \cdot \overline{B}$$

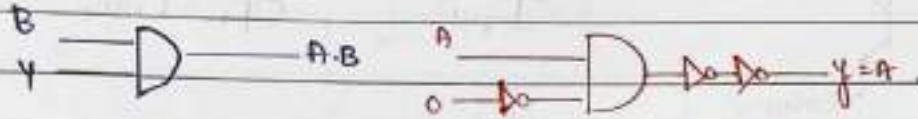
Q3. For the given truth table.

| A | B | f |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

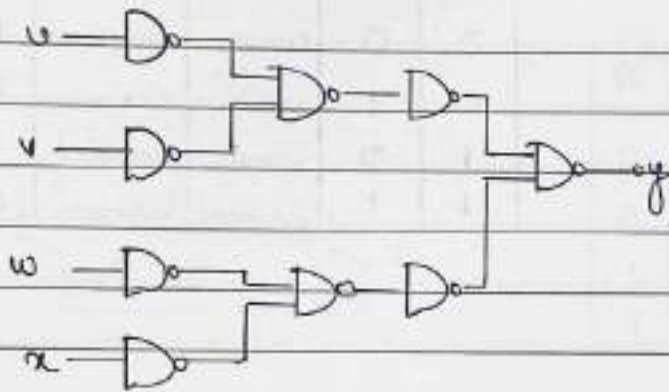
Select the respective Combinational diagram



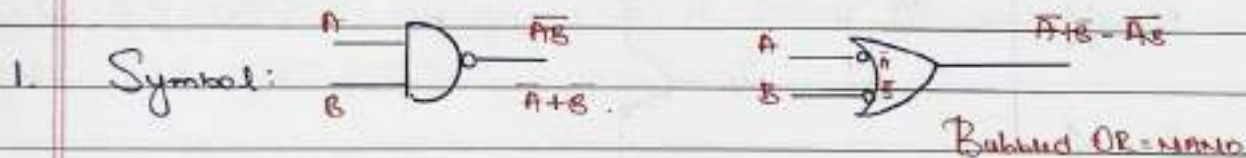
Q4. A logical circuit is as shown below, which of the following circuit can be used to get the desired expression.



Q5. The logic circuit shown below, is equivalent to.



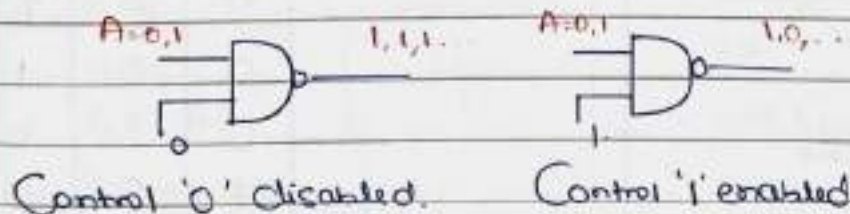
### NAND Gate.



2. Truth Table.

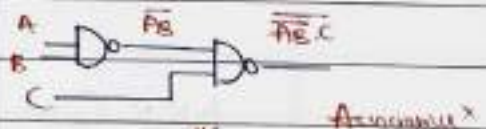
| A | B | y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

3. Enable/Disable.

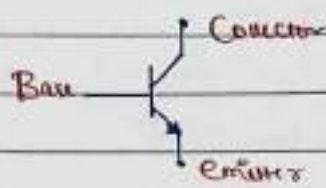


4. Commutative Law. ✓

5. Associative Law. ✗



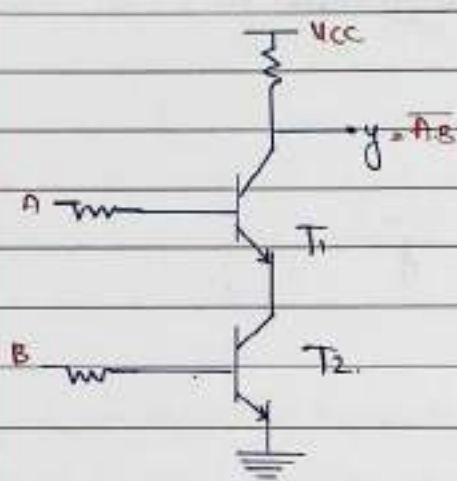
NAND follows commutative law but does not follow associative law.

Transistor

Input = 1

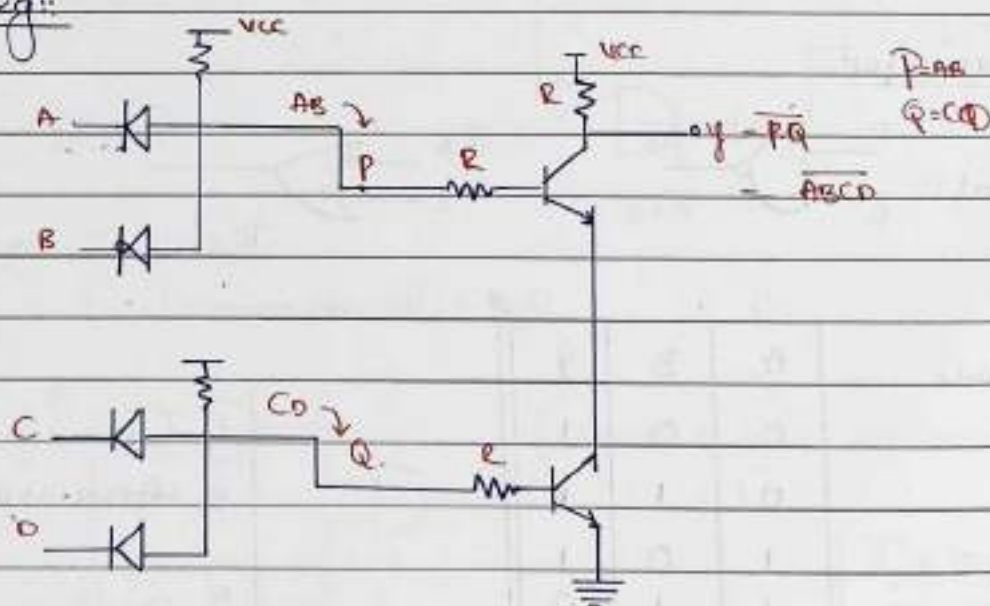
ON = Saturation  $\rightarrow$  Short Circuit

Input = 0

OFF = Cutoff  $\rightarrow$  Open Circuit

| A | B | T <sub>1</sub> | T <sub>2</sub> | y |
|---|---|----------------|----------------|---|
| 0 | 0 | Cutoff         | Cutoff         | 1 |
| 0 | 1 | Cutoff         | Saturation     | 1 |
| 1 | 0 | Saturation     | Cutoff         | 1 |
| 1 | 1 | Saturation     | Saturation     | 0 |

eg.:

NOR Gate

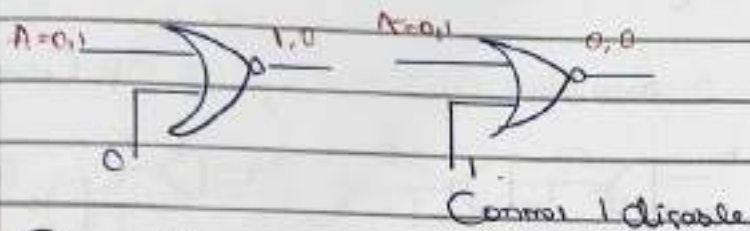
1. Symbol:

Bubbled AND = NOR

2. Truth Table:

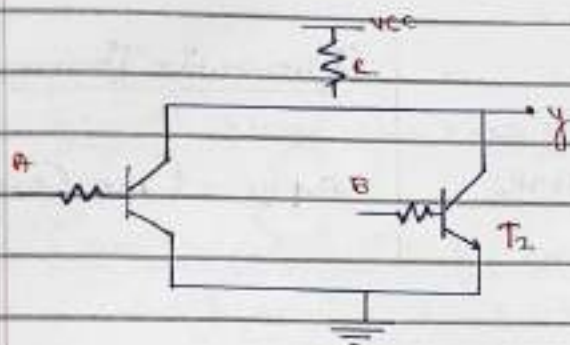
| A | B | y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

## 3. Enable/Disable



4. Commutative Law ✓

5. Associative Law ✗



| A | B | T <sub>1</sub> | T <sub>2</sub> | Y |
|---|---|----------------|----------------|---|
| 0 | 0 | Cutoff         | Cutoff         | 1 |
| 0 | 1 | Cutoff         | Saturation     | 0 |
| 1 | 0 | Saturation     | Cutoff         | 0 |
| 1 | 1 | Saturation     | Saturation     | 0 |

Note: NAND, NOR are called "Universal logic".

Q1. Which of the following option is called universal logic?

Ans. Both NAND and NOR. (option given)

Q2. Which of the following is called Universal logic? [MCQ]

Ans. NAND (if both are given choose NAND)

Q3. Which of the following Option(s) is/are called universal logic? [MCQ]

Ans.  $(A+B)$ ,  $(\bar{A} \cdot \bar{B})$ ,  $(A+B)$ ,  $A \cdot \bar{B}$ .

 $\bar{A} \cdot \bar{B}$  $\bar{A} + \bar{B}$  $\bar{A} + B$  $A + \bar{B}$  $\bar{A} \cdot B$ ,  $A \cdot \bar{B}$ 

Multiplexer

Decoder + OR logic

Universal  
Logic

X-OR Gate



X-NOR Gate

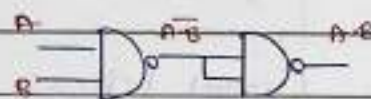


## NAND as Universal logic

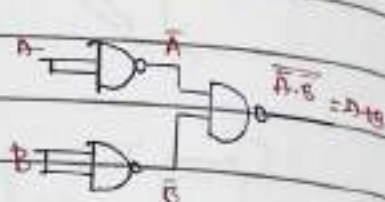
### 1. NOT Gate



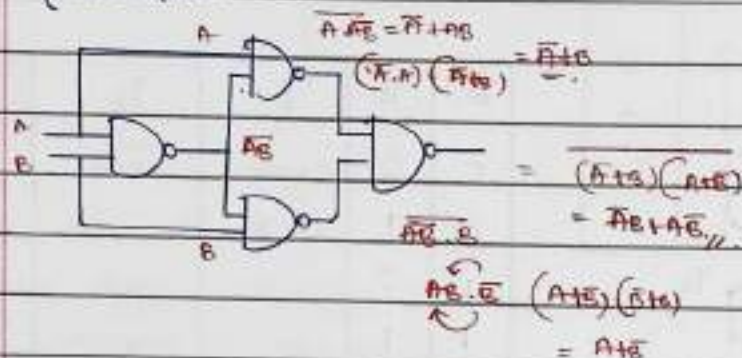
## 2. AND Gate



3. OR Care



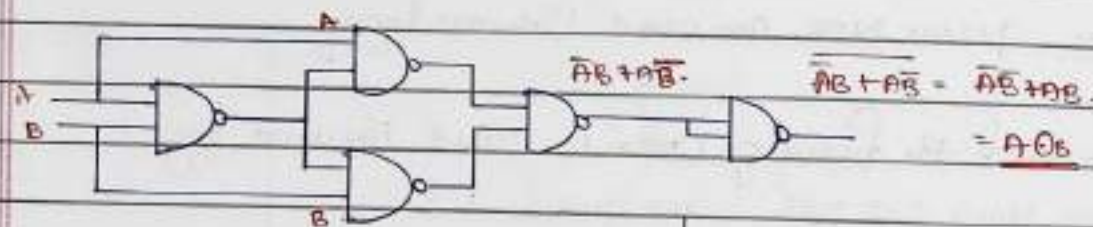
4. Хор Сан



### Distribution Theorem

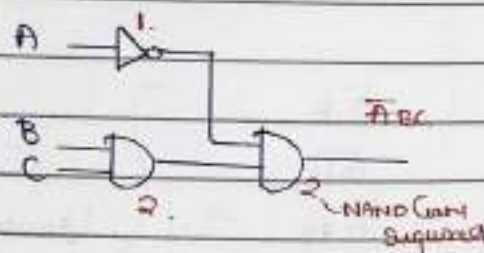
$$A + BC = (A + B)(A + C)$$

### 5. X-NOR Gate



|             |       | NAND | NOR |
|-------------|-------|------|-----|
| Table of    | NOT   | 1    | 1   |
| no. of      | AND   | 2    | 3   |
| NAND and    | OR    | 3    | 2   |
| NOR gate    | X-OR  | 4    | 5   |
| required to | X-NOR | 5    | 4   |
| make the    | NAND  | 1    | 4   |
| given Gate  | NOR   | 4    | 1   |

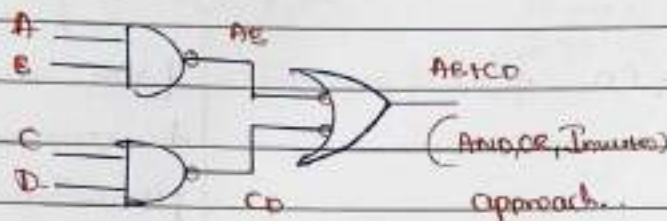
Q1. Find the minimum no. of two input NAND Gate required to implement the logic given below

$$f(A, B, C) = \bar{A}BC$$


$\therefore 1+2+2 = \underline{5}$  required

Q2 Find the minimum no. of two input NAND Gate required to implement the logic given below.

$$f(A, B, C, D) = AB + CD$$



Create using AOI approach and make it with NAND by adding inverter ('o') to balance out! approach.

∴ 3 NAND Gate required.

No. of NAND and NOR Gate Required:

Case (1):  $A \cdot B \cdot \bar{C} \cdot D \cdot E \dots$

$$\rightarrow \text{NAND} = (2n - 2) + k$$

$n$  = total no. of variables

$k$  = no. of complement variables

$$\rightarrow \text{NOR} = (3n - 3) + k$$

$n$  = no. of variables

$k$  = no. of complement variables

eg:  $f = A \cdot \bar{B} \cdot C$

$n = 3, k = 1$

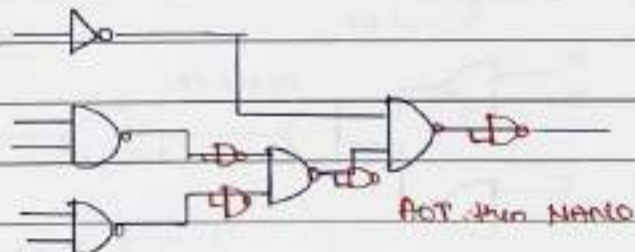
$$\text{NAND} = (2n - 2) + k \\ = (2 \times 3 - 2) + 1 = 5$$

$$\text{NOR} = (3n - 3) + k \\ = (3 \times 3 - 3) + 1 \\ = 6 - 1 = 5$$

eg:  $f = A \cdot B \cdot C \cdot D \cdot E$

$n = 5, k = 1$

$$\text{NAND} = (2 \times 5 - 2) + 1 \\ = 9$$



Case (2):  $A + B + \bar{C} + D \dots$

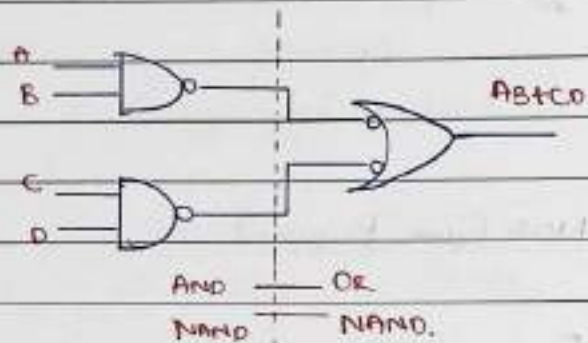
$$\text{NAND} = (3n - 3) - k$$

$$\text{NOR} = (2n - 2) + k$$

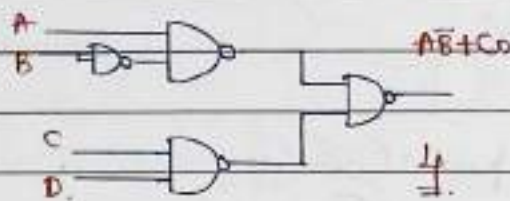
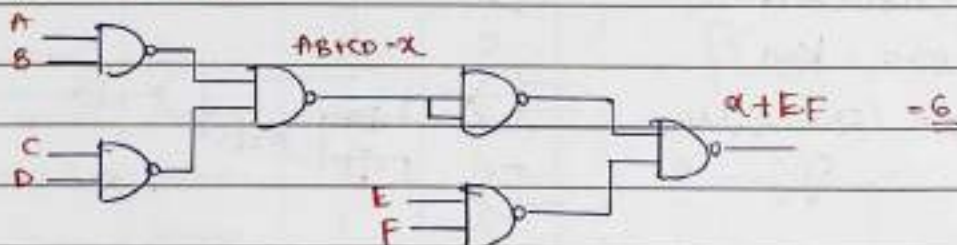
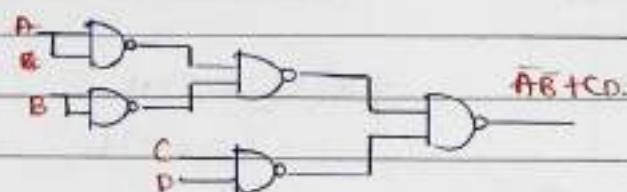
eg:  $f = A + B, n = 2, k = 1$

$$\text{NAND} = (3n - 3) - k \\ = (3 \times 2 - 3) - 1 = 2$$

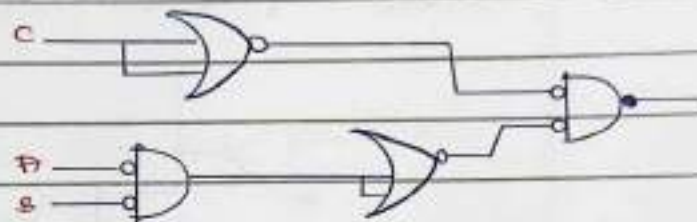
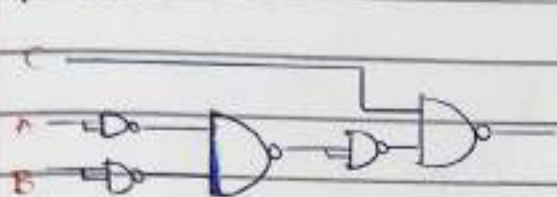
$$\text{NOR} = (2n - 2) + k \\ = (2 \times 2 - 2) + 1 \\ = 3$$

eg:  $\bar{A} + B$ Case (3)  $f = AB + CD$  [Sum of product]

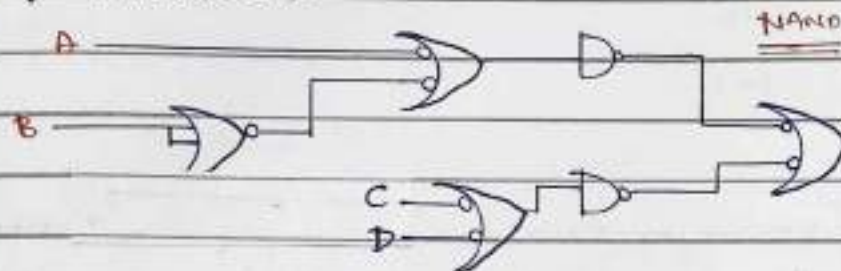
Note: Whenever in the problem the minimum number of NAND Gate are asked then write in SOP form and implement it by AOT and this is exactly equal to NAND NAND implementation.

Q  $f = AB + CD$ Q  $f = AB + CD + E$   $f = AB + CD + E$  (AB + CD)Q  $f = AB + CD + EF$ Q  $\bar{A}\bar{B} + CD$ 

Q1.  $f = \bar{A}\bar{B}C$



Q2.  $f = \bar{A} + \bar{B} + \bar{C} + \bar{D}$



NAND

$(20-3)+4$

$= (20+4)-3$

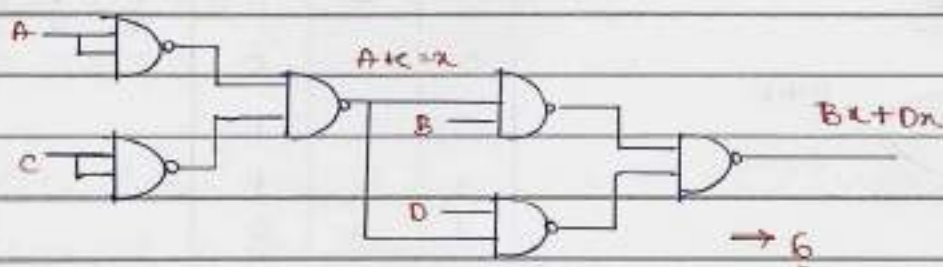
$= 24-3$

$= 21$

Q3.  $f = AB + BC + CD + DA$

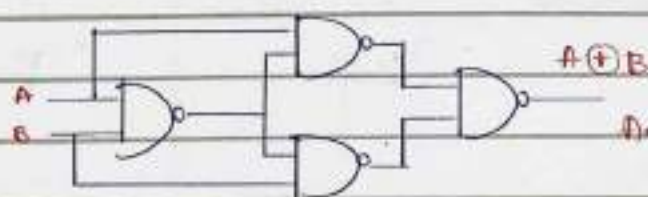
$f = B(A+C) + D(A+C)$

$f = Bx + Dx$ ,  $A+C=x$



Case (4) Most Important.

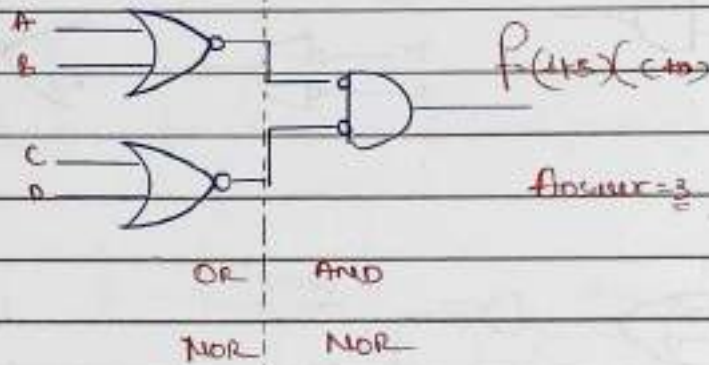
Q4.  $\bar{A}B + A\bar{B} \rightarrow \text{XOR}$



$A \oplus B$

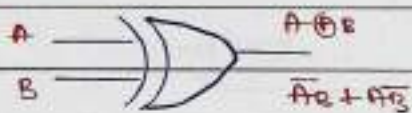
Answer: 4

Case (5)  $F = (A+B) \cdot (C+D) \rightarrow$  Product of Sum Form



Note: Whenever minimum number of Nor Gate are asked, write the function in Pos form and implement by using AND which is AND-OR Implementation which is exactly equal to the NOR implementation.

XOR Gate



| A | B | Y | Truth Table |
|---|---|---|-------------|
| 0 | 0 | 0 |             |
| 0 | 1 | 1 |             |
| 1 | 0 | 1 |             |
| 1 | 1 | 0 |             |

$$A \oplus A = 0$$

$$A \oplus 0 = A$$

$$A \oplus 1 = \bar{A}$$

$$A \oplus \bar{A} = 1$$

eg.:

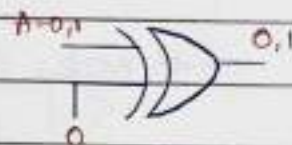
$$A \oplus B = C$$

$$A \oplus C = B$$

$$B \oplus C = A$$

$$A \oplus B \oplus C = 0$$

Enable/Disable



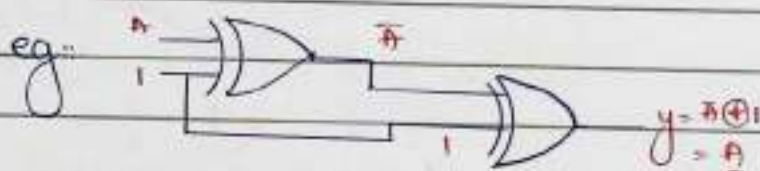
"Buffer"



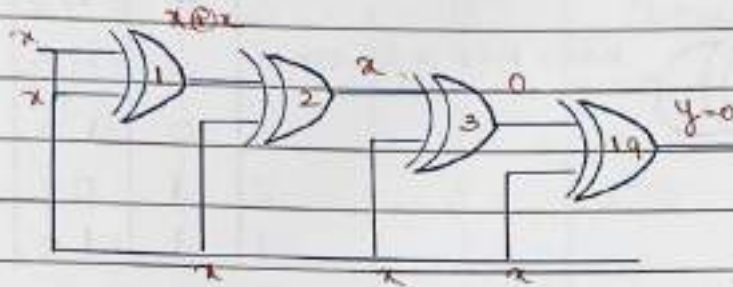
"Inverter"

$\rightarrow A \oplus A = 0$  (For even)

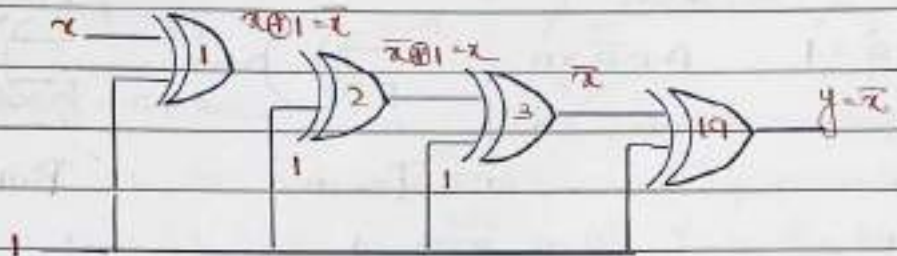
$A \oplus A \oplus A = A$  (For odd)



Q1. Find the output y.



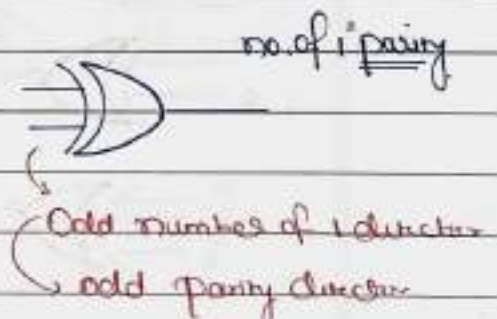
Q2. Find the output y.



- XOR Gate follows Commutative as well as associative law.
- $A \oplus B = B \oplus A$  → Commutative law
- $A \oplus B \oplus C = (A \oplus B) \oplus C$  → associative law.

- X-OR Gate output will be high when odd number of 1's are connected in the inputs.

| ABC | $A \oplus B \oplus C$ | $(A \oplus B) \oplus C$ |
|-----|-----------------------|-------------------------|
| 000 | 0                     | 0                       |
| 001 | 1                     | 1                       |
| 010 | 1                     | 1                       |
| 011 | 0                     | 0                       |
| 100 | 1                     | 1                       |
| 101 | 0                     | 0                       |
| 110 | 0                     | 0                       |
| 111 | 1                     | 1                       |



X-NOR Gate:

Symbol:   $A \oplus B = A \odot B = \bar{A}B + A\bar{B}$

| A | B | $A \odot B$ |
|---|---|-------------|
| 0 | 0 | 1           |
| 0 | 1 | 0           |
| 1 | 0 | 0           |
| 1 | 1 | 1           |

Truth Table

\*  $A=B \quad y=1$

\*  $A \neq B \quad y=0$

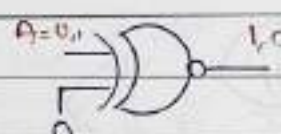
\* It is also called as "even parity detector" / "Equivalence logic" / "Equal detector" / "Coincidence logic".

\*  $A \odot A = 1$

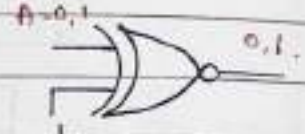
$A \odot 1 = A$

$A \odot \bar{A} = 0$

$A \odot 0 = \bar{A}$

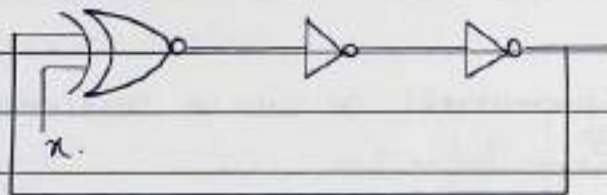


"Inverter".



"Buffer".

eg.:



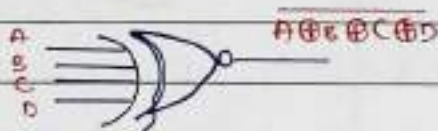
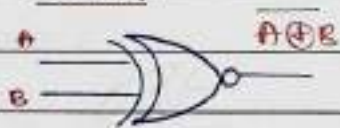
$X=0$  - Astable multivibrator

$X=1$  - Bistable multivibrator

\* X-NOR Gate follows commutative as well as associative law.

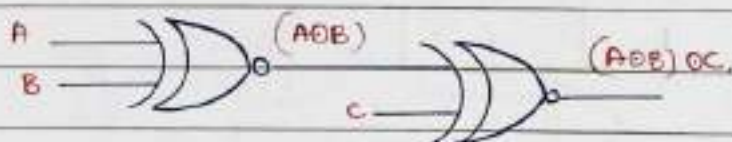
\* For even number of variable and even number of inputs.

Case 1:



} Output will be high for even no. of inputs are high.

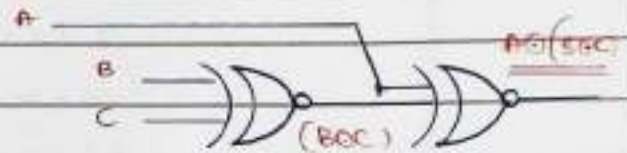
Case 2:



|    | A | B | C | $(A \odot B) \odot C$ | $A \odot (B \odot C)$ |
|----|---|---|---|-----------------------|-----------------------|
| 1. | 0 | 0 | 0 | 0                     | 0                     |
| 2. | 0 | 0 | 1 | 1                     | 1                     |
| 3. | 0 | 1 | 0 | 1                     | 1                     |
| 4. | 0 | 1 | 1 | 0                     | 0                     |
| 5. | 1 | 0 | 0 | 1                     | 1                     |
| 6. | 1 | 0 | 1 | 0                     | 0                     |
| 7. | 1 | 1 | 0 | 0                     | 0                     |
| 8. | 1 | 1 | 1 | 1                     | 1                     |

$$(A \odot B) \odot C = A \odot (B \odot C)$$

→ Associative Law



$$A \odot B = \overline{A \oplus B}$$

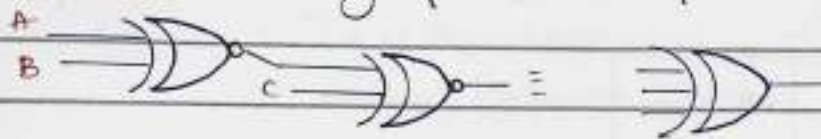
$$A \odot B \odot C \odot D = \overline{A \oplus B \oplus C \oplus D}$$

for even! Variables

$$A \odot B \odot C = A \oplus B \oplus C$$

$$(((A \odot B) \odot C) \odot D) \odot E = A \oplus B \oplus C \oplus D \oplus E$$

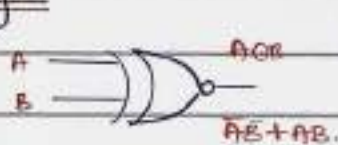
\* Output will be high for odd no. of 1's in the input.



\* Output will be high for even no. of 1's in input

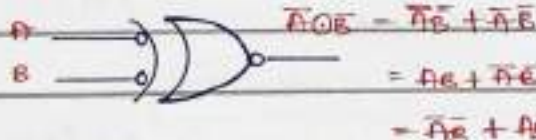


Tragic



$$\overline{A \oplus B} = \overline{A \oplus B}$$

$$\overline{A \oplus B} = \overline{A \oplus B}$$



Same!

$$A \odot B = \overline{A \oplus B} = \overline{A \oplus B} = \overline{A \oplus B} = \overline{A \oplus B}$$

$$A \oplus B = \overline{A \odot B} = \overline{A \odot B} = \overline{A \odot B} = \overline{A \odot B}$$

Q1. The boolean function given:  $f(A, B) = A \oplus B \oplus AB$   
Which Statement is/are correct?

$$f(A, B) = (A \oplus B) \oplus AB$$

$$= 1 \oplus AB$$

$$A \oplus B = 1 = \bar{A}B + A\bar{B}$$

$$= \bar{1} \cdot AB + 1 \cdot \bar{AB}$$

$$\bar{A \oplus B} = \bar{1} = A \oplus B = \bar{A}B + A\bar{B}$$

$$= [\bar{A}B + AB] \cdot AB + [\bar{A}B + AB] \cdot [\bar{A} + \bar{B}]$$

$$= \bar{A}B \cdot AB + AB \cdot AB + \bar{A}B \cdot \bar{A} + \bar{A}B \cdot \bar{B} + AB \cdot \bar{A} + AB \cdot \bar{B}$$

$$= AB + \bar{A}B + AB$$

$$= AB + \bar{A}B + \bar{A}B$$

$$= A[B + \bar{B}] + \bar{A}B$$

$$= A \cdot 1 + \bar{A}B \Rightarrow A + \bar{A}B = (A + \bar{A})(A + B) = A + B \quad (\text{OR})$$

Ans:  $\rightarrow$  It is a OR Gate

$\rightarrow$  It requires 3 NAND Gate to implement the function.

| A B | AB | $A \oplus B \oplus C$ |
|-----|----|-----------------------|
| 0 0 | 0  | 0                     |
| 0 1 | 0  | 1                     |
| 1 0 | 0  | 1                     |
| 1 1 | 1  | 1                     |

$$f = A + B$$

Q2. Minimized expression with to  $Y = A \oplus (A + B)$

$$A + B = 1$$

$$A \oplus 1$$

$$\bar{A} + A\bar{1}$$

$$\bar{A}(A+B) + A[\bar{A} + \bar{B}]$$

$$\bar{A}A + \bar{A}B + A\bar{A} + A\bar{B}$$

$$= \bar{A}B$$

$\rightarrow$  Associative law

does not follow

by  $\bar{x}$ -or Gate and  
OR Gate.

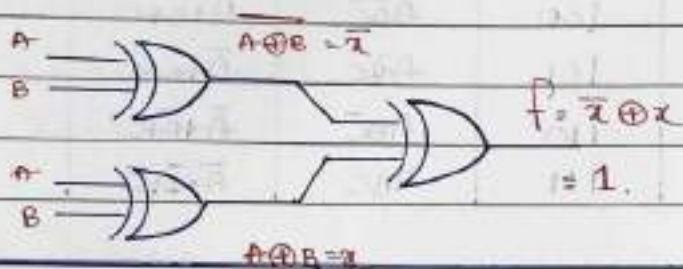
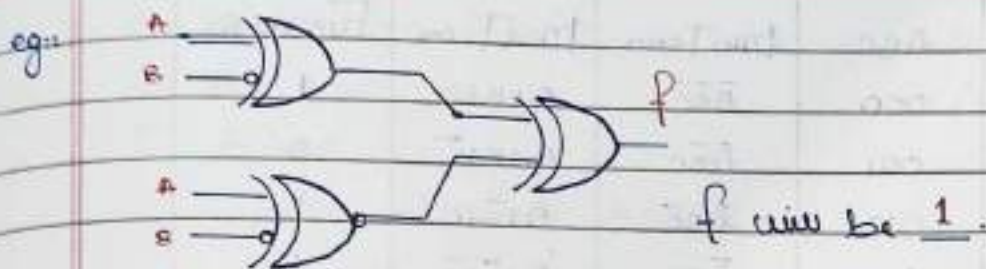
$$A \oplus (A + B)$$

$$(A \oplus A) + (A \oplus B)$$

$$0 + A \oplus B$$

$$A \oplus B$$

| A B | A | $\bar{A+B}$<br>X | $A \oplus X$ | $A \oplus (A+B)$<br>$A \oplus 1$ |
|-----|---|------------------|--------------|----------------------------------|
| 0 0 | 0 | 0                | 0            |                                  |
| 0 1 | 0 | 1                | 1            |                                  |
| 1 0 | 1 | 1                | 0            |                                  |
| 1 1 | 1 | 1                | 0            |                                  |



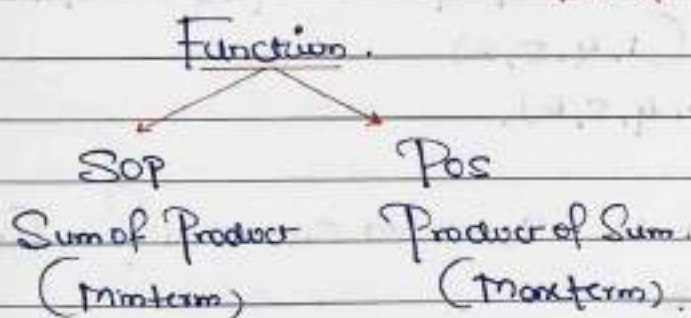
## Law of Boolean Algebra

Boolean function: It is the combination of inputs on which output depends.

eg:  $f(A, B) = \underline{AB} + \underline{\bar{A}\bar{B}}$

No. of variables = 2

No. of terms = 2, Literals = 4



## Standard Canonical Form

Each term should contain all the variables

eg:  $f(A, B) = \bar{A}B + A\bar{B}$  ✓  
 $f(A, B) = A + \bar{A}B$

| Decimal | ABC | Min Term                | Max Term                  | Functions |
|---------|-----|-------------------------|---------------------------|-----------|
| 0       | 000 | $\bar{A}\bar{B}\bar{C}$ | $A+B+C$                   | 1         |
| 1       | 001 | $\bar{A}\bar{B}C$       | $A+B+\bar{C}$             | 0         |
| 2       | 010 | $\bar{A}B\bar{C}$       | $A+\bar{B}+C$             | 1         |
| 3       | 100 | $\bar{A}BC$             | $A+\bar{B}+\bar{C}$       | 1         |
| 4       | 100 | $A\bar{B}\bar{C}$       | $\bar{A}+B+C$             | 0         |
| 5       | 101 | $A\bar{B}C$             | $\bar{A}+B+\bar{C}$       | 0         |
| 6       | 110 | $AB\bar{C}$             | $\bar{A}+\bar{B}+C$       | 0         |
| 7       | 111 | $ABC$                   | $\bar{A}+\bar{B}+\bar{C}$ | 1         |

Standard Canonical SOP form:

$$\begin{aligned}
 f(A,B,C) &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + ABC \\
 &= m_0 + m_2 + m_3 + m_7 \\
 &= \sum m(0, 2, 3, 7) \\
 &= \sum (0, 2, 3, 7)
 \end{aligned}$$

Standard Canonical Pos' form:

$$\begin{aligned}
 F(A,B,C) &= (A+B+C)(\bar{A}+B+C)(\bar{A}+B+\bar{C})(\bar{A}+\bar{B}+C) \\
 &= \pi_{11} \cdot \pi_{19} \cdot \pi_{12} \cdot \pi_{16} \\
 &= \pi M(1, 4, 5, 6) \\
 &= \pi (1, 4, 5, 6)
 \end{aligned}$$

Q1. No. of terms present in Standard Canonical form (SOP)?

Ans.  $f(A,B,C) = A+B+C$

$$\begin{aligned}
 &= A(\bar{B}+B)(\bar{C}+C) + (\bar{A}+A)BC \\
 &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC + \bar{A}BC
 \end{aligned}$$

$\therefore$  5 terms are present.

① Distribution Theorem

$$\begin{aligned}
 &(A+B)(A+C) \\
 &= A.A + A.C + AB + BC \\
 &= A + AC + AB + BC
 \end{aligned}$$

$$= A[1+C+B] + BC$$

$$= A \cdot 1 + BC$$

$$= A + BC$$

$$A + BC = (A+B) \cdot (A+C)$$

$$\text{eg: } A + \overline{A}B$$

$$(A + \overline{A})(A + B)$$

$$= 1(A + B)$$

$$= A + B$$

$$\text{Q2. } f(A, B) = \overline{A} + \overline{A}B$$

$$= (\overline{A} + A)(\overline{A} + B)$$

$$= \overline{A} + B$$

### ② Consensus Theorem

$$f(A, B, C) = AB + \overline{A}C + BC$$

$$= AB + \overline{A}C + (\overline{A}B + A)BC$$

$$= AB + \overline{A}C + \overline{A}BC + ABC$$

$$= AB[1+C] + \overline{A}C[1+B]$$

$$= AB + \overline{A}C$$

$BC \rightarrow$  Redundant term

\* 3 variable function

\* each term consist of 2 variables.

\* each variable repeated twice except one

\* One variable repeated in form of complement.

$$\text{eg: } AB + \overline{B}C + AC$$

$$\Rightarrow AB + \overline{B}C$$

$AC \rightarrow$  redundant.

### ③ Transpose Theorem

$$(A+B)(\overline{A}+C)$$

$$= A\overline{A} + AC + \overline{A}B + BC$$

$$= AC + \overline{A}B + BC$$

$$= AC + \overline{A}B$$

$$(A+B)(\overline{A}+C)$$

$$= AC + \overline{A}B$$

### ④ De Morgan's Law

$$\overline{ABC} = \overline{A} + \overline{B} + \overline{C}$$

$$\overline{A+B+C} = \overline{A} \cdot \overline{B} \cdot \overline{C}$$

Q1. Find min. no of NAND Gate required to implement  $f(A, B, C) = A + ABC + A\overline{B}C$ .

$$A[1+BC+\overline{B}C] = A$$

$\therefore$  0 NAND Gate Required

Q2. Find the minimum no. of NAND gate required to implement the boolean function

Ans 
$$f(A, B, C) = A + ABC + ABC$$

$$= A [1 + BC + BC]$$

$$= A$$

$$\therefore 0 \text{ NAND Gates required}$$

Q3. Minimize the expression  $f(A, B) = A + AB$

Ans 
$$= A [1 + B]$$

$$= A$$

Q4. Minimize the expression  $f(A, B) = \bar{A}\bar{B} + \bar{A}B + AB$

Ans 
$$= \bar{A} [\bar{B} + B] + AB$$

$$= \bar{A} + AB = (\bar{A} + A)(\bar{A} + B)$$

$$= \bar{A} + B$$

Q5. Minimize the expression  $f(A, B) = \bar{A}\bar{B} + \bar{A}B + A\bar{B} + AB$

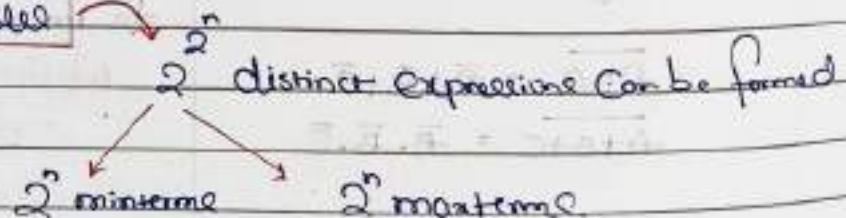
Ans 
$$= \bar{A} [\bar{B} + B] + A [\bar{B} + B]$$

$$= \bar{A} + A \Rightarrow 1$$

| n=1       |     | n=2              |                     |           |                             |
|-----------|-----|------------------|---------------------|-----------|-----------------------------|
| A         | } 4 | $\bar{A}\bar{B}$ | $\bar{A} + \bar{B}$ | A         | 0                           |
| $\bar{A}$ |     | $\bar{A}B$       | $\bar{A} + B$       | $\bar{A}$ | 1                           |
| 1         |     | $A\bar{B}$       | $A + \bar{B}$       | B         | $\bar{A}B + A\bar{B}$       |
| 0         |     | $AB$             | $A + B$             | $\bar{B}$ | $\bar{A}\bar{B} + A\bar{B}$ |
|           |     | minterms         |                     | maxterms  |                             |

16 different terms

"n" variables



Q6. Minimize the expression  $f(A, B) = \bar{A}B + A\bar{B} = A \oplus B$ .

Ans. Already minimized

Q7. Minimize the expression:  $f(A, B) = AB + \bar{A}C + BC$

function written in wrong format.

$$\begin{aligned} \text{eg: } f(A, B, C) &= \overset{001}{\bar{A}\bar{B}C} + \overset{010}{\bar{A}B\bar{C}} + \overset{100}{A\bar{B}\bar{C}} + \overset{111}{ABC} \\ &= \sum (1, 2, 4, 7) \\ &= A \oplus B \oplus C \quad (\text{Already minimized}) \end{aligned}$$

$$\begin{aligned} \text{eg: } f(B, A) &= \overset{01}{\bar{A}B} + \overset{10}{A\bar{B}} + \overset{11}{AB} = \sum m(1, 2, 3) \times \\ &\quad \uparrow \uparrow \\ &= B\bar{A} + \bar{B}A + BA = \sum m(1, 2, 3) \checkmark \end{aligned}$$

note the order, function should be written in same order.

Q8. Minimize the expression:  $f(A, B, C) = \bar{A}\bar{B} + \bar{A}C + \bar{B}\bar{C}$

Ans.  $= \bar{A}C + \bar{B}\bar{C}$

Q9. Minimize the expression:  $f(A, B, C) = (A+B)(A+C)(\bar{B}+C)$

Ans.  $= (A+B)(\bar{B}+C)$

Q10. Write the function for truth table and minimize it.

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

SOP form:

$$Y = \bar{A}\bar{B} + \bar{A}B\bar{C} + A\bar{B}C + ABC$$

$$Y = \bar{A}\bar{B} + A\bar{B} + AB$$

$$Y = \bar{A}\bar{B} + A[\bar{B}+B]$$

$$Y = A + \bar{A}\bar{B} = (A + \bar{A})(A + \bar{B})$$

$$Y = A + \bar{B}$$

Pos:

$$Y = (A+B+1) \cdot (A+B\bar{C}) \cdot (\bar{A}+1)$$

$$= 1(A+B) \cdot 1 \cdot 1$$

$$= A + B$$

Q11. Write the function for truth table and minimize it.

Ans.

| A | B | Y         |
|---|---|-----------|
| 0 | 0 | C         |
| 0 | 1 | $\bar{C}$ |
| 1 | 0 | 1         |
| 1 | 1 | 1         |

Sol:

$$Y: \bar{A}\bar{B}C + \bar{A}B\bar{C} + AB + AB$$

Q12. Two way switch is an example of which logic?

Ans. X-OR

Q13. If we have 4 variable, then total different expressions will be?

Ans.

$$2^{2^n} = 2^{2^4} = 2^{16} = 2^6 \times 2^{10} \\ = \underline{64K}$$

Q14.  $A + BC + \bar{A}C$  is equal to :

$$A + BC + \bar{A}C$$

$$= (A + \bar{A})(A + C) + BC$$

$$= A + C + BC$$

$$= A + C [1 + B]$$

$$= \underline{A + C} \text{ (not in option!)}$$

eg:  $f = AB + \bar{A}\bar{B} + ABC + \bar{A}\bar{B}\bar{C}$

$$= AB[1 + C] + \bar{A}\bar{B}[1 + \bar{C}]$$

$$= AB + \bar{A}\bar{B} \text{ — (Semi minimised) } \rightarrow AB + \bar{A}\bar{B} + \bar{A}\bar{B}C$$

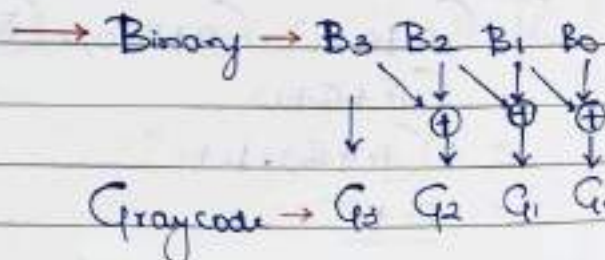
Modified Venned Diagram

→ It is also known as 'K-map'.

Minimization by K-map

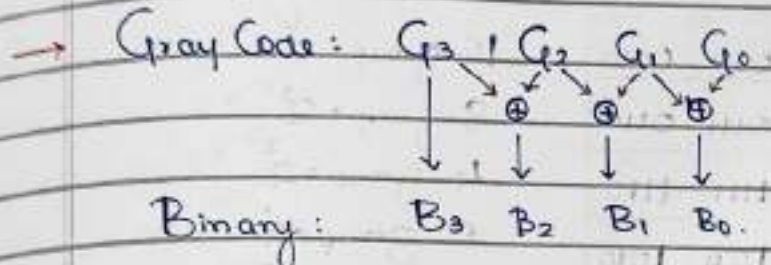
\* Based on graycode

\* Graycode:



eg: Binary: 0110101  
Gray code: 0101111

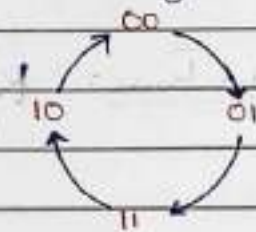
eg: Binary: 111001011  
Gray code: 100101110



eg: Gray Code: 1101011  
Binary: 1001101

Gray Code: Gray code is the code in which successive numbers differ by 1 bit.

| Decimal | Binary | Gray Code |
|---------|--------|-----------|
| 0       | 00     | 00        |
| 1       | 01     | 01        |
| 2       | 10     | 11        |
| 3       | 11     | 10        |



1. Unity Hamming distance code
2. Cycle code
3. Reflecting code

| Decimal | Binary | Gray Code |
|---------|--------|-----------|
| 0       | 000    | 000       |
| 1       | 001    | 001       |
| 2       | 010    | 011       |
| 3       | 011    | 010       |
| 4       | 100    | 110       |
| 5       | 101    | 111       |
| 6       | 110    | 101       |
| 7       | 111    | 100       |

\* eg:  $P(A, B)$

MSB (Most Significant bit)      LSB (Least Significant bit)

|             | $\bar{B}$                | $B$                |
|-------------|--------------------------|--------------------|
| $\bar{A}$ 0 | $\bar{A}\bar{B}$<br>00 @ | $\bar{A}B$<br>01 @ |
| $A$ 1       | $A\bar{B}$<br>10 @       | $AB$<br>11 @       |

→  $f(A, B, C)$

|             | $\bar{B}\bar{C}$ | $\bar{B}C$ | $BC$     | $\bar{B}\bar{C}$ |
|-------------|------------------|------------|----------|------------------|
| $\bar{A}$ 0 | 000<br>0         | 001<br>1   | 011<br>3 | 010<br>2         |
| $A$ 1       | 100<br>4         | 101<br>5   | 111<br>7 | 110<br>6         |

$f(A, B, C, D)$ \* Group can be formed in order of  $2^n$ .

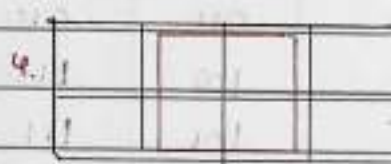
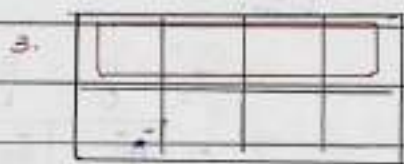
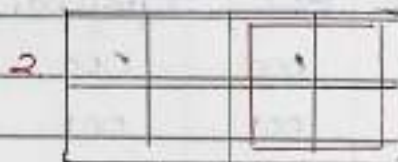
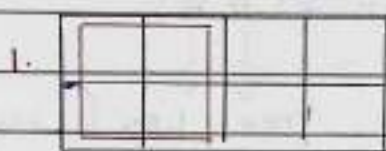
|                  |    | $\bar{C}\bar{D}$ | $\bar{C}D$ | $CD$ | $C\bar{D}$ |
|------------------|----|------------------|------------|------|------------|
|                  |    | 00               | 01         | 11   | 10         |
| $\bar{A}\bar{B}$ | 00 | 0000             | 0001       | 0011 | 0010       |
| $\bar{A}B$       | 01 | 0100             | 0101       | 0111 | 0110       |
| $A\bar{B}$       | 11 | 1100             | 1101       | 1111 | 1110       |
| $AB$             | 10 | 1000             | 1001       | 1011 | 1010       |

16 group =  $2^4 \rightarrow$  4 variables reduce8 group =  $2^3 \rightarrow$  3 variables reduce4 group =  $2^2 \rightarrow$  2 variables reduce2 group =  $2^1 \rightarrow$  1 variable reduce1 group =  $2^0 \rightarrow$  0 variable reduceRule of Minimization

Terms will minimize

Cy Baba Rule: Kam se kam group banana hai and bade se bade group banana hai.

Variable will minimize.

Quad (making group)

And many similar combinations possible

etc

Q1.  $f(A, B) = \bar{A}\bar{B} + \bar{A}B + AB = \sum m(0, 1, 3)$

|           |   | $\bar{B}$ | $B$ |
|-----------|---|-----------|-----|
|           |   | 0         | 1   |
| $\bar{A}$ | 0 | 1         | 1   |
| $A$       | 1 | 0         | 1   |

$$f(A, B) = \bar{A}\bar{B} \cdot 1 + \bar{A}B \cdot 1 + A\bar{B} \cdot 0 + AB \cdot 1$$

$$\Rightarrow \bar{A}B$$

Q2.  $f(A,B) = \bar{A}\bar{B} + \bar{A}B + A\bar{B} + AB = \sum m(0,1,2,3)$

| $A \backslash B$ | 0 | 1 |
|------------------|---|---|
| $\bar{A}$ 0      | 1 | 1 |
| A 1              | 1 | 1 |

 $\Rightarrow 1$ 

Q3.  $f(A,B,C) = \sum m(0,1,3,6,7)$

| $A \backslash BC$ | 00 | 01 | 11 | 10 |
|-------------------|----|----|----|----|
| $\bar{A}$ 0       | 1  | 1  | 1  |    |
| A 1               |    |    | 1  | 1  |

$\bar{A}\bar{B}$  (group 00),  $\bar{A}B$  (group 01, 11),  $AB$  (group 11, 10),  $\bar{A}C$  (group 00, 01),  $\bar{A}B$  (group 01, 11),  $\bar{A}C$  (group 11, 10)

| $A \backslash BC$ | 00 | 01 | 11 | 10 |
|-------------------|----|----|----|----|
| $\bar{A}$ 0       | 1  | 1  | 1  |    |
| A 1               |    |    | 1  | 1  |

$$= \bar{A}\bar{B} + \bar{A}B + \bar{A}C$$

Semi minimized expression.

| $A \backslash BC$ | 00 | 01 | 11 | 10 |
|-------------------|----|----|----|----|
| $\bar{A}$ 0       | 1  | 1  | 1  |    |
| A 1               |    |    | 1  | 1  |

$\bar{A}\bar{B}$  (group 00),  $\bar{A}B$  (group 01, 11),  $AB$  (group 11, 10),  $\bar{A}C$  (group 00, 01),  $\bar{A}B$  (group 01, 11),  $\bar{A}C$  (group 11, 10)

$$= \bar{A}\bar{B} + \bar{A}B + \bar{A}C$$

## Question Practice Session

Q1. If  $x \odot y = \bar{x} + y$  and  $z = x \odot y$  Then  $z \odot y$  will be:

$$\begin{aligned} z \odot y &= \bar{z} + y = \overline{x \odot y} + y \\ &= \overline{\bar{x} + y} + y = \bar{x} \cdot \bar{y} + y \\ &= \bar{x} \bar{y} + y \\ &= (\bar{x} y) (\bar{y} + y) \\ &= \underline{\underline{\bar{x} y}} \end{aligned}$$

Q2. If  $A * B = AB + \bar{A} \bar{B}$  and  $C = A * B$ . Then which one is correct?

$$A \odot B = AB + \bar{A} \bar{B}$$

$$C = A \odot B$$

$$B \odot C = A$$

$$(-A \odot B) \odot (A \odot B)$$

$$x \odot x = 1$$

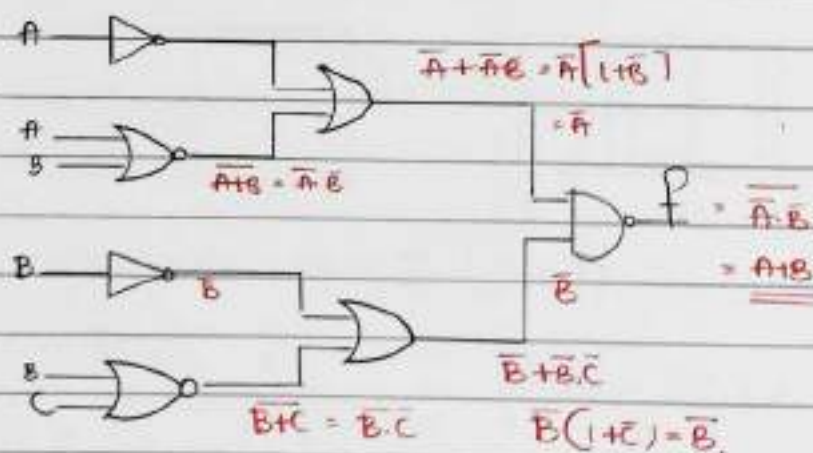
$$(i) A = B * C \checkmark$$

$$(ii) B = A * C \checkmark$$

$$(iii) A \odot B \odot C = 1 \times$$

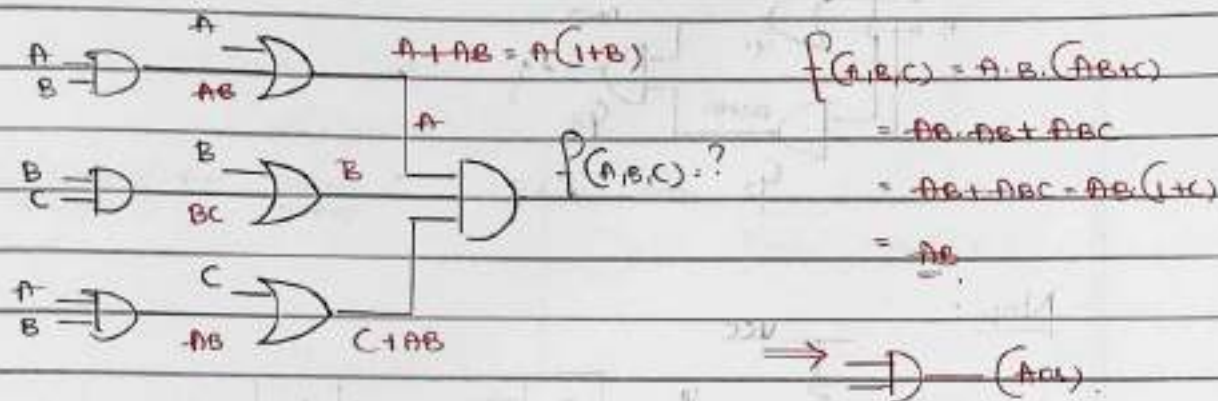
$$(iv) A = B \times$$

Q3. The output for the given logic circuit will be:

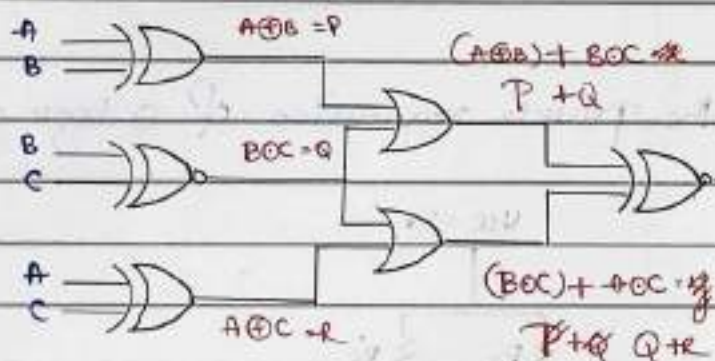


$$\rightarrow \underline{\underline{F = A+B}}$$

Q4. Consider the given logic circuit with the inputs  $A, B$  and  $C$  then  $f(A, B, C)$  will be.



Q5. The output for the given logic circuit will be:



$$f = \bar{P}\bar{Q}\bar{R} + P\bar{Q}\bar{R} + P\bar{Q}R + P\bar{Q}R + P\bar{Q}R + P\bar{Q}R + P\bar{Q}R + P\bar{Q}R$$

$$f = \bar{P}\bar{Q}\bar{R} + Q[P+R] + PR$$

$$f = \bar{P}\bar{Q}\bar{R} + Q + PR$$

$$= \bar{P}\bar{Q}\bar{R} + Q + PR$$

$$= (\bar{P}\bar{Q} + Q) + PR$$

$$f = \bar{P}\bar{Q} + PR + Q \Rightarrow (P\bar{Q}) + Q$$

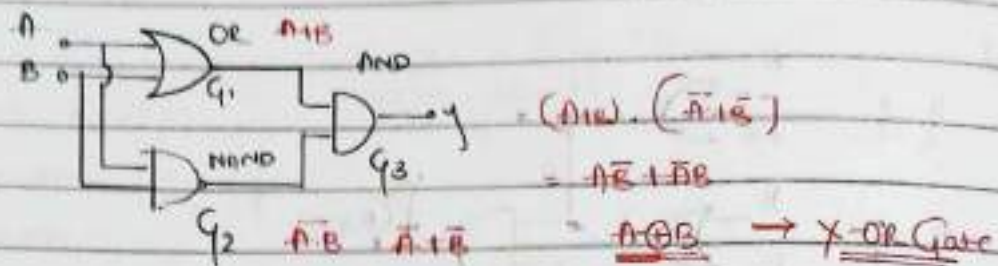
| A \ BC | 00 | 01 | 11 | 10 |
|--------|----|----|----|----|
| 0      | 1  |    | 1  |    |
| 1      | 1  |    | 1  |    |

$$f = \bar{P}\bar{Q} + PR$$

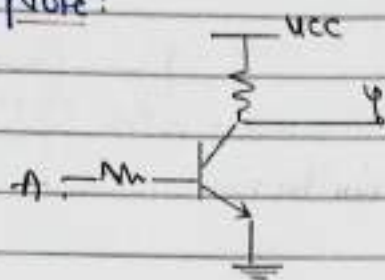
$$= \underline{\underline{B\bar{C}}}$$

| ABC | P<br>A+B | Q<br>B+C | R<br>A+C | x | y | x⊕y |
|-----|----------|----------|----------|---|---|-----|
| 000 | 0        | 0        | 0        | 1 | 1 | 1   |
| 001 | 0        | 0        | 1        | 0 | 1 | 0   |
| 010 | 1        | 0        | 0        | 1 | 0 | 0   |
| 011 | 1        | 1        | 1        | 1 | 1 | 1   |
| 100 | 1        | 0        | 0        | 1 | 0 | 0   |
| 101 | 1        | 0        | 1        | 0 | 1 | 0   |
| 110 | 0        | 1        | 1        | 0 | 1 | 0   |
| 111 | 0        | 1        | 0        | 1 | 1 | 1   |

Q6. The following logic gate circuit equivalent to



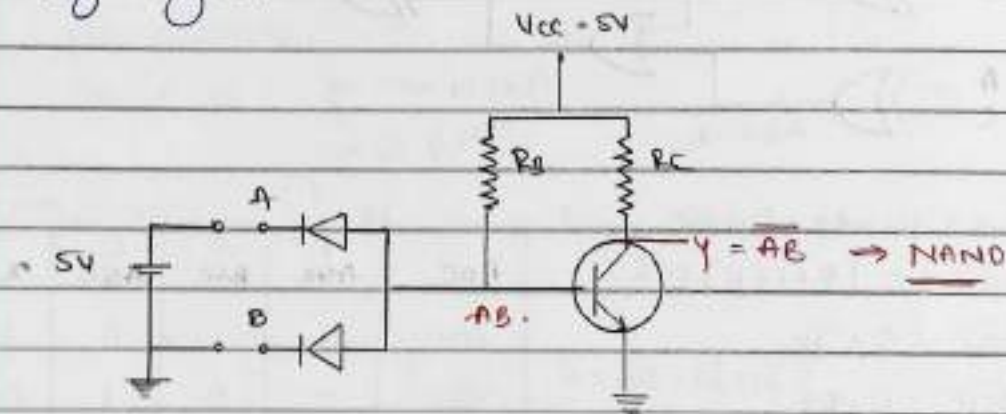
Note:



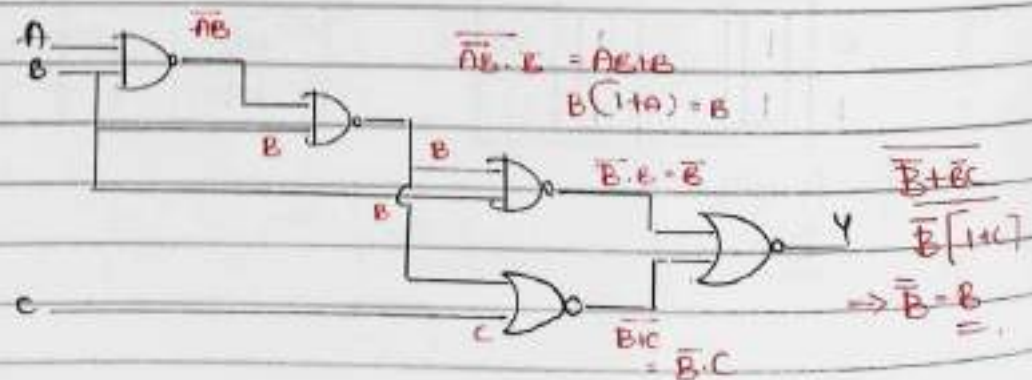
| A | Transistor | Y |
|---|------------|---|
| 0 | cutoff     | 1 |
| 1 | Saturation | 0 |

NOT Gate

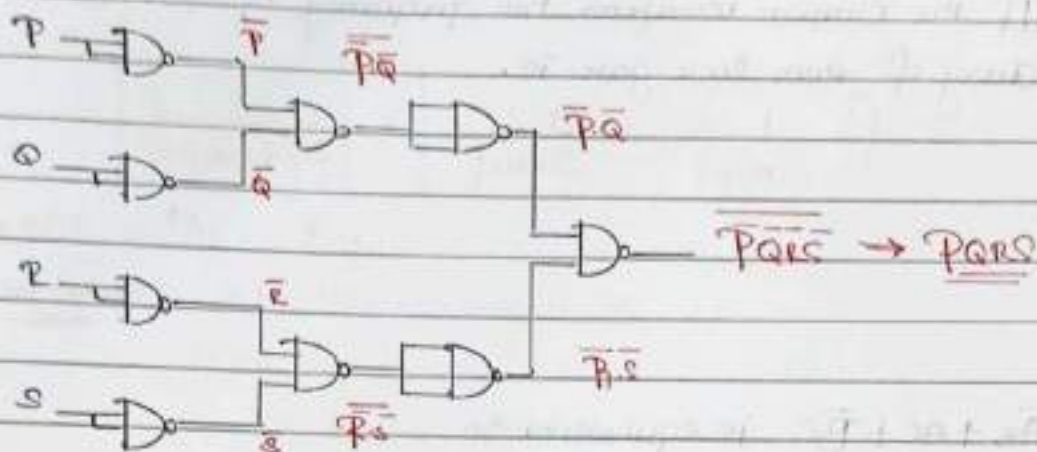
Q7. Figure shows the particle realization of a logic gate, Identify the logic gate.



Q8. For the logic circuit shown, the simplified boolean expression for the output y is.

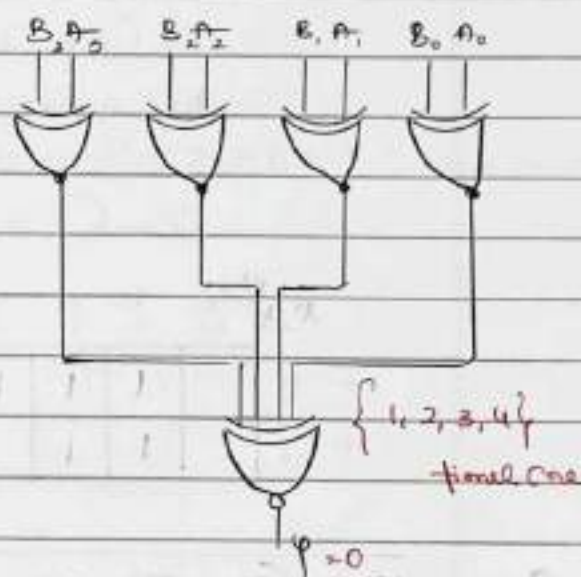


Q13. For the circuit shown in fig. the boolean expression for the output  $y$  in terms of inputs  $P, Q, R$  and  $S$  is.



Q14.  $A_3, A_2, A_1, A_0, B_3, B_2, B_1, B_0$  is shown in figure. To get output  $y=0$ , Choose one pair of correct input numbers.

$\Rightarrow 0010, 1011$

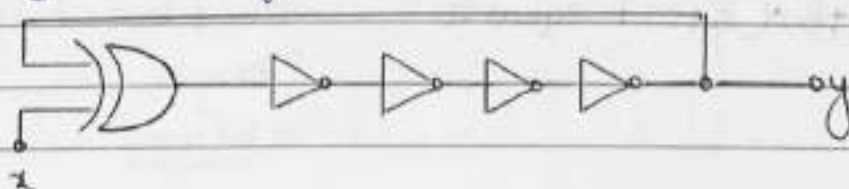


Q15. The minimum number of two input NAND gate required to implement  $y = abcd$  is.

$$y = abcd$$

$$NAND = (2n-2) + k = (2 \times 4 - 2) + 0 = 6$$

Q16. All the logic gates in the circuit shown below, have equal finite propagation delay.



Q17. The circuit can be used as a clock generator, if  $X=1$

Q18. If the output waveform has frequency of 10 MHz the propagation delay of each logic gate is.

$$f = \frac{1}{2N\tau_d} = \frac{1}{2 \times 11 \times \tau_d} = \frac{1}{2 \times 5 \times 10 \times 10^{-6}}$$

$$= \frac{10^{-7}}{10} = 10^{-8} = 10 \times 10^{-9}$$

$$= \underline{\underline{10 \text{ ns}}}$$

Q19.  $\bar{A}B + AC + \bar{B}C$  is equivalent to

$$\begin{aligned} f &= \bar{A}B + AC + \bar{B}C \\ &= \bar{A}B(\bar{C}+C) + A(\bar{B}+B)C + (\bar{A}+A)BC \\ &= \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}C + ABC + \bar{A}B\bar{C} + A\bar{B}C \\ &= \sum m(2, 3, 5, 7, 1) \\ &= \sum m(1, 2, 3, 5, 7) \end{aligned}$$

| A \ BC         |                        |                  |                  |            |
|----------------|------------------------|------------------|------------------|------------|
|                | $\bar{B}\bar{C}$<br>00 | $\bar{B}C$<br>01 | $B\bar{C}$<br>11 | $BC$<br>10 |
| $\bar{A}$<br>0 |                        | 1                | 1                | 1          |
| A<br>1         |                        | 1                | 1                |            |

$\Rightarrow C + \bar{A}B$

Q20.  $\overline{A\bar{B} + (\bar{A}B + \bar{B}C + A\bar{B}D + A\bar{B}\bar{D})}$  is equal to

$$\begin{aligned} &\overline{A\bar{B} + \bar{A}B + \bar{B}C + A\bar{B}D + A\bar{B}\bar{D}} \\ &= \overline{A\bar{B} [1 + D + \bar{D}] + \bar{A}B + \bar{B}C} \\ &= \overline{A\bar{B} + \bar{A}B + \bar{B}C} \\ &= \overline{A\bar{B} \cdot \bar{A}B \cdot \bar{B}C} \\ &= (\bar{A}+A)(\bar{A}+\bar{B})(\bar{B}+C) \\ &= (\bar{A}\bar{A} + \bar{A}B + A\bar{B} + BA)(\bar{B}+C) \\ &= (\bar{A}\bar{B} + AB)(\bar{B}+C) \\ &= \bar{A}\bar{B}\bar{B} + \bar{A}\bar{B}C + A\bar{B}\bar{B} + ABC \\ &= \bar{A}\bar{B}[1+\bar{C}] + \bar{A}\bar{B}C \\ &\Rightarrow \underline{\underline{A\bar{B} + \bar{A}\bar{B}C}} \end{aligned}$$

Q21.  $\overline{(\bar{A}+B)(\bar{B}+C)}$  is equal to :

$$\begin{aligned} &\overline{\bar{A}+B} + \overline{\bar{B}+C} \\ &= \bar{A}B + \bar{B}C \\ &\Rightarrow \underline{\underline{B(A+C)}} \end{aligned}$$

Q22.  $\bar{A}\bar{B} + AC + \bar{B}C$  is equivalent to

$$\bar{A}\bar{B}(\bar{C}+C) + A(\bar{B}+B) + (\bar{A}+A)\bar{B}C$$

$$\Sigma m(0, 1, 5, 7)$$

| A \ BC | 00 | 01 | 11 | 10 |
|--------|----|----|----|----|
| 0      | 1  | 1  |    |    |
| 1      |    | 1  | 1  |    |

$$\Rightarrow \underline{\bar{A}\bar{B} + AC}$$

Q23.  $(A+B)(A+C)(A+\bar{C})$  is equivalent to

$$(A+BC)(A+\bar{C})$$

$$= A + BC\bar{C}$$

$$\Rightarrow \underline{A}$$

HW

Q24. A logical function is given as

$$F(A, B, C, D) = B\bar{C}[A + B\bar{C}D + \bar{B}CD + \bar{A}B\bar{C} + \bar{A}B\bar{C}] \text{ is equivalent to}$$

$$Q25. f(A, B, C) = \Sigma m(0, 1, 3, 5, 6, 7)$$

| A \ BC | 00 | 01 | 11 | 10 |
|--------|----|----|----|----|
| 0      | 1  | 1  | 1  |    |
| 1      |    | 1  | 1  | 1  |

$$= \bar{A}\bar{B} + AB + C$$

$$Q26. f(A, B, C) = \Sigma m(0, 2, 4, 6)$$

| A \ BC | 00 | 01 | 11 | 10 |
|--------|----|----|----|----|
| 0      | 1  |    |    | 1  |
| 1      | 1  |    |    | 1  |

$$C \Rightarrow \bar{C}$$

$$Q27. f(A, B, C) = \Sigma m(0, 3, 5, 6)$$

| A \ BC | 00 | 01 | 11 | 10 |
|--------|----|----|----|----|
| 0      | 1  |    | 1  |    |
| 1      |    | 1  |    | 1  |

Q28  $f(A, B, C, D) = \sum (0, 2, 4, 6, 10, 11, 13, 15)$

| $AB \backslash CD$  | 00 | 01 | 11 | 10 |
|---------------------|----|----|----|----|
| $\bar{A}\bar{B}$ 00 | 1  |    |    | 1  |
| $\bar{A}B$ 01       | 1  |    |    | 1  |
| $AB$ 11             |    | 1  | 1  |    |
| $A\bar{B}$ 10       |    |    | 1  | 1  |

 $\bar{A}B\bar{D}$  $A\bar{B}C$ 

$$\Rightarrow \bar{A}\bar{D} + \bar{A}BD + A\bar{B}C$$

Q29  $f(A, B, C, D) = \sum m(0, 1, 2, 4, 6, 9, 10, 11, 12, 13, 15)$

| $AB \backslash CD$  | 00 | 01 | 11 | 10 |
|---------------------|----|----|----|----|
| $\bar{A}\bar{B}$ 00 | 1  | 1  |    | 1  |
| $\bar{A}B$ 01       | 1  |    |    | 1  |
| $AB$ 11             | 1  | 1  | 1  |    |
| $A\bar{B}$ 10       |    | 1  | 1  | 1  |

 $\bar{A}\bar{B}C$  $\bar{A}B\bar{C}$  $AB$  $A\bar{B}C$ 

$$\Rightarrow \bar{A}\bar{D} + \bar{A}D + \bar{A}B\bar{C} + \bar{A}B\bar{C} + \bar{A}B\bar{C}$$

Q30  $f(A, B, C, D) = \sum m(1, 5, 6, 7, 11, 12, 13, 15)$

| $AB \backslash CD$  | 00 | 01 | 11 | 10 |
|---------------------|----|----|----|----|
| $\bar{A}\bar{B}$ 00 |    | 1  |    |    |
| $\bar{A}B$ 01       |    | 1  | 1  | 1  |
| $AB$ 11             | 1  | 1  | 1  |    |
| $A\bar{B}$ 10       |    |    | 1  |    |

$$\Rightarrow \bar{A}B\bar{C} + \bar{A}B\bar{C}D + \bar{A}B + \bar{A}CD$$

Q31  $f(A, B, C, D) = \sum m(1, 2, 5, 6, 8, 10, 12, 13, 14, 15)$

| $AB \backslash CD$  | 00 | 01 | 11 | 10 |
|---------------------|----|----|----|----|
| $\bar{A}\bar{B}$ 00 |    | 1  |    | 1  |
| $\bar{A}B$ 01       |    | 1  |    | 1  |
| $AB$ 11             | 1  | 1  | 1  | 1  |
| $A\bar{B}$ 10       | 1  |    |    | 1  |

 $\bar{A}\bar{B}D$ 

$$\Rightarrow AB + \bar{C}D + \bar{A}\bar{B} + \bar{A}\bar{B}C$$

Q.32.  $f(A, B, C, D) = \sum m(1, 5, 6, 7, 11, 12, 13, 15)$

| $AB \backslash CD$ | 00 | 01 | 11 | 10 |
|--------------------|----|----|----|----|
| 00                 | 1  | 1  | 1  | 1  |
| 01                 | 1  | 1  | 1  | 1  |
| 10                 | 1  | 1  | 1  | 1  |
| 11                 | 1  |    | 1  | 1  |

$$\Rightarrow \bar{A} + B + C + \bar{D}$$

eg.:

| $AB \backslash CD$ | 00 | 01 | 11 | 10 |
|--------------------|----|----|----|----|
| 00                 | 1  | 1  |    | 1  |
| 01                 |    | 1  | 1  |    |
| 11                 | 1  | 1  |    | 1  |
| 10                 | 1  |    |    | 1  |

$$\Rightarrow \bar{B}\bar{D} + \bar{B}C\bar{D} + \bar{A}B\bar{D} + A\bar{C}\bar{B} + \bar{A}\bar{B}C$$

$$\Rightarrow \bar{B}\bar{D} + \bar{B}C\bar{D} + \bar{A}B\bar{D} + A\bar{C}\bar{D} + \bar{A}\bar{C}D$$

### Don't Care Condition:

Combination of inputs on which the output may or may not depend are called Don't Care Conditions.

$$f(A, B) = \bar{A}B + \bar{A}B$$

Q.1.  $f(A, B, C) = \sum m(0, 1, 6, 7) + \sum d(3, 5)$

| $A \backslash BC$ | 00 | 01 | 11 | 10 |
|-------------------|----|----|----|----|
| 0                 | 1  | 1  | X  |    |
| 1                 |    |    | 1  | 1  |

$\Rightarrow A\bar{B} + \bar{A}B$

$f(A, B, C) = \sum m(0, 1, 2) + \sum d(3, 6)$

| $A \backslash BC$ | 00 | 01 | 11 | 10 |
|-------------------|----|----|----|----|
| 0                 | 1  | 1  | X  | 1  |
| 1                 |    |    |    | X  |

eg.:

| $A \backslash BC$ | 00 | 01 | 11 | 10 |
|-------------------|----|----|----|----|
| 0                 | 1  | X  | X  | X  |
| 1                 |    |    | X  | X  |

$\Rightarrow \bar{A}$

| $A \backslash BC$ | 00 | 01 | 11 | 10 |
|-------------------|----|----|----|----|
| 0                 | X  | 1  | 1  |    |
| 1                 |    | X  | 1  | X  |

$\Rightarrow C$

Q2.  $f(A, B, C, D) = \sum m(0, 2, 4, 6, 7, 8, 10, 11, 12, 14, 15) + \sum d(1, 3)$

| $\overline{AB} \backslash CD$ | 00 | 01 | 11 | 10 |
|-------------------------------|----|----|----|----|
| 00                            | 1  | x  | x  | 1  |
| 01                            | 1  |    | 1  | 1  |
| 11                            | 1  |    | 1  | 1  |
| 10                            | 1  |    | 1  | 1  |

$\Rightarrow C + \overline{D}$

eg.:

| $\overline{AB} \backslash CD$ | 00 | 01 | 11 | 10 |
|-------------------------------|----|----|----|----|
| 00                            | 1  | 1  | x  | 1  |
| 01                            |    |    | 1  | x  |
| 11                            |    | x  | 1  |    |
| 10                            | x  | x  |    | 1  |

$$\Rightarrow \overline{A}\overline{B} + \overline{B}\overline{D} + BCD$$

| $\overline{AB} \backslash CD$ | 00 | 01 | 11 | 10 |
|-------------------------------|----|----|----|----|
| 00                            | 1  | 1  |    |    |
| 01                            | 1  | x  | x  | 1  |
| 11                            |    | x  | x  | 1  |
| 10                            |    |    | 1  |    |

$$\Rightarrow \overline{A}\overline{B}\overline{C} + BC + ACD$$

Q3.  $F(A, B, C) = \prod m(0, 1, 3, 6, 7) = \sum m(2, 4, 5)$

M1:

| $A \backslash BC$ | 00 | 01 | 11 | 10 |
|-------------------|----|----|----|----|
| 0                 | 0  | 0  | 0  | 1  |
| 1                 | 1  | 1  | 0  | 0  |

Pos.  $\begin{cases} F = (A+B) \cdot (\overline{A}+\overline{B}) \cdot (A+\overline{C}) \\ F = (A+B) \cdot (\overline{A}+\overline{B}) \cdot (\overline{B}+\overline{C}) \end{cases}$

SOP  $\begin{cases} f = A\overline{B} + \overline{A}B\overline{C} \end{cases}$

$\downarrow$  B+C  
B+C  
00

M2 (Method 2)

| $A \backslash BC$ | 00 | 01 | 11 | 10 |
|-------------------|----|----|----|----|
| 0                 | 0  | 0  | 0  | 1  |
| 1                 | 1  | 1  | 0  | 0  |

$$\overline{F} = \overline{A}\overline{B} + AB + \overline{A}\overline{C}$$

$$F = \overline{\overline{A}\overline{B} + AB + \overline{A}\overline{C}}$$

$$F = (A+B) \cdot (\overline{A}+\overline{B}) \cdot (A+\overline{C})$$

Q4.  $f(A, B, C, D) = \sum m(0, 1, 3, 6, 7, 10, 11)$  minimize it in Pos form  
 $= \prod M(2, 4, 5, 8, 9, 12, 13)$

| $AB \backslash CD$ | $C+D$<br>00 | $C+D$<br>01 | $\bar{C}+\bar{D}$<br>11 | $\bar{C}+D$<br>10 |
|--------------------|-------------|-------------|-------------------------|-------------------|
| $A+B$ 00           | 1           | 1           | 1                       | 0                 |
| $A+B$ 01           | 0           | 0           | 1                       | 1                 |
| $\bar{A}+B$ 11     | 0           | 0           | 0                       | 0                 |
| $\bar{A}+B$ 10     | 0           | 0           | 1                       | 1                 |

$$F = (\bar{B}+C)(\bar{A}+B)(\bar{A}+C)(A+B+\bar{C}+D)$$

| $AB \backslash CD$ | $C+D$<br>00 | $C+D$<br>01 | $\bar{C}+\bar{D}$<br>11 | $\bar{C}+D$<br>10 |
|--------------------|-------------|-------------|-------------------------|-------------------|
| $A+B$ 00           | 0           | 0           |                         |                   |
| $A+B$ 01           | 0           |             | 0                       | X                 |
| $\bar{A}+B$ 11     |             | X           | X                       |                   |
| $\bar{A}+B$ 10     |             | 0           | 0                       | X                 |

$$F = (A+C+D) \cdot (B+C+D)(\bar{B}+\bar{C}+D)(\bar{A}+D)$$

Variable inside the K-map

eg:

| $A \backslash B$ | $\bar{B}$ | $B$ |
|------------------|-----------|-----|
| $\bar{A}$ 0      | C         | C   |
| $A$ 1            |           |     |

$\underline{\underline{A \cdot C}}$

$$\rightarrow \bar{A}\bar{B}C + \bar{A}BC$$

$$= \bar{A}\bar{B}C + \bar{A}BC = \bar{A}C(\bar{B}+B) = \underline{\underline{\bar{A}C}}$$

eg:

| $A \backslash B$ | $\bar{B}$ | $B$ |
|------------------|-----------|-----|
| $\bar{A}$ 0      | 1         | C   |
| $A$ 1            |           |     |

$$f(A, B, C) = \bar{A}\bar{B} \cdot 1 + \bar{A}B \cdot C$$

$$= \bar{A}[\bar{B} + B \cdot C]$$

$$= \bar{A}[(\bar{B}+B) \cdot (\bar{B}+C)]$$

$$= \underline{\underline{\bar{A}(\bar{B}+C)}}$$

Method 1

Boolean Algebra

Method 4

Case 1:

| $A \backslash B$ | $\bar{B}$ | $B$ |
|------------------|-----------|-----|
| $\bar{A}$ 0      | 1         | 0   |
| $A$ 1            | 0         | 0   |

$C=0$

$$= \bar{A}\bar{B}C$$

Case 2:

| $A \backslash B$ | $\bar{B}$ | $B$ |
|------------------|-----------|-----|
| $\bar{A}$ 0      | 1         | 1   |
| $A$ 1            | 0         | 0   |

$C=1$

$$\rightarrow \bar{A}C$$

$$\bar{A}\bar{B}C + \bar{A}C$$

$$\Rightarrow \bar{A}[C + \bar{B}C]$$

$$\Rightarrow \bar{A}[(\bar{B}+C)]$$

$$\Rightarrow \underline{\underline{\bar{A}(\bar{B}+C)}}$$

Method 2

$$f(A, B, C) = \bar{A}\bar{B}(C+1) + \bar{A}BC$$

$$= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}BC$$

$$= \sum m(0, 1, 3) = \pi m(2, 4, 5, 6, 7)$$

| $\bar{A} \backslash B \rightarrow \bar{B} \quad B$ | 00 | 01 | 11 | 10 |
|----------------------------------------------------|----|----|----|----|
| $\bar{A} \quad 0$                                  | 1  | 1  | 1  | 0  |
| $A \quad 1$                                        | 0  | 0  | 0  | 0  |

$$\text{Sop } f = \bar{A}\bar{B} + \bar{A}C$$

$$= \underline{\bar{A}(\bar{B}+C)}$$

| $\bar{A} \backslash B \rightarrow \bar{B} \quad B$ | 00 | 01 | 11 | 10 |
|----------------------------------------------------|----|----|----|----|
| $\bar{A} \quad 0$                                  | 1  | 1  | 1  | 0  |
| $A \quad 1$                                        | 0  | 0  | 0  | 0  |

$$\text{Pos } = \underline{\bar{A}(B+C)}$$

Method 3

| $\bar{A} \backslash B \rightarrow \bar{B} \quad B$ | 0 | 1 |
|----------------------------------------------------|---|---|
| $\bar{A} \quad 0$                                  | 1 | C |
| $A \quad 1$                                        |   |   |

$\rightarrow \underline{\bar{A}(\bar{B}+C)}$

Q1.

| $\bar{A} \backslash B \rightarrow \bar{B} \quad B$ | 0 | 1 |
|----------------------------------------------------|---|---|
| $\bar{A} \quad 0$                                  | 1 | C |
| $A \quad 1$                                        |   | C |

$$\underline{BC + \bar{A}(\bar{B}+C)}$$

| $\bar{A} \backslash B \rightarrow \bar{B} \quad B$ | 0 | 1 |
|----------------------------------------------------|---|---|
| $\bar{A} \quad 0$                                  |   | 1 |
| $A \quad 1$                                        |   |   |

$$\bar{A}\bar{B}\bar{C}$$

| $\bar{A} \backslash B \rightarrow \bar{B} \quad B$ | 0 | 1 |
|----------------------------------------------------|---|---|
| $\bar{A} \quad 0$                                  | 1 | 1 |
| $A \quad 1$                                        |   | 1 |

$$(\bar{A}+B)C$$

$$\bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}BC$$

$$\Rightarrow \bar{A}(\bar{B}\bar{C}+C) + \bar{A}BC$$

$$\Rightarrow \underline{\bar{A}(\bar{B}+C) + \bar{A}BC}$$

Q2.

| $\bar{A} \backslash B \rightarrow \bar{B} \quad B$ | 0 | 1 |
|----------------------------------------------------|---|---|
| $\bar{A} \quad 0$                                  |   | 1 |
| $A \quad 1$                                        | C | C |

$$\underline{AC + \bar{A}B + BC}$$

| $\bar{A} \backslash B \rightarrow \bar{B} \quad B$ | 0 | 1 |
|----------------------------------------------------|---|---|
| $\bar{A} \quad 0$                                  |   | 1 |
| $A \quad 1$                                        | 0 | 0 |

$$(\bar{A}\bar{B}\bar{C})$$

| $\bar{A} \backslash B \rightarrow \bar{B} \quad B$ | 0 | 1 |
|----------------------------------------------------|---|---|
| $\bar{A} \quad 0$                                  |   | 1 |
| $A \quad 1$                                        | 1 | 1 |

$$(\bar{A}+B)C$$

$$\bar{A}\bar{B}\bar{C} + AC + BC$$

$$\Rightarrow \underline{AC + B(\bar{A}\bar{C}+C) = AC + B(\bar{A}+C)}$$

## Implicants and Prime Implicants

1. Implicants: The total number of min-terms in the boolean expression are called implicants. Or in kmap the total no. of 1 is called implicants.
2. Prime Implicants: Total number of possibilities of formation of group are called prime implicants.
3. Essential Prime Implicants / Selective PE: If there are two answers for a boolean expression then the terms common in both the answers are called essential prime implicants.
4. Reduced Prime Implicants: The terms that are not essential prime implicants are called reduced prime implicants.

Q1.  $f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C$   
 $= \sum m(0, 1, 3, 6, 7)$

| A \ B     | C         |     |           |     |
|-----------|-----------|-----|-----------|-----|
|           | $\bar{C}$ | $C$ | $\bar{C}$ | $C$ |
| $\bar{A}$ | 1         | 1   | 1         |     |
| A         |           |     | 1         | 1   |

$$\Rightarrow \bar{A}\bar{B} + A\bar{B} + \bar{A}C$$

Implicants = 5

Prime Implicants = 4

Essential prime implicants =

SPI = 1

RPI = 1.

| A \ B     | C         |     |           |     |
|-----------|-----------|-----|-----------|-----|
|           | $\bar{C}$ | $C$ | $\bar{C}$ | $C$ |
| $\bar{A}$ | 1         | 1   | 1         |     |
| A         |           |     | 1         | 1   |

$$\Rightarrow \bar{A}\bar{B} + A\bar{B} + BC$$

$$PI \{ \bar{A}\bar{B}, A\bar{B}, \bar{A}C, BC \}$$

Q2.  $f(A, B, C) = \sum m(2, 3, 4, 5, 7)$   
 $PI \{ \bar{A}\bar{B}, \bar{A}B, BC, AC \}$

Q5.  $f(A, B, C, D) = \sum m(1, 5, 6, 7, 11, 12, 13, 15)$

| $AB \backslash CD$ | 00 | 01 | 11 | 10 |
|--------------------|----|----|----|----|
| 00                 |    | 1  |    |    |
| 01                 |    | 1  | 1  | 1  |
| 11                 | 1  | 1  | 1  |    |
| 10                 |    |    | 1  |    |

$P_2 \{ \bar{A}CD, ABC, ACD, \bar{A}BC, BD \}$

Implicants = 8

$PI = 5$

$EPI = 4$

$SPI = 0$

$RPI = 1$

$\Rightarrow \bar{A}CD + \bar{A}BC + ACD + ABC$

$EPI \{ \bar{A}CD + \bar{A}BC + ACD + ABC \}$

Q6

| $A \backslash B$ | 00 | 01 | 11 | 10 |
|------------------|----|----|----|----|
| 0                | 1  | 1  | 1  |    |
| 1                |    | 1  | 1  | 1  |

$PI \{ \bar{A}\bar{B} + AB + C \}$

$\Rightarrow \bar{A}\bar{B} + AB + C$

Implicants = 6

$PI = 3$

$EPI = 3$

$SPI = 0$   $RPI = 0$

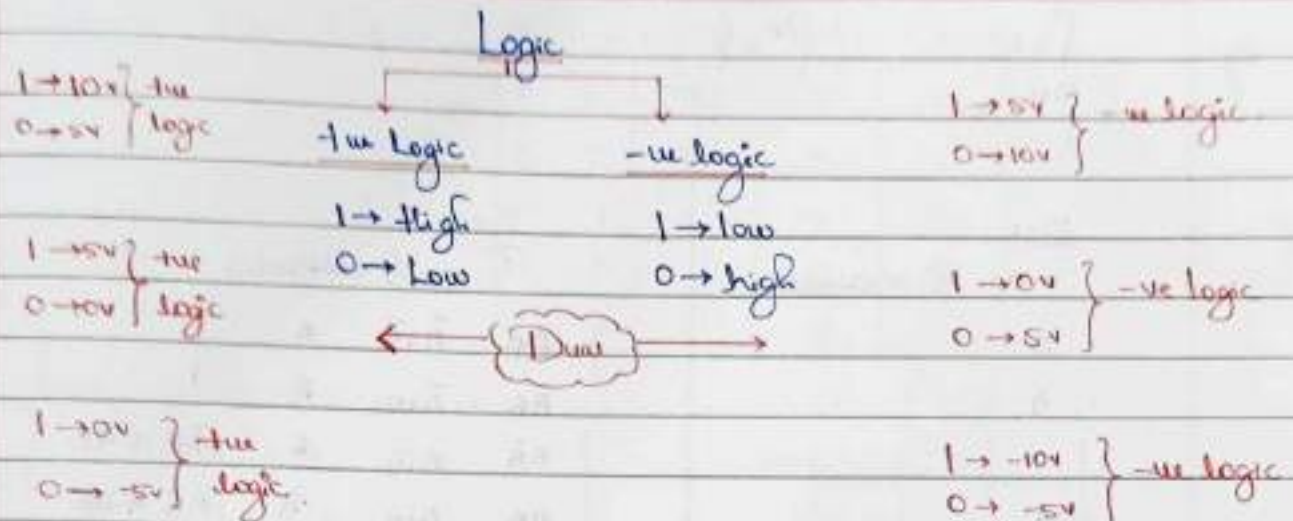
HW

Q7.

| $AB \backslash CD$ | $\bar{C}\bar{D}$ | $\bar{C}D$ | $CD$ | $C\bar{D}$ |
|--------------------|------------------|------------|------|------------|
| 00                 | 1                | 1          |      |            |
| 01                 |                  | 1          | 1    |            |
| 11                 |                  |            | 1    | 1          |
| 10                 |                  |            |      | 1          |

Q8.

| $AB \backslash CD$ | $\bar{C}\bar{D}$ | $\bar{C}D$ | $CD$ | $C\bar{D}$ |
|--------------------|------------------|------------|------|------------|
| 00                 | 1                | 1          |      |            |
| 01                 | 1                | 1          |      | 1          |
| 11                 |                  | 1          | 1    | 1          |
| 10                 |                  |            | 1    | 1          |



And Case (+ve)

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

Logic

(conversion)

↓

Dual

| A | B | Y |
|---|---|---|
| 1 | 1 | 1 |
| 1 | 0 | 1 |
| 0 | 1 | 1 |
| 0 | 0 | 0 |

OR

Dual  $\rightarrow$  +ve logic  $\longleftrightarrow$  -ve logic

- |                                            |                                       |                   |
|--------------------------------------------|---------------------------------------|-------------------|
| 1. $1 \longleftrightarrow 0$               | 0 $\longleftrightarrow$ 1             | $f = A \cdot 1$   |
| 2. AND $\longleftrightarrow$ OR            | $A \longleftrightarrow A$             | $f^0 = A + 0 = A$ |
| 3. $\cdot \longleftrightarrow +$           | $\bar{A} \longleftrightarrow \bar{A}$ |                   |
| 4. NAND $\longleftrightarrow$ NOR          |                                       |                   |
| 5. X-OR $\longleftrightarrow$ X-NOR        |                                       |                   |
| 6. Buffer $\longleftrightarrow$ Buffer     |                                       |                   |
| 7. Inverter $\longleftrightarrow$ Inverter |                                       |                   |

|                                                                               |                                                                        |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------|
| eg: $f = AB + CD + EF$<br>$f^0 = (A+B) \cdot (C+D) \cdot (\bar{E} + \bar{F})$ | $f = \overline{ABC + DEF}$<br>$f^0 = \overline{(A+B+C) \cdot (D+E+F)}$ |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------|

eg:  $f = AB + CD$       $f^0 = (A+B) \cdot (C+D)$

$f^{00} = AB + CD$

$f^{00} = f$

Q9.

| AB \ CD             | 00 | 01 | 11 | 10 |
|---------------------|----|----|----|----|
| $\bar{A}\bar{B}$ 00 | 1  | 1  | 1  | 1  |
| $\bar{A}B$ 01       |    | 1  | 1  |    |
| $AB$ 11             | 1  |    |    | 1  |
| $A\bar{B}$ 10       | 1  |    |    | 1  |

$\Rightarrow \bar{B}\bar{D} + \bar{B}D + \bar{A}\bar{D}$  (Ans 1)  
 $\Rightarrow \bar{A}\bar{B} + \bar{A}D + \bar{B}\bar{D}$  (Ans 2)

Implicants = 4  
 PI = 4  
 EPI = 2  
 SPI = 1  
 RPI = 1

Q10. Let a function  $F$  which has 3 input variables  $(x, y, z)$ . The function  $F$  will be high only when atleast two of the input variables are set to high. Draw the K-map for the given function. Let the number of PI in K-map = 'a' and the no. of EPI in K-map = 'b'. Find the quadratic mean of 'a' and 'b'.

Ans:

|   | x | y | z | f |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 |
| 2 | 0 | 1 | 0 | 0 |
| 3 | 0 | 1 | 1 | 1 |
| 4 | 1 | 0 | 0 | 0 |
| 5 | 1 | 0 | 1 | 1 |
| 6 | 1 | 1 | 0 | 1 |
| 7 | 1 | 1 | 1 | 1 |

$$f(x, y, z) = \sum m(3, 5, 6, 7)$$

| x \ yz      | 00 | 01 | 11 | 10 |
|-------------|----|----|----|----|
| $\bar{x}$ 0 |    |    | 1  |    |
| x 1         |    | 1  | 1  | 1  |

$$a = \text{PI} = 3$$

$$b = \text{EPI} = 3$$

$$f = xy + yz + xz$$

Quadratic mean:

$$= \sqrt{\frac{a^2 + b^2}{2}} = \sqrt{\frac{3^2 + 3^2}{2}} = \underline{\underline{3}}$$

Q11. Find the number of prime implicants & essential prime implicants in the given K.

| PQ \ RS | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      | 1  |    |    |    |
| 01      |    | 1  |    | 1  |
| 11      | 1  |    |    |    |
| 10      |    | 1  |    | 1  |

$$\text{PI} = 6$$

$$\text{EPI} = 6$$

Q3.

$$f(A,B,C) = \sum m(0,1,3,4,6,7)$$

|        |    |    |    |    |
|--------|----|----|----|----|
| A \ BC | 00 | 01 | 11 | 10 |
| 0      | 1  | 1  | 1  |    |
| 1      | 1  |    | 1  | 1  |

 $\Rightarrow \bar{B}\bar{C} + \bar{A}C + AB$ 

Implicants = 6

PI = 6

EPI = 0 (term common in both)

SPI = 3

RPI = 3

|        |    |    |    |    |
|--------|----|----|----|----|
| A \ BC | 00 | 01 | 11 | 10 |
| 0      | 1  | 1  | 1  |    |
| 1      | 1  |    | 1  | 1  |

 $\Rightarrow \bar{A}B + BC + \bar{A}\bar{C}$ 

$$PI \{ \bar{A}B + BC + \bar{A}\bar{C} + \bar{B}\bar{C} + \bar{A}C + AB \}$$

Q2  
(Continued)

|        |    |    |    |    |
|--------|----|----|----|----|
| A \ BC | 00 | 01 | 11 | 10 |
| 0      |    |    | 1  | 1  |
| 1      | 1  | 1  | 1  |    |

 $\Rightarrow \bar{A}\bar{B} + \bar{A}B + AC$ 

Implicants = 5

PI = 4

EPI = 2

SPI = 1

RPI = 1

|        |    |    |    |    |
|--------|----|----|----|----|
| A \ BC | 00 | 01 | 11 | 10 |
| 0      |    |    | 1  | 1  |
| 1      | 1  | 1  | 1  |    |

 $\Rightarrow \bar{A}\bar{B} + \bar{A}B + BC$ 

$$PI \{ \bar{A}\bar{B} + \bar{A}B + AC + BC \}$$

Q4.  $f(A,B,C) = \sum m(0,2,3,4,5,7)$ 

|        |    |    |    |    |
|--------|----|----|----|----|
| A \ BC | 00 | 01 | 11 | 10 |
| 0      | 1  |    | 1  | 1  |
| 1      | 1  | 1  | 1  |    |

 $\Rightarrow \bar{B}\bar{C} + AC + \bar{A}B$ 

Implicants = 6

PI = 6

EPI = 0

SPI = 3

RPI = 3

|        |    |    |    |    |
|--------|----|----|----|----|
| A \ BC | 00 | 01 | 11 | 10 |
| 0      | 1  |    | 1  | 1  |
| 1      | 1  | 1  | 1  |    |

 $\Rightarrow \bar{A}\bar{C} + BC + \bar{A}B$ 

$$PI \{ \bar{B}\bar{C} + AC + \bar{A}B + \bar{A}\bar{C} + BC + \bar{A}B \}$$

eg.  $P = A$   
 $P^D = A$   $P^D = P$   $\rightarrow$  Self Dual.

| $n=1$<br>(no. of variables)                                            | $n=2$<br>(no. of variables)                                                                                                                           |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\left\{ \begin{array}{c} A \\ \bar{A} \\ 0 \\ 1 \end{array} \right\}$ | $\left\{ \begin{array}{cc} \bar{A}\bar{B} & \bar{A} + \bar{B} \\ \bar{A}B & \bar{A} + B \\ A\bar{B} & A + \bar{B} \\ AB & A + B \end{array} \right\}$ |
| Self dual = 2                                                          | $\left\{ \begin{array}{cc} A & 0 \\ \bar{A} & 1 \\ B & \bar{A}B + A\bar{B} \\ \bar{B} & \bar{A}\bar{B} + A\bar{B} \end{array} \right\}$               |
|                                                                        | Self dual = 4                                                                                                                                         |

\*  $n$  variable  $\rightarrow 2^{2^n}$  Expressions  $\rightarrow 2^{2^n - 1}$  Self dual expressions.

### Combinational Circuit:

\* A circuit without feedback or memory are called Combinational Circuit.

\* Static circuit.

### Designing of Combinational Circuit:

Step 1: Find the number of inputs and outputs.

Step 2: Write the truth table

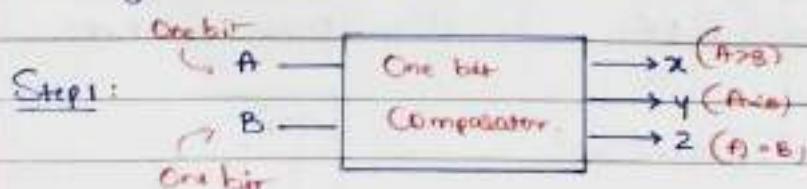
Step 3: Write the logical expression

Step 4: Minimize the logical expression.

Step 5: Hardware implementation

### Comparator

1. Design a one bit Comparator.



Step 2: Truth Table

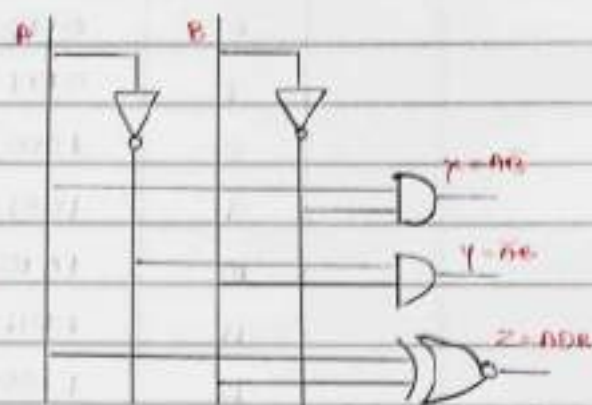
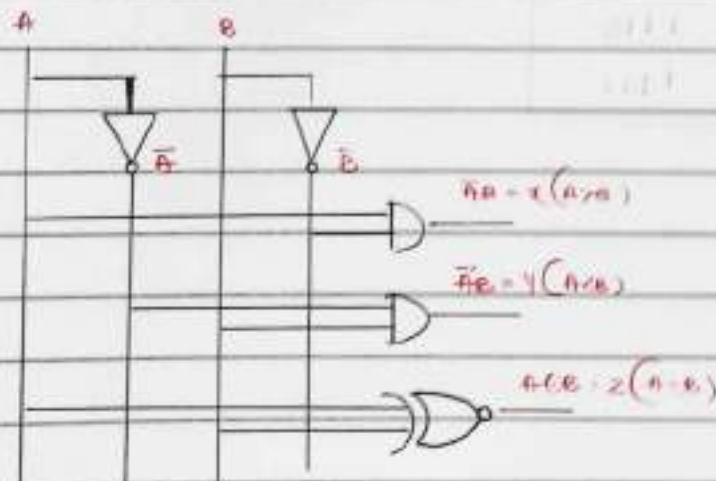
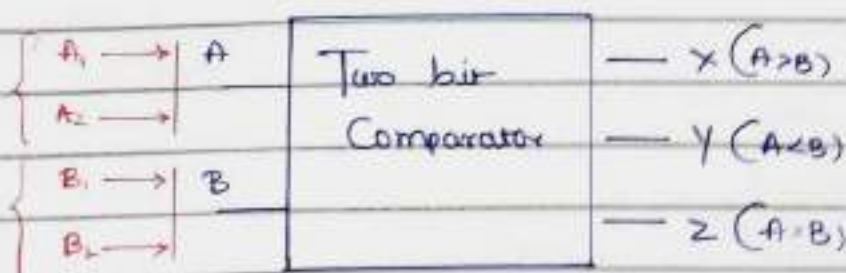
|   |   | $(A > B)$ | $(A < B)$ | $(A = B)$ |
|---|---|-----------|-----------|-----------|
| A | B | X         | Y         | Z         |
| 0 | 0 | 0         | 0         | 1         |
| 0 | 1 | 0         | 1         | 0         |
| 1 | 0 | 1         | 0         | 0         |
| 1 | 1 | 0         | 0         | 1         |

Step 3: Logical expression:

$$X(A > B) = \bar{A}\bar{B}$$

$$Y(A < B) = \bar{A}B$$

$$Z(A = B) = \bar{A}\bar{B} + AB = A \odot B$$

Step 4: Minimization (Already minimized)Steps: Hardware implementation.Two Bit Comparator:

| Decimal | Binary | Step 2      | A <sub>1</sub> | A <sub>0</sub> | B <sub>1</sub> | B <sub>0</sub> | X | Y | Z |
|---------|--------|-------------|----------------|----------------|----------------|----------------|---|---|---|
| 0       | 0000   | Truth Table | 0              | 0              | 0              | 0              | 0 | 0 | 1 |
| 1       | 0001   |             | 0              | 0              | 0              | 1              | 0 | 1 | 0 |
| 2       | 0010   |             | 0              | 0              | 1              | 0              | 0 | 1 | 0 |
| 3       | 0011   |             | 0              | 0              | 1              | 1              | 0 | 1 | 0 |
| 4       | 0100   |             | 0              | 1              | 0              | 0              | 1 | 0 | 0 |
| 5       | 0101   |             | 0              | 1              | 0              | 1              | 0 | 0 | 1 |
| 6       | 0110   |             | 0              | 1              | 1              | 0              | 0 | 1 | 0 |
| 7       | 0111   |             | 0              | 1              | 1              | 1              | 0 | 1 | 0 |
| 8       | 1000   |             | 1              | 0              | 0              | 0              | 1 | 0 | 0 |
| 9       | 1001   |             | 1              | 0              | 0              | 1              | 1 | 0 | 0 |
| 10      | 1010   |             | 1              | 0              | 1              | 0              | 0 | 0 | 1 |
| 11      | 1011   |             | 1              | 0              | 1              | 1              | 0 | 1 | 0 |
| 12      | 1100   |             | 1              | 1              | 0              | 0              | 1 | 0 | 0 |
| 13      | 1101   |             | 1              | 1              | 0              | 1              | 1 | 0 | 0 |
| 14      | 1110   |             | 1              | 1              | 1              | 0              | 1 | 0 | 0 |
| 15      | 1111   |             | 1              | 1              | 1              | 1              | 0 | 0 | 1 |

Step 3 Logical expression:

$$X(A \oplus B) = \sum m(4, 8, 9, 12, 13, 14)$$

$$Y(A \oplus B) = \sum m(1, 2, 3, 6, 7, 11)$$

$$Z(A \oplus B) = \sum m(0, 5, 10, 15)$$

Step 4 Minimization

for X

| A <sub>1</sub> A <sub>0</sub> \ B <sub>1</sub> B <sub>0</sub> | B <sub>1</sub> B <sub>0</sub> = 00 | B <sub>1</sub> B <sub>0</sub> = 01 | B <sub>1</sub> B <sub>0</sub> = 11 | B <sub>1</sub> B <sub>0</sub> = 10 |
|---------------------------------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| A <sub>1</sub> A <sub>0</sub> = 00                            |                                    |                                    |                                    |                                    |
| A <sub>1</sub> A <sub>0</sub> = 01                            | 1                                  |                                    |                                    |                                    |
| A <sub>1</sub> A <sub>0</sub> = 11                            | 1                                  | 1                                  |                                    | 1                                  |
| A <sub>1</sub> A <sub>0</sub> = 10                            | 1                                  | 1                                  |                                    |                                    |

for Y

| A <sub>1</sub> A <sub>0</sub> \ B <sub>1</sub> B <sub>0</sub> | B <sub>1</sub> B <sub>0</sub> = 00 | B <sub>1</sub> B <sub>0</sub> = 01 | B <sub>1</sub> B <sub>0</sub> = 11 | B <sub>1</sub> B <sub>0</sub> = 10 |
|---------------------------------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| A <sub>1</sub> A <sub>0</sub> = 00                            |                                    | 1                                  | 1                                  | 1                                  |
| A <sub>1</sub> A <sub>0</sub> = 01                            |                                    |                                    | 1                                  | 1                                  |
| A <sub>1</sub> A <sub>0</sub> = 11                            |                                    |                                    |                                    |                                    |
| A <sub>1</sub> A <sub>0</sub> = 10                            |                                    |                                    | 1                                  |                                    |

Semiminimised

$$\begin{aligned}
 X(A > B) &= A_1 \bar{B}_1 + \bar{A}_1 A_0 \bar{B}_1 \bar{B}_0 + A_1 A_0 B_1 \bar{B}_0 \\
 &= A_1 \bar{B}_1 + (\bar{A}_1 \bar{B}_1 + A_1 B_1) A_0 \bar{B}_0 \\
 &= A_1 \bar{B}_1 + (A_1 \oplus B_1) A_0 \bar{B}_0
 \end{aligned}$$

Pr. 2  
A<sub>1</sub> A<sub>0</sub> B<sub>1</sub> B<sub>0</sub>

|    | 00 | 01 | 11 | 10 |
|----|----|----|----|----|
| 00 | 1  |    |    |    |
| 01 |    | 1  |    |    |
| 11 |    |    | 1  |    |
| 10 |    |    |    | 1  |

Minimised

$$\begin{aligned}
 X(A > B) &= A_1 \bar{B}_1 + A_0 \bar{B}_1 \bar{B}_0 + A_1 A_0 \bar{B}_0 \\
 &= A_1 \bar{B}_1 + (A_1 + B_1) A_0 \bar{B}_0
 \end{aligned}$$

$$Z(A = B) = (A_1 \odot B_1) (A_0 \odot B_0)$$

Semiminimised

$$\begin{aligned}
 Y(A < B) &= \bar{A}_1 B_1 + \bar{A}_1 \bar{A}_0 \bar{B}_1 B_0 + A_1 \bar{A}_0 B_1 B_0 \\
 &= \bar{A}_1 B_1 + (\bar{A}_1 \bar{B}_1 + A_1 B_1) \bar{A}_0 B_0 \\
 &= \bar{A}_1 B_1 + (A_1 \oplus B_1) \bar{A}_0 B_0
 \end{aligned}$$

Minimised

$$Y(A < B) = \bar{A}_1 B_1 + (\bar{A}_1 + B_1) \bar{A}_0 B_0$$

HW.Steps

Hardware Implementation

Comparators

| One bit Comparator      | Two bit Comparators   | Three bit Comparators   |
|-------------------------|-----------------------|-------------------------|
| Total Conditions = 4    | Total conditions = 16 | Total combinations = 64 |
| Equal Condition = 2     | Equal Condition = 4   | Equal Condition = 8     |
| Unequal Condition = 2   | Unequal " = 12        | Unequal " = 56          |
| Greater = less Cond = 1 | Greater = less " = 6  | Greater = less " = 28   |

'N' bit Comparators

$$\text{Total Condition} = 2^{2n}$$

$$\text{Equal Condition} = 2^n$$

$$\text{Unequal Condition} = 2^{2n} - 2^n$$

$$\text{Greater = less} = 2^{2n} - 2^n / 2$$



$$X(A > B) = A_1 \bar{B}_1 + (A_1 \odot B_1) A_0 \bar{B}_0$$

$$Y(A < B) = \bar{A}_1 B_1 + (A_1 \odot B_1) \bar{A}_0 B_0$$

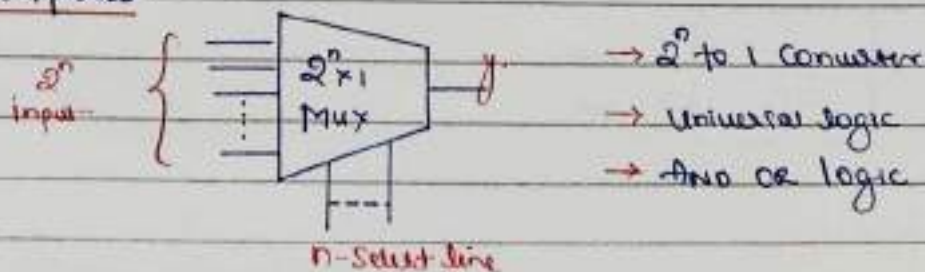
$$Z(A = B) = (A_1 \odot B_1) \cdot (A_0 \odot B_0)$$



$$X(A > B) = A_2 \bar{B}_2 + (A_2 \odot B_2) A_1 \bar{B}_1 + (A_2 \odot B_2) (A_1 \odot B_1) A_0 \bar{B}_0$$

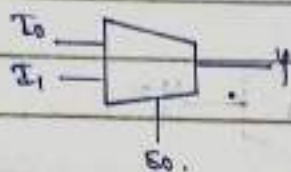
$$Y(A < B) = \bar{A}_2 B_2 + (A_2 \odot B_2) \bar{A}_1 B_1 + (A_2 \odot B_2) (A_1 \odot B_1) \bar{A}_0 B_0$$

$$Z(A = B) = (A_2 \odot B_2) (A_1 \odot B_1) (A_0 \odot B_0)$$

Multiplexer

Q1. Design a 2x1 Mux.

Step 1:



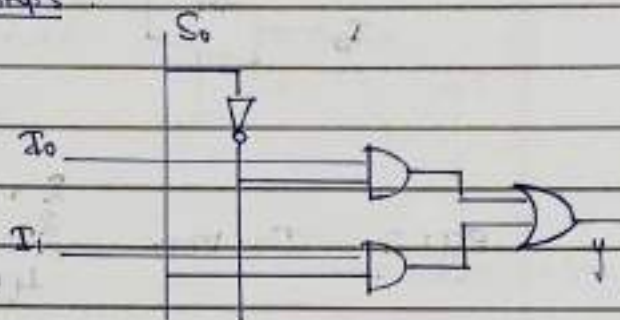
Step 2:

| S <sub>0</sub> | Y              |
|----------------|----------------|
| 0              | I <sub>0</sub> |
| 1              | I <sub>1</sub> |

Step 3:  $y = \bar{S}_0 I_0 + S_0 I_1$

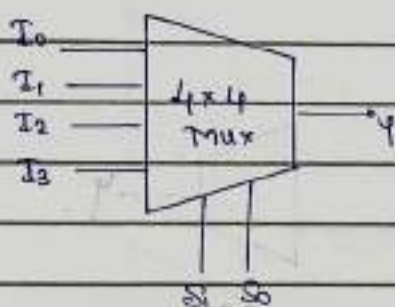
Step 4: Already Minimized

Step 5:



Q2 Design a 4x1 Mux.

St. 1



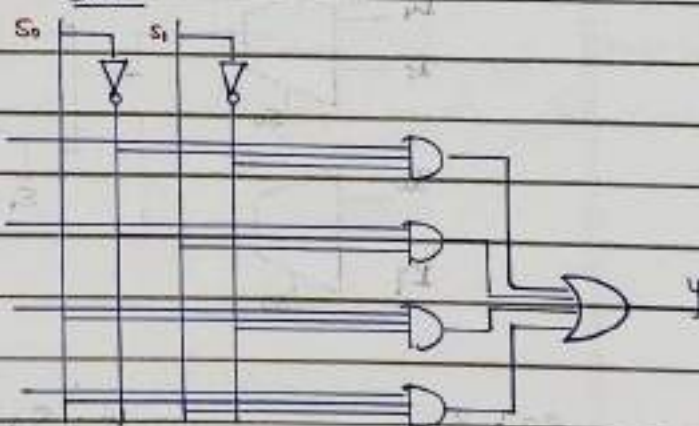
Step 2:

| S <sub>1</sub> | S <sub>0</sub> | Y              |
|----------------|----------------|----------------|
| 0              | 0              | I <sub>0</sub> |
| 0              | 1              | I <sub>1</sub> |
| 1              | 0              | I <sub>2</sub> |
| 1              | 1              | I <sub>3</sub> |

Step 3:  $y = \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$

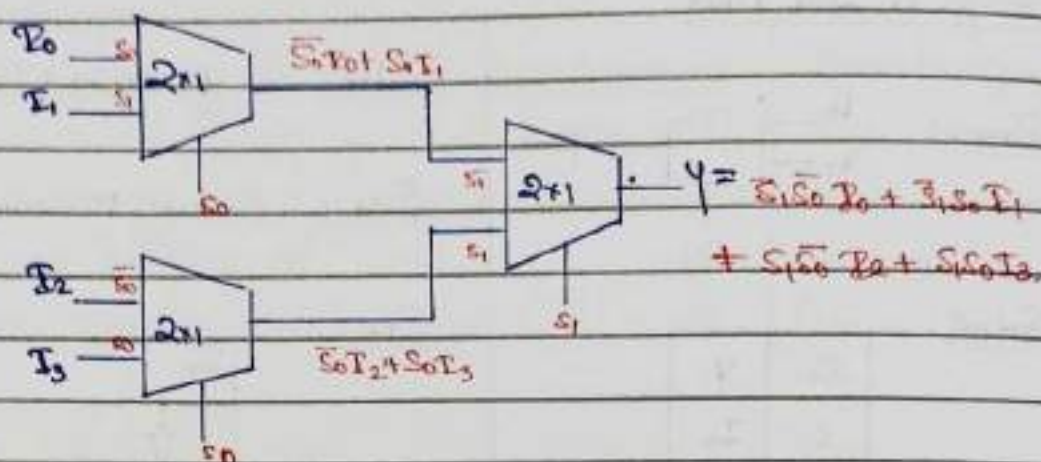
Step 4: No minimization

Step 5:

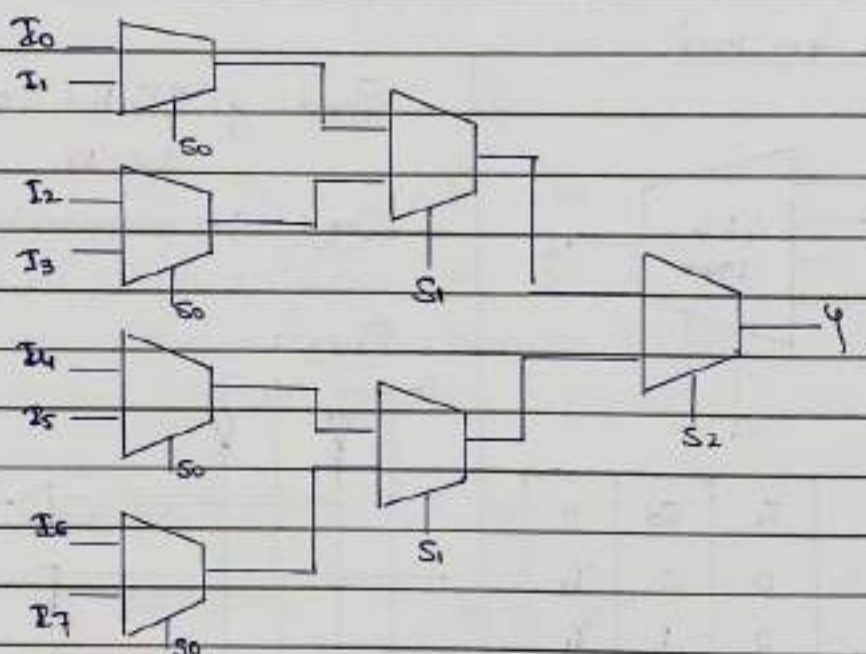
Type: 1 Designing of higher order mux by using lower order mux.

eg. 2x1 mux  $\frac{4^2}{2} + \frac{2^1}{2}$  4x1 mux

$2 + 1 = 3$



eg: 2      2x1 mux       $\frac{8}{2} + \frac{4}{2} + \frac{20}{2}$       8x1 mux  
 $4 + 2 + 1 = 7$



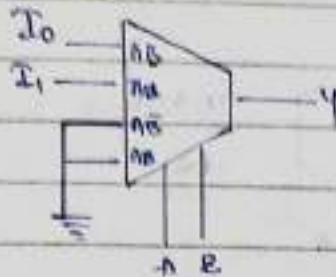
eg: 3      2x1 mux       $\frac{16}{2} + \frac{8}{2} + \frac{4}{2} + \frac{2}{2}$       16x1 mux  
 $8 + 4 + 2 + 1 = 15$

2x1 mux      63      64x1 mux

2x1 mux       $2^n - 1$        $2^n \times 1$  mux

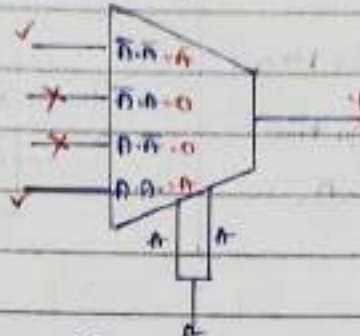
Q1.  $4 \times 1$  Mux  $\xrightarrow{\frac{8}{4} + \frac{8}{4}}$   $8 \times 1$  Mux  
 2 select line  $2+1=3$  3 select line

Method 1



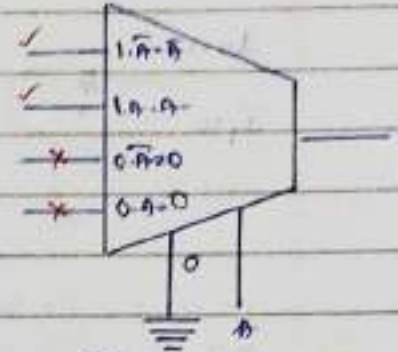
Select line = 2

Method 2



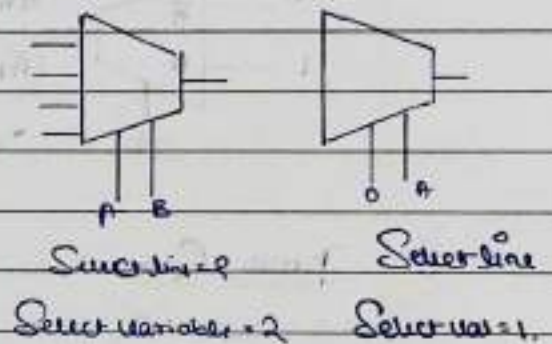
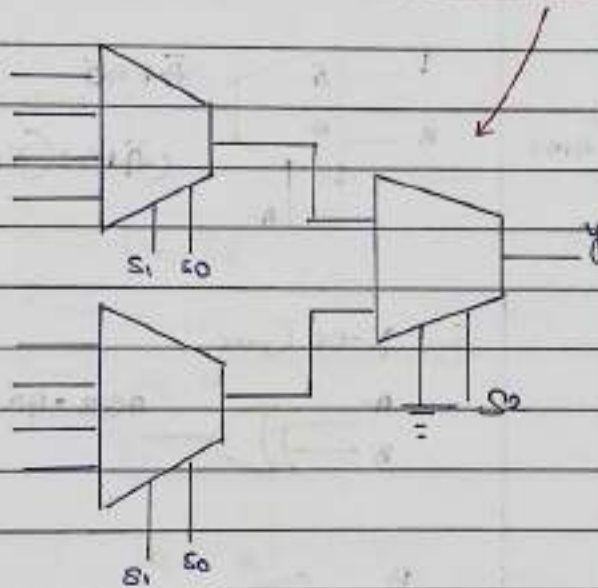
Select line = 1

Method 3

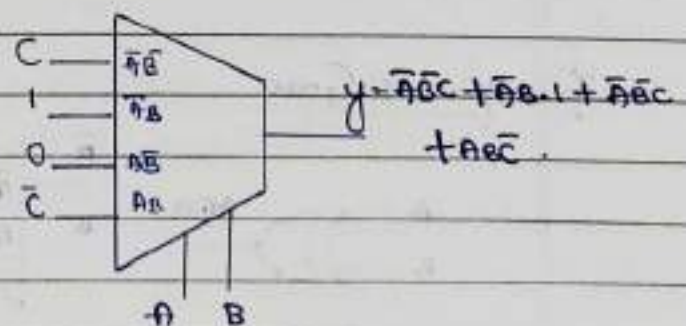
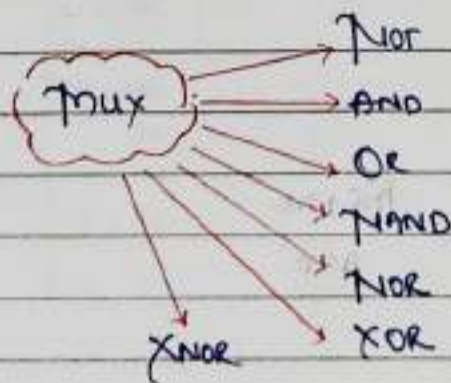


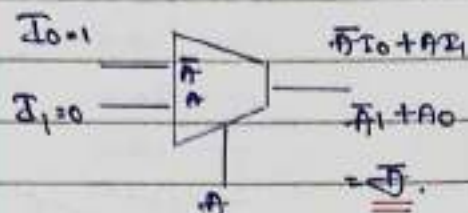
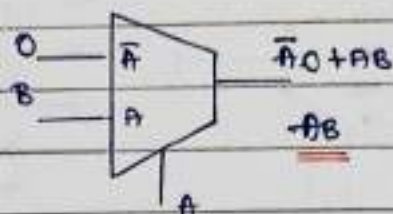
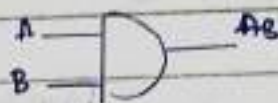
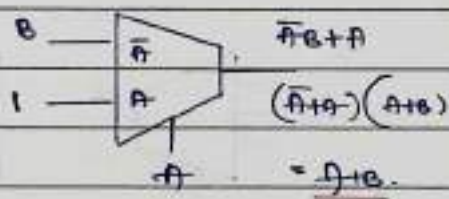
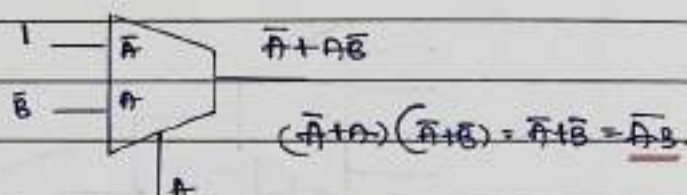
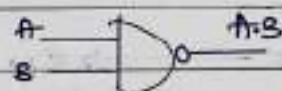
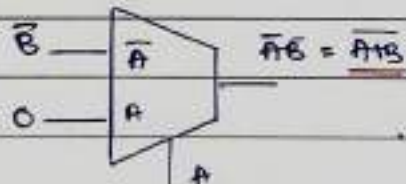
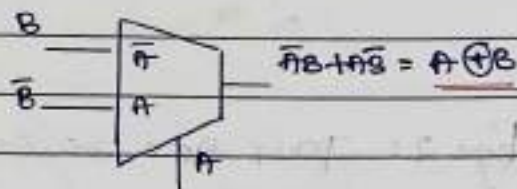
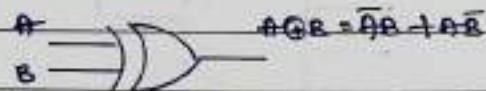
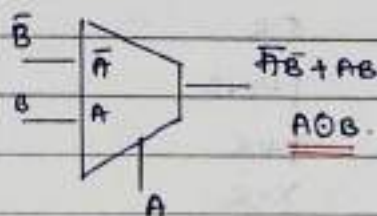
Select variable = One

$4 \times 1$  Mux  $\rightarrow 8 \times 1$  Mux  
 3 Select variables



Type 2: Mux As a universal logic.



1. Not Gate2. And Gate3. OR Gate4. NAND Gate5. NOR Gate6. X-OR Gate7. X-NOR Gate

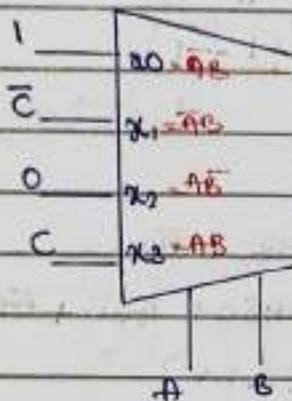
Note: To design NOT, AND, OR  $\rightarrow$  One 2x1 mux required  
To design NAND, NOR, XOR, XNOR,  $\rightarrow$  Two 2x1 mux required.

### Type 3 Minimization.

Q. Find the output f:

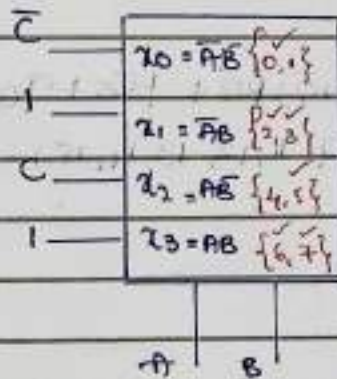
| $\begin{matrix} \nearrow \\ \text{A} \end{matrix}$ | 00 | 01 | 11 | 10 |
|----------------------------------------------------|----|----|----|----|
| 0                                                  | 1  | 1  |    | 1  |
| 1                                                  |    |    | 1  |    |

$$\Rightarrow \bar{A}\bar{C} + \bar{A}\bar{B} + \bar{A}BC$$



$$\begin{aligned} f(A, B, C) &= \bar{A}\bar{B}.1 + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}.0 + \bar{A}\bar{B}.C \\ &= \bar{A}\bar{B} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C \\ &= \bar{A}\bar{B}(\bar{C} + C) + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C \\ &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C \\ &= \sum m(0, 1, 2, 7) \end{aligned}$$

Q2 Find output of "f".

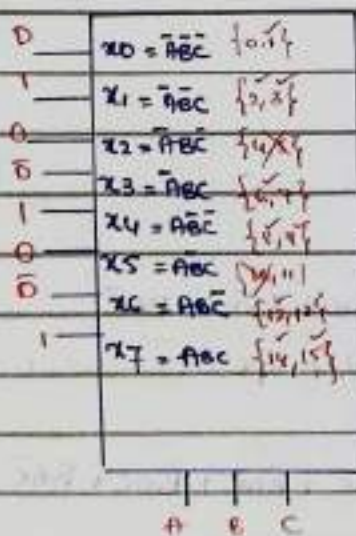


$$\begin{aligned} f(A, B, C) &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B} + \bar{A}\bar{B}C + \bar{A}\bar{B} \\ &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}(\bar{C} + C) + \bar{A}\bar{B}C + \bar{A}\bar{B}(\bar{C} + C) \\ &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C \\ &= \sum m(0, 2, 3, 5, 6, 7) \end{aligned}$$

| $\begin{matrix} \nearrow \\ \text{A} \end{matrix}$ | 00 | 01 | 11 | 10 |
|----------------------------------------------------|----|----|----|----|
| 0                                                  | 1  |    | 1  | 1  |
| 1                                                  |    | 1  | 1  | 1  |

$$\Rightarrow B + \bar{A}\bar{C} + AC$$

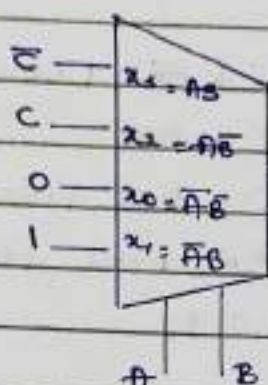
Q3.



$$\begin{aligned} f(A, B, C, D) &= \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}D \\ &\quad + \bar{A}\bar{B}C \\ &= \sum m(1, 2, 3, 6, 8, 9, 12, 14, 15) \end{aligned}$$

| AB \ C | 00 | 01 | 11 | 10 |
|--------|----|----|----|----|
| 00     |    | 1  | 1  | 1  |
| 01     |    |    |    | 1  |
| 11     | 1  |    | 1  | 1  |
| 10     | 1  | 1  |    |    |

(spans)

Q.4 find  $f = ?$ 

| AB \ C | 00 | 01 | 11 | 10 |
|--------|----|----|----|----|
| 0      |    |    | 1  | 1  |
| 1      |    | 1  |    | 1  |

$$\Rightarrow \underline{\underline{\bar{A}B + BC + A\bar{B}C}}$$

$$f(A, B, C) = A\bar{B}\bar{C} + A\bar{B}C + \bar{A}B$$

$$= A\bar{B}\bar{C} + A\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC$$

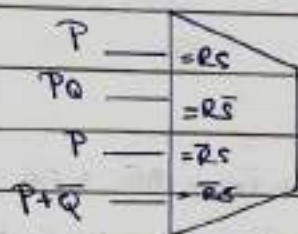
$$= \sum m(2, 3, 5, 6)$$

$$\rightarrow P(\bar{Q} + Q)\bar{R}$$

$$P\bar{Q}\bar{R} = 1000$$

$$PQ\bar{R} = 1100$$

Q.5



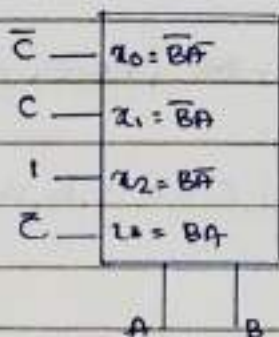
$$(P, Q, R, S) = (P + \bar{Q})\bar{R}S + P\bar{R}S + PQ\bar{R}S + P\bar{R}S$$

$$= \sum m(0, 8, 9, 11, 12, 13, 14, 15)$$

| PQ \ RS | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      | 1  |    |    |    |
| 01      |    |    |    |    |
| 11      | 1  | 1  | 1  | 1  |
| 10      | 1  | 1  | 1  |    |

$$\Rightarrow P\bar{Q} + P\bar{Q} + \bar{Q}\bar{R}S$$

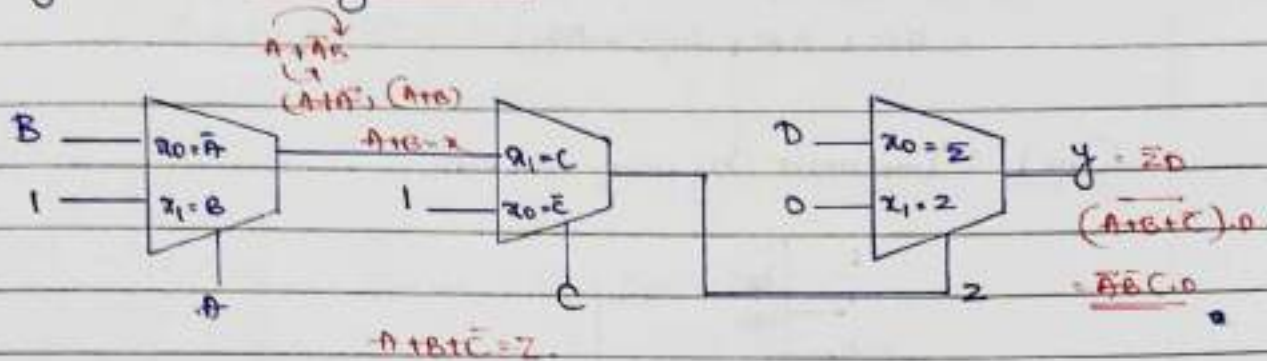
Q.6



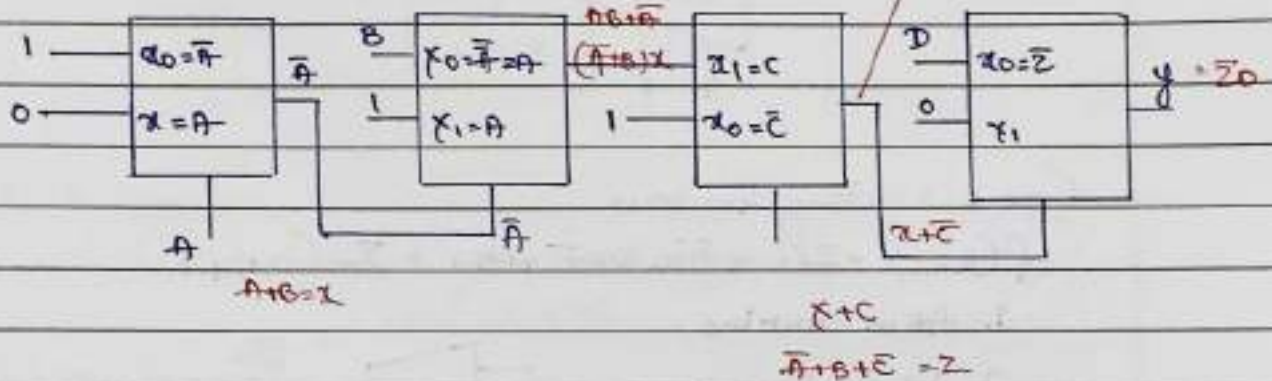
$$f(A, B, C) = \bar{B}\bar{A}\bar{C} + \bar{B}\bar{A}C + \bar{B}A + B\bar{A}\bar{C}$$

$$= \sum m(0, 2, 4, 5, 6) \text{ (1) method}$$

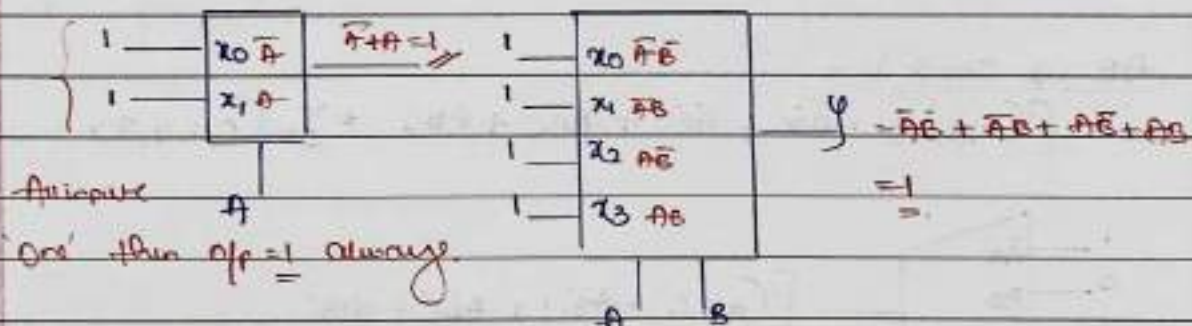
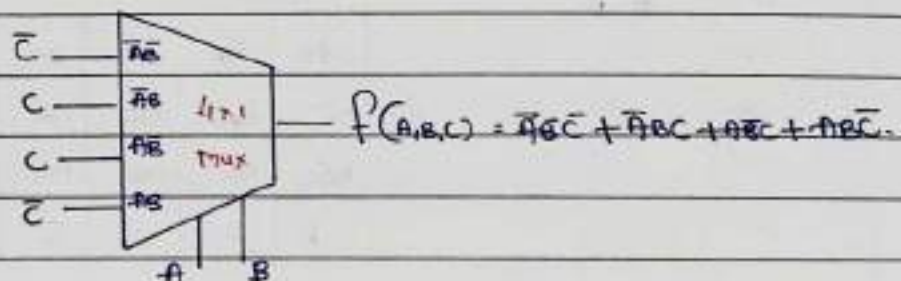
$$f(A, B, C) = \bar{B}\bar{A}\bar{C} + \bar{B}\bar{A}C + \bar{B}A\bar{C} + \bar{B}AC + B\bar{A}\bar{C} \text{ (2) method}$$

Type 4: Cascading of Mux:

Q1. Find the output of y.  $\bar{A} + AB = (\bar{A} + A)(\bar{A} + B) = \bar{A} + B$

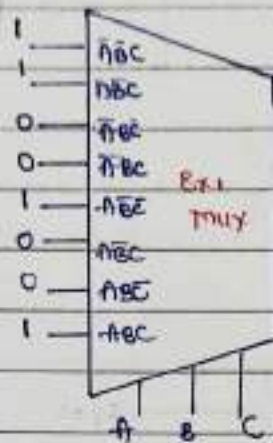


$$y = \bar{z}D = (\bar{A} + B + \bar{C})D = \bar{A}\bar{B}\bar{C}D$$

Note:Answer = 1Type 5: Implementation of Function:

Ex.  $f(A, B, C) = \sum m(0, 1, 4, 7)$   
 $= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$

Case I : Implement by using 8x1 mux



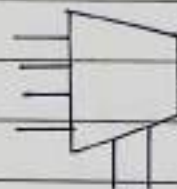
$$f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$$

$$= \sum m(0, 1, 4, 7)$$

Case II : Using 4x1 Mux

$$f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC = \sum m(0, 1, 4, 7)$$

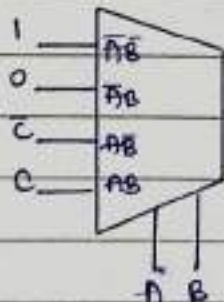
1.  $\bar{A}\bar{B}$  as select line
2.  $\bar{B}C$  as select line
3.  $AC$  as select line



$$f(A, B, C) = \sum m(0, 1, 4, 7)$$

1.  $\bar{A}\bar{B}$  as select line

$$f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC = \sum m(0, 1, 4, 7)$$



$$f(A, B, C) = \bar{A}\bar{B}.1 + A\bar{B}\bar{C} + ABC$$

$$= \bar{A}\bar{B}(\bar{C} + C) + A\bar{B}\bar{C} + ABC$$

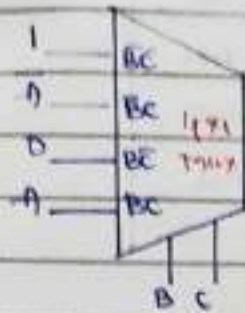
$$= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$$

|           | $\bar{A}\bar{B}$          | $\bar{A}B$          | $A\bar{B}$          | $AB$          |
|-----------|---------------------------|---------------------|---------------------|---------------|
| $\bar{C}$ | $\bar{A}\bar{B}\bar{C}$ ① | $\bar{A}B\bar{C}$ ② | $A\bar{B}\bar{C}$ ④ | $AB\bar{C}$ ⑤ |
| $C$       | $\bar{A}\bar{B}C$ ③       | $\bar{A}BC$ ⑥       | $A\bar{B}C$ ⑦       | $ABC$ ⑧       |
|           | 1                         | 0                   | $\bar{C}$           | $C$           |

2. BC as Select line

$$f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + ABC$$

$$= \sum m(0, 1, 4, 7)$$



$$f(A, B, C) = \bar{B}\bar{C} + \bar{A}BC + ABC$$

$$= (\bar{A} + A)\bar{B}\bar{C} + \bar{A}BC + ABC$$

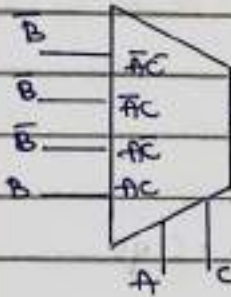
$$= \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + \bar{A}BC + ABC$$

|           | $\bar{B}\bar{C}$          | $\bar{B}C$          | $BC$                | $BC$          |
|-----------|---------------------------|---------------------|---------------------|---------------|
| $\bar{A}$ | $\bar{A}\bar{B}\bar{C}$ ① | $\bar{A}\bar{B}C$ ② | $\bar{A}B\bar{C}$ ③ | $\bar{A}BC$ ④ |
| $A$       | $A\bar{B}\bar{C}$ ⑤       | $A\bar{B}C$ ⑥       | $AB\bar{C}$ ⑦       | $ABC$ ⑧       |
|           | 1                         | $\bar{A}$           | 0                   | A             |

3. AC is select line

$$f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$= \sum m(0, 1, 4, 7)$$



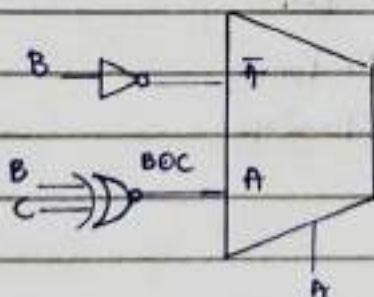
$$f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

|           | $\bar{A}\bar{C}$          | $\bar{A}C$          | $AC$                | $AC$          |
|-----------|---------------------------|---------------------|---------------------|---------------|
| $\bar{B}$ | $\bar{A}\bar{B}\bar{C}$ ① | $\bar{A}B\bar{C}$ ② | $A\bar{B}\bar{C}$ ③ | $A\bar{B}C$ ④ |
| B         | $\bar{A}B\bar{C}$ ⑤       | $A\bar{B}\bar{C}$ ⑥ | $A\bar{B}C$ ⑦       | $ABC$ ⑧       |
|           | $\bar{B}$                 | $\bar{B}$           | $\bar{B}$           | B             |

Case II: Using 2:1 Mux(i) A as Select line

$$f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$= \sum m(0, 1, 4, 7)$$



$$\bar{A}\bar{B} + A(\bar{B}\bar{C} + BC)$$

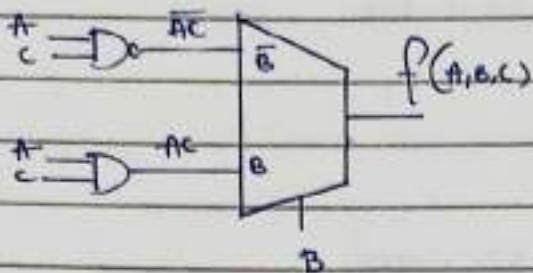
$$\bar{A}\bar{B}(\bar{C} + C) + \bar{A}BC$$

$$+ ABC$$

$$= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$= f(A, B, C) + ABC$$

|                  | $\bar{A}$                 | A                   |
|------------------|---------------------------|---------------------|
| $\bar{B}\bar{C}$ | $\bar{A}\bar{B}\bar{C}$ ① | $A\bar{B}\bar{C}$ ④ |
| $\bar{B}C$       | $\bar{A}B\bar{C}$ ②       | $A\bar{B}C$ ⑤       |
| $BC$             | $A\bar{B}\bar{C}$ ③       | $A\bar{B}C$ ⑥       |
| $BC$             | $\bar{A}BC$ ⑦             | $ABC$ ⑧             |
|                  | $\bar{B}$                 | B                   |

(iii) B as select line

$$f(A, B, C) = \sum m(0, 1, 4, 7)$$

|                  | $\bar{B}$                    | B                     |
|------------------|------------------------------|-----------------------|
| $\bar{A}\bar{C}$ | $\bar{A}\bar{B}\bar{C}$ (0)  | $\bar{A}B\bar{C}$ (2) |
| $\bar{A}C$       | $\bar{A}\bar{B}C$ (1)        | $\bar{A}BC$ (3)       |
| $AC$             | $AB\bar{C}$ (4)              | $AB\bar{C}$ (6)       |
| $AC$             | $ABC$ (5)                    | $ABC$ (7)             |
|                  | $\bar{A}C + \bar{A}\bar{B}C$ | $AC$                  |

$$= \bar{A}(\bar{C} + C) + AC$$

$$= \bar{A} + AC$$

$$= \bar{A}C$$

HW(ii) C as select line  $f(A, B, C) = \sum m(0, 1, 4, 7)$ 

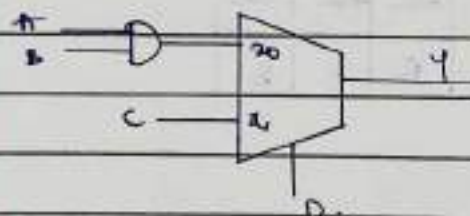
Q1.  $f(A, B, C, D) = \sum m(0, 1, 3, 5, 7, 9, 12, 15)$

1. ABD is select line2. ACD is select line

→ 8x1 mux, 8x1 mux, 4x1 mux

2x1 mux

( AB, AC, AD as select line  
A as select line,

Type-6: Delay in Mux

$$T_{AND} = 1 \mu s, T_{MUX} = 2 \mu s$$

Case 1: D=0

$$T = T_{AND} + T_{MUX} = 1 \mu s + 2 \mu s$$

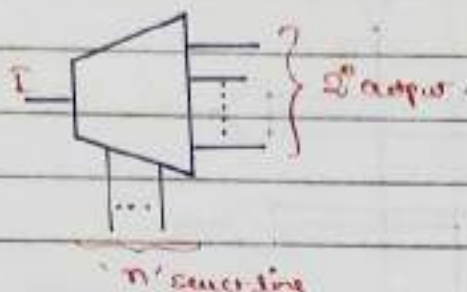
$$T = \underline{\underline{3 \mu s}}$$

Case 2: D=1

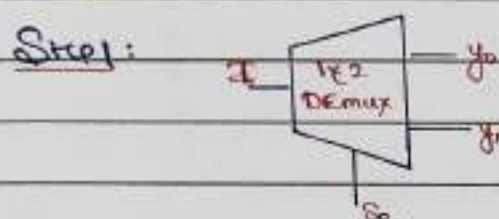
$$T = T_{MUX}$$

$$T = \underline{\underline{2 \mu s}}$$

DE-mux: DE-mux is called And Logic



Q1. Design 1x2 DE-mux



Step 2:

| $S_0$ | $y_0$ | $y_1$ |
|-------|-------|-------|
| 0     | I     | 0     |
| 1     | 0     | I     |

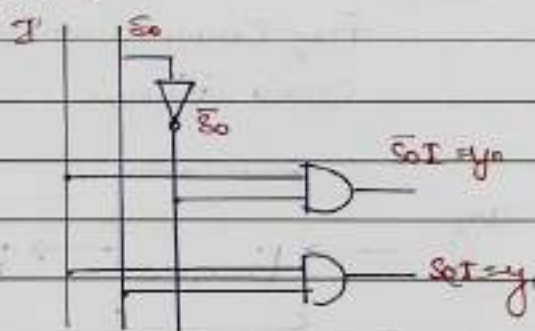
Step 3:

$$y_0 = \bar{S}_0 I$$

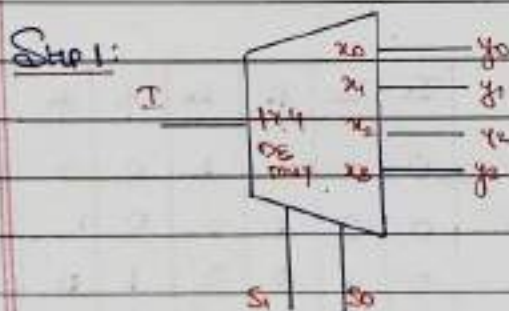
$$y_1 = S_0 I$$

Step 5:

Step 4:



Q2. Design a 1x4 DEMUX?



Step 2:

| $S_1$ | $S_0$ | $y_0$ | $y_1$ | $y_2$ | $y_3$ |
|-------|-------|-------|-------|-------|-------|
| 0     | 0     | I     | 0     | 0     | 0     |
| 0     | 1     | 0     | I     | 0     | 0     |
| 1     | 0     | 0     | 0     | I     | 0     |
| 1     | 1     | 0     | 0     | 0     | I     |

Step 3:

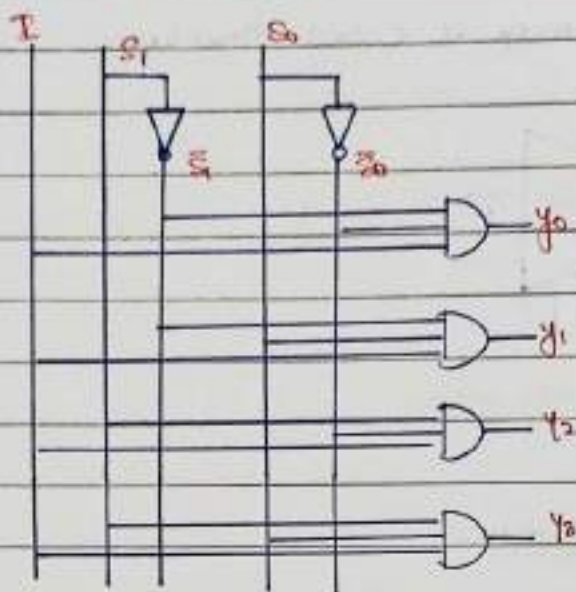
$$y_0 = \bar{S}_1 \bar{S}_0 I$$

$$y_1 = \bar{S}_1 S_0 I$$

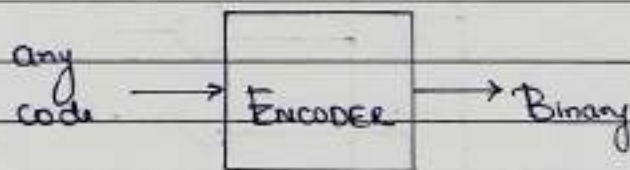
$$y_2 = S_1 \bar{S}_0 I$$

$$y_3 = S_1 S_0 I$$

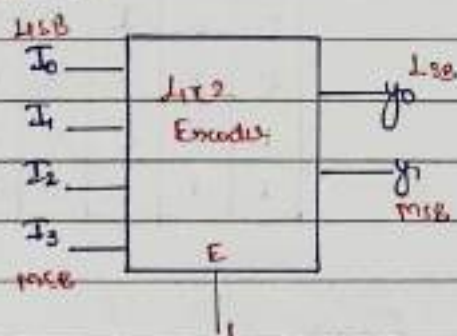
Step 4:

Steps:Q. NoQ. Design a  $1 \times 8$  DE MUXQ. Design a  $1 \times 16$  DE MUX.Encoder:

Any Circuit which is used to convert any code into Binary is called Encoder.



- 1)  $4 \times 2$  Encoder (Quad to Binary)
- 2)  $8 \times 3$  Encoder (Octa to Binary)
- 3)  $16 \times 4$  Encoder (Hexa to Binary)

Q. Design a  $4 \times 2$  Encoder.Step 1:Step 2:

| $I_3$ | $I_2$ | $I_1$ | $I_0$ | $Y_1$ | $Y_0$ |
|-------|-------|-------|-------|-------|-------|
| 0     | 0     | 0     | 1     | 0     | 0     |
| 0     | 0     | 1     | 0     | 0     | 1     |
| 0     | 1     | 0     | 0     | 1     | 0     |
| 1     | 0     | 0     | 0     | 1     | 1     |

Step 3:  $y_0 = I_1 + I_2$   
 $y_1 = I_2 + I_3$

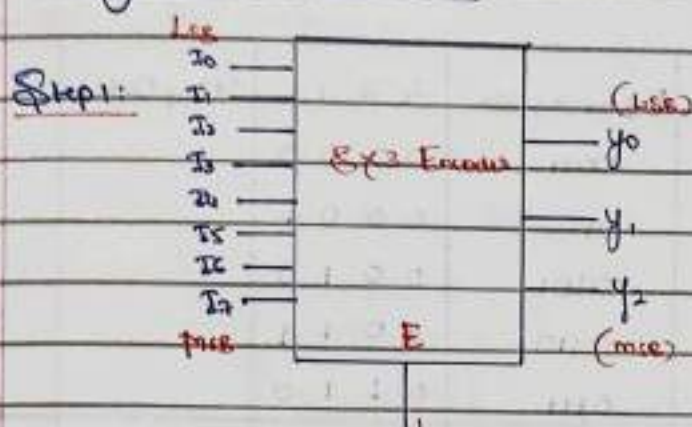
Step 5:

$$\begin{matrix} I_1 \\ I_3 \end{matrix} \Rightarrow y_0 \text{ (LSB)}$$

Step 4:

$$\begin{matrix} I_2 \\ I_3 \end{matrix} \Rightarrow y_1 \text{ (MSB)}$$

Q. Design a 8x3 Encoder.



HW

Q. Design a 16x4 Encoder

Step 2:

| $I_7$ | $I_6$ | $I_5$ | $I_4$ | $I_3$ | $I_2$ | $I_1$ | $I_0$ | $y_2$ | $y_1$ | $y_0$ |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     |
| 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 1     |
| 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 1     | 0     |
| 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 1     | 1     |
| 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 1     | 0     | 0     |
| 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 1     |
| 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 0     |
| 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 1     |

Step 3:  $y_0 = I_1 + I_3 + I_5 + I_7$   
 $y_1 = I_2 + I_3 + I_6 + I_7$   
 $y_2 = I_4 + I_5 + I_6 + I_7$

Step 5:

$$\begin{matrix} I_1 \\ I_3 \\ I_5 \\ I_7 \end{matrix} \Rightarrow y_0$$

$$\begin{matrix} I_2 \\ I_3 \\ I_6 \\ I_7 \end{matrix} \Rightarrow y_1$$

$$\begin{matrix} I_4 \\ I_5 \\ I_6 \\ I_7 \end{matrix} \Rightarrow y_2$$

Step 4:

Non: Priority Encoder

(L.O.B)

L.O.B Priority Encoder

| (True) | $I_3$ | $I_2$ | $I_1$ | $I_0$ (L.O.B) | $Y_1$ | $Y_0$ |
|--------|-------|-------|-------|---------------|-------|-------|
|        | X     | X     | X     | 1             | 0     | 0     |
|        | X     | X     | 1     | 0             | 0     | 1     |
|        | X     | 1     | 0     | 0             | 1     | 0     |
|        | 1     | 0     | 0     | 0             | 1     | 1     |

Weighted, Self-Complemented

|  | Decimal | B.C.D | Excess-3 code | 4 3 2 1 | Gray Code |
|--|---------|-------|---------------|---------|-----------|
|  | 0       | 0000  | 0011          | 0 0 0 0 |           |
|  | 1       | 0001  | 0100          | 0 0 0 1 |           |
|  | 2       | 0010  | 0101          | 0 0 1 0 |           |
|  | 3       | 0011  | 0110          | 0 0 1 1 |           |
|  | 4       | 0100  | 0111          | 0 1 1 0 |           |
|  | 5       | 0101  | 1000          | 1 0 0 1 |           |
|  | 6       | 0110  | 1001          | 1 1 0 0 |           |
|  | 7       | 0111  | 1010          | 1 1 0 1 |           |
|  | 8       | 1000  | 1011          | 1 1 1 0 |           |
|  | 9       | 1001  | 1100          | 1 1 1 1 |           |

Decimal → Weighted code  
Self Complemented

$10^3 \ 10^2 \ 10^1 \ 1$   
 $a_3 \ a_2 \ a_1 \ a_0$

$r \rightarrow$  Base (radix)

Complement

$(r-1)'s$

$r's$  Complement

$r=2$

1's

2's

$r=8$

7's

8's

$r=10$

9's

10's

B.C.D → Weighted Code  
Not a Self Complement Code.

→ Binary Coded Decimal

→ Each decimal numbers are represented by 4 bits.

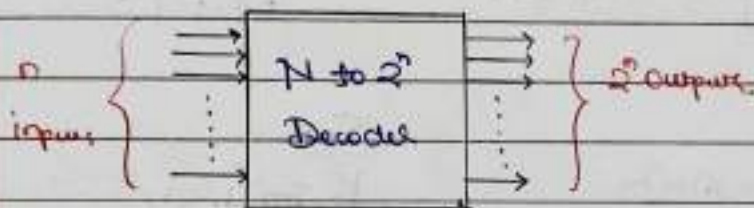
Eg:  $\left\{ \begin{array}{l} \text{Decimal} - 0 \\ \text{BCD} - 0000 \end{array} \right\}$   $\left\{ \begin{array}{l} \text{Decimal} - 9 \\ \text{BCD} - 1001 \end{array} \right\}$   $\left\{ \begin{array}{l} \text{Decimal} - 29 \\ \text{BCD} - 0010001 \end{array} \right\}$

Excess-3 code:

- It is not a weighted code
- Self complemented code

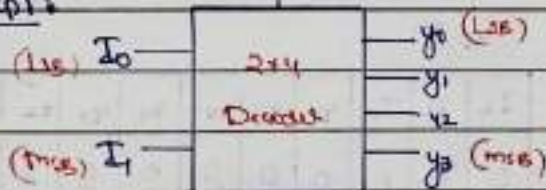
Decoder: Circuit which is used to convert Binary into any other code.

1.  $2 \times 4$  Decoder
2.  $3 \times 8$  Decoder
3.  $4 \times 16$  Decoder.



$2 \times 4$  Decoder

Step 1:



Step 2:

| $I_1$ | $I_0$ | $y_3$ | $y_2$ | $y_1$ | $y_0$ |
|-------|-------|-------|-------|-------|-------|
| 0     | 0     | 0     | 0     | 0     | 1     |
| 0     | 1     | 0     | 0     | 1     | 0     |
| 1     | 0     | 0     | 1     | 0     | 0     |
| 1     | 1     | 1     | 0     | 0     | 0     |

Step 3:  $y_0 = \bar{I}_1 \bar{I}_0$

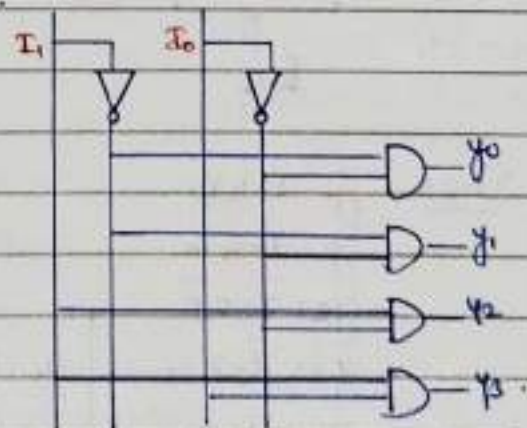
$y_1 = \bar{I}_1 I_0$

$y_2 = I_1 \bar{I}_0$

$y_3 = I_1 I_0$

Step 4: Minimization

Step 4:



Example:

eg:

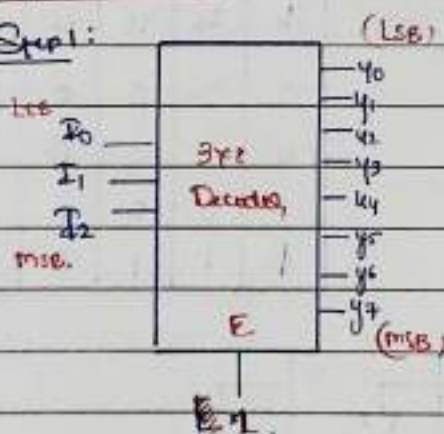
|           |      |                            |                   |  | A | B | P |
|-----------|------|----------------------------|-------------------|--|---|---|---|
| (L.S.B) B | AB 0 | $\overline{A}\overline{B}$ | $\Sigma m(1,2,3)$ |  | 0 | 0 | 0 |
|           | AB 1 | $\overline{A}B$            | $\Sigma m(0,2,3)$ |  | 0 | 1 | 1 |
| (M.S.B) A | AB 2 | $A\overline{B}$            | $\Sigma m(0,1,3)$ |  | 1 | 0 | 1 |
|           | AB 3 | $AB$                       | $\Sigma m(0,1,2)$ |  | 1 | 1 | 1 |

Pos =

$$P = (A+B) = \pi m(0)$$

Sop

$$P = \Sigma m(1,2,3)$$

3x8 DecoderStep 1:Step 2:

| I <sub>2</sub> | I <sub>1</sub> | I <sub>0</sub> | y <sub>0</sub> | y <sub>1</sub> | y <sub>2</sub> | y <sub>3</sub> | y <sub>4</sub> | y <sub>5</sub> | y <sub>6</sub> | y <sub>7</sub> |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 1              |
| 0              | 0              | 1              | 0              | 0              | 0              | 0              | 0              | 0              | 1              | 0              |
| 0              | 1              | 0              | 0              | 0              | 0              | 0              | 0              | 1              | 0              | 0              |
| 0              | 1              | 1              | 0              | 0              | 0              | 0              | 1              | 0              | 0              | 0              |
| 1              | 0              | 0              | 0              | 0              | 0              | 1              | 0              | 0              | 0              | 0              |
| 1              | 0              | 1              | 0              | 0              | 1              | 0              | 0              | 0              | 0              | 0              |
| 1              | 1              | 0              | 0              | 1              | 0              | 0              | 0              | 0              | 0              | 0              |
| 1              | 1              | 1              | 1              | 0              | 0              | 0              | 0              | 0              | 0              | 0              |

Step 3:

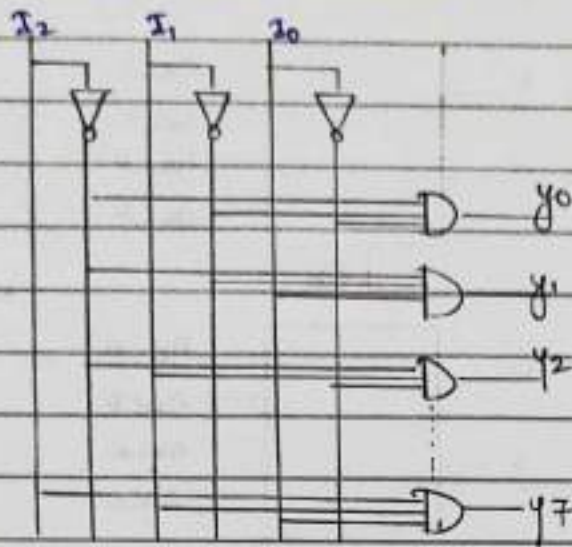
$$\begin{aligned} y_0 &= \overline{I_2} \overline{I_1} \overline{I_0} \\ y_1 &= \overline{I_2} \overline{I_1} I_0 \\ y_2 &= \overline{I_2} I_1 \overline{I_0} \\ y_3 &= \overline{I_2} I_1 I_0 \\ y_4 &= I_2 \overline{I_1} \overline{I_0} \\ y_5 &= I_2 \overline{I_1} I_0 \\ y_6 &= I_2 I_1 \overline{I_0} \\ y_7 &= I_2 I_1 I_0 \end{aligned}$$

$$\begin{aligned} y_6 &= I_2 I_1 \overline{I_0} \\ y_7 &= I_2 I_1 I_0 \end{aligned}$$

Step 4: Minimization

Steps: Hardware ImplementationHW

Design 4x16 Decoder

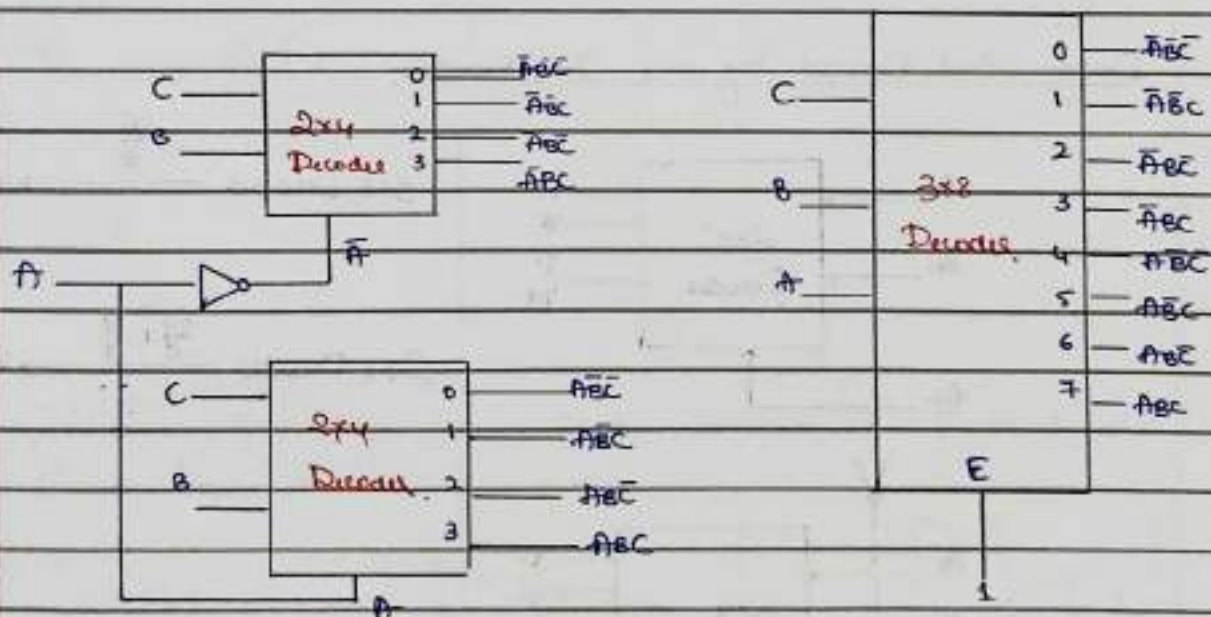


Q.1  $2 \times 4$  Decoder  $\xrightarrow{\substack{B=2 \\ 4}} 3 \times 8$  Decoder

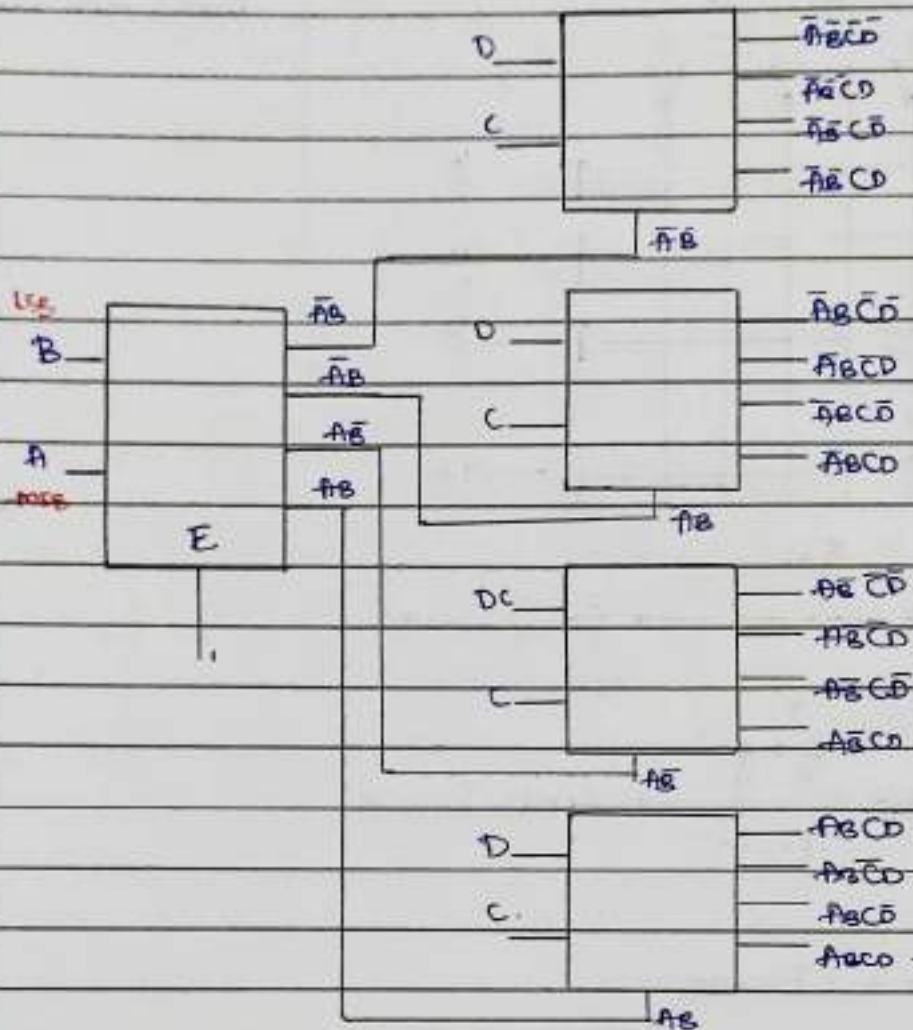
$\uparrow$  input       $\uparrow$  output

Q.2  $2 \times 4$  Decoder  $\xrightarrow{\substack{16 \\ 4 + 12}} 4 \times 16$  Decoder

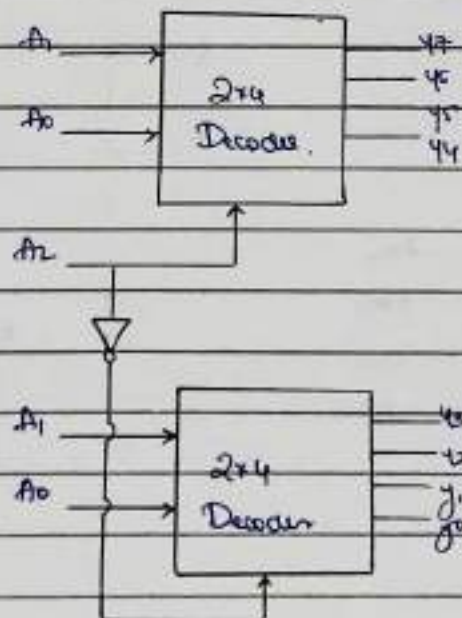
$4 + 1 = 5$

 $2 \times 4 \rightarrow 3 \times 8$  Decoder

2x4 Decoder → 4x16 Decoder

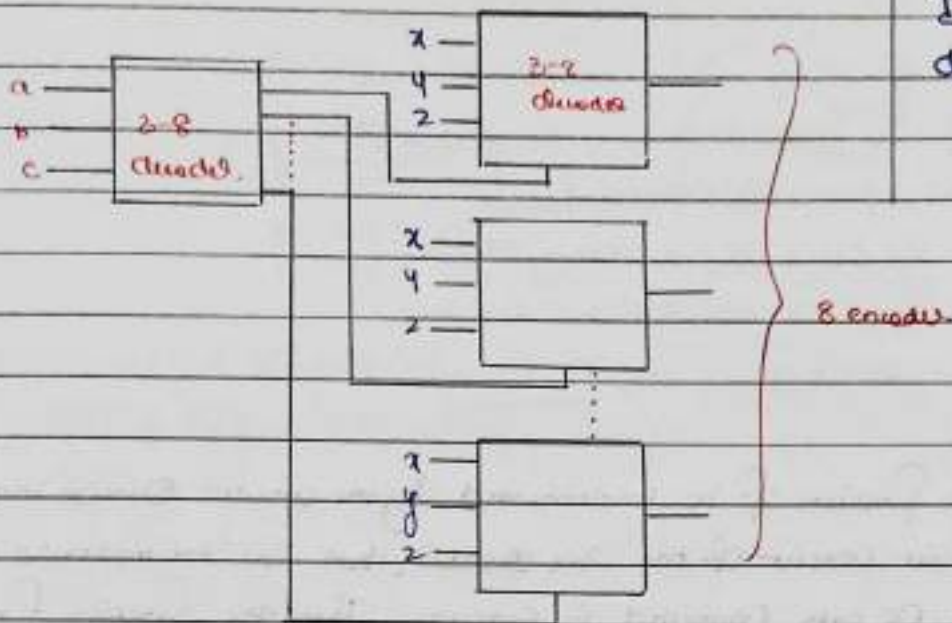


Design 3x8 Decoder by using 2x4 Decoder



3x8 Decoder  $\frac{16^0}{8} = 2 \rightarrow 4 \times 16$  Decoder

3x8 Decoder  $\frac{64^0}{8} + \frac{8^0}{8} = 9 \rightarrow 6 \times 64$  Decoder

3x8 Decoder  $\longrightarrow$  8x64 Decoder

H.W.  
Design Binary to decimal decoder.

Q.1 A logic circuit consist of 2x4 decoders as shown in the figure. The output are as follows:

$D_0 = 1$  when  $A_0 = 0, A_1 = 0$

$D_1 = 1$  when  $A_0 = 1, A_1 = 0$

$D_2 = 1$  "  $A_0 = 0, A_1 = 1$

$D_3 = 1$  "  $A_0 = 1, A_1 = 1$

The name of  $(x, y, z)$  is.

Ans: 1

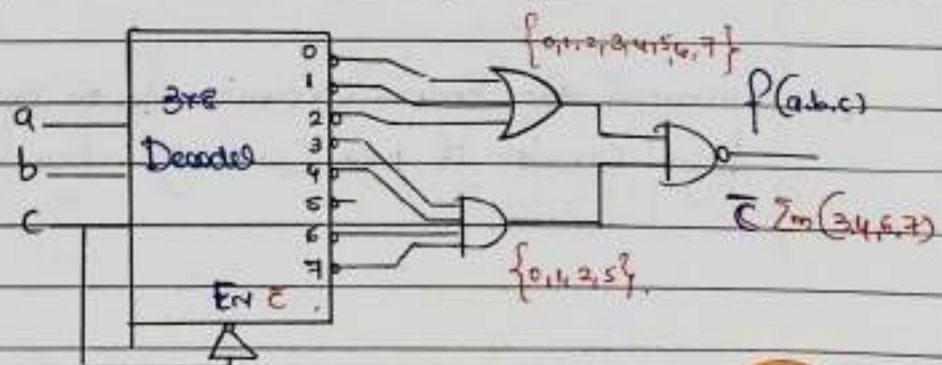


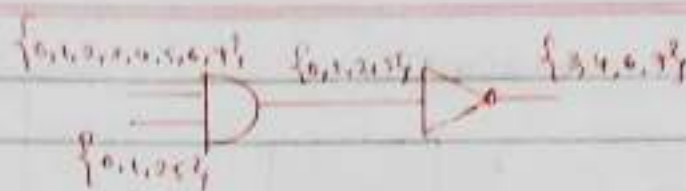
| $A_1$ | $A_0$ | $y$       |
|-------|-------|-----------|
| 0     | 0     | $D_0 = 1$ |
| 0     | 1     | $D_1$     |
| 1     | 0     | $D_2$     |
| 1     | 1     | $D_3$     |

Q2 The boolean expression  $f(a, b, c)$  in its canonical form for the decoder circuit shown below is

Ans:

$$\Rightarrow \Sigma_m(4, 6)$$





$$F = \sum m(3, 4, 5, 6)$$

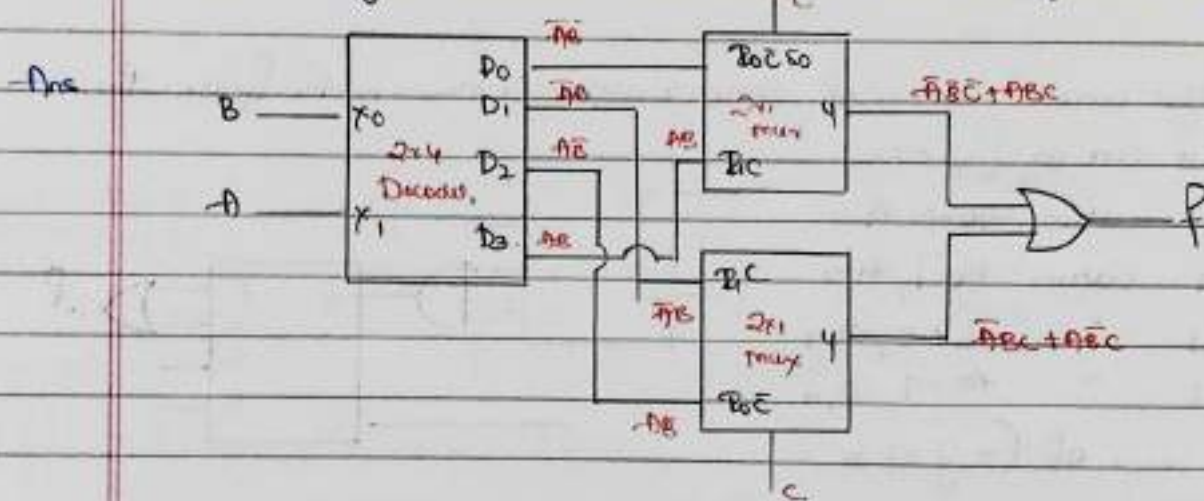
$$= \bar{C} [ABC + A\bar{B}C + AB\bar{C} + A\bar{B}\bar{C}]$$

$$= \bar{C} [ABC + A\bar{B}C + AB\bar{C} + A\bar{B}\bar{C}]$$

$$= \bar{A}BC\bar{C} + A\bar{B}C\bar{C} + AB\bar{C}\bar{C} + A\bar{B}\bar{C}\bar{C}$$

$$= \bar{A}BC + A\bar{B}C + AB\bar{C} + A\bar{B}\bar{C} \rightarrow \sum m(3, 4, 5, 6)$$

Q3. A logic function  $f$  is implemented by the circuit shown in the figure below. The circuit consists of one 2x4 decoder, two 2x1 multiplexers and a two input OR gate connected in cascade. Then the function  $f$  is equal to



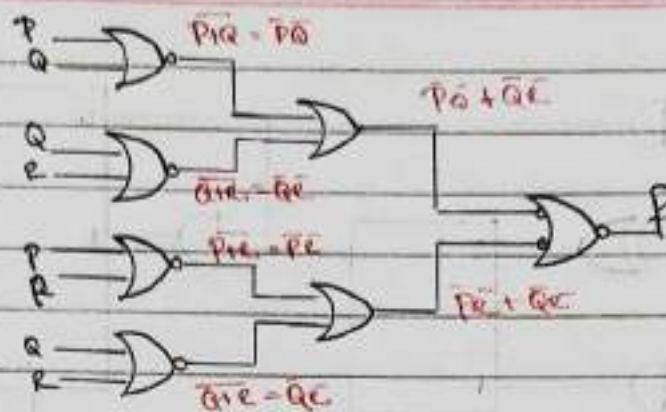
$$F = \bar{A}\bar{B}\bar{C} + A\bar{B}C + \bar{A}BC + A\bar{B}\bar{C}$$

$$= \sum m(0, 2, 4, 7)$$

$$\rightarrow \bar{B}\bar{C} + BC + \bar{B}C$$

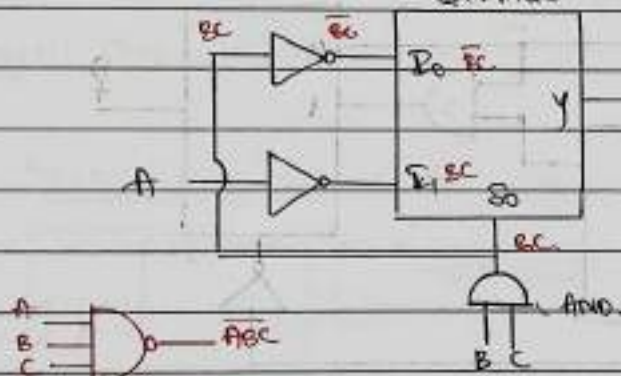
| A \ B | 00 | 01 | 11 | 10 |
|-------|----|----|----|----|
| 0     | 1  |    | 1  |    |
| 1     | 1  |    | 1  |    |

Q2. What is the boolean expression for the output  $f$  of the combinational logic circuit of NOR Gates given below.



$$\begin{aligned}
 f &= [\bar{P}\bar{Q} + \bar{Q}\bar{R}] \cdot [\bar{P}\bar{R} + \bar{Q}\bar{R}] \\
 &= \bar{P}\bar{Q}\bar{R} + \bar{P}\bar{Q}\bar{R} + \bar{P}\bar{Q}\bar{R} + \bar{Q}\bar{R} \\
 &= \bar{P}\bar{Q}\bar{R} + \bar{Q}\bar{R} \\
 &= \bar{Q}\bar{R}[1 + \bar{P}] \\
 &= \bar{Q}\bar{R} \Rightarrow \underline{\underline{\bar{Q}\bar{R}}}
 \end{aligned}$$

Q4. The Combinational Circuit given below implements which of the following Mux



$$= \bar{B}\bar{C} \cdot \bar{A}C + \bar{A}BC$$

$$= \bar{B}\bar{C} + \bar{A}BC$$

$$= \bar{A} + \bar{A}x$$

$$= (\bar{A} + \bar{A})(\bar{A} + x)$$

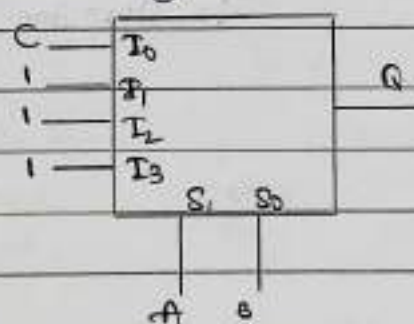
$$= \bar{B}\bar{C} + \bar{A}$$

$$= \bar{B}\bar{C} + \bar{A}$$

$$\left\{ \begin{array}{l} \bar{B}\bar{C} = x \end{array} \right.$$

∴ NAND Gate

Q5. The Combinational logic circuit shown in the given figure has an output Q which is



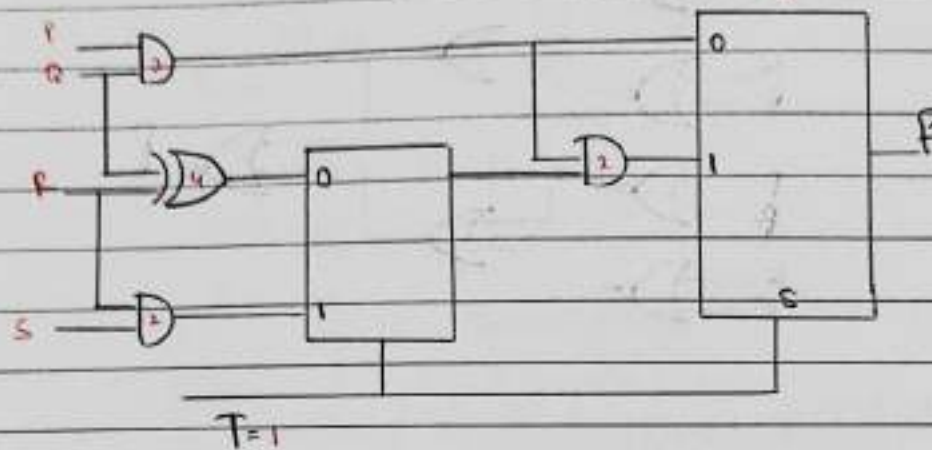
$$= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C}$$

$$= \sum m(1, 2, 3, 4, 5, 6, 7)$$

| A | 00 | 01 | 11 | 10 |
|---|----|----|----|----|
| 0 | 0  | 1  | 1  | 1  |
| 1 | 1  | 1  | 1  | 1  |

$$\Rightarrow \underline{\underline{A + B + C}}$$

Q6.

Case 1:  $T=0$ 

$$T = T_{\text{AND}} + T_{\text{MUX}}$$

$$T = 2 + 1 = \underline{3\text{ns}}$$

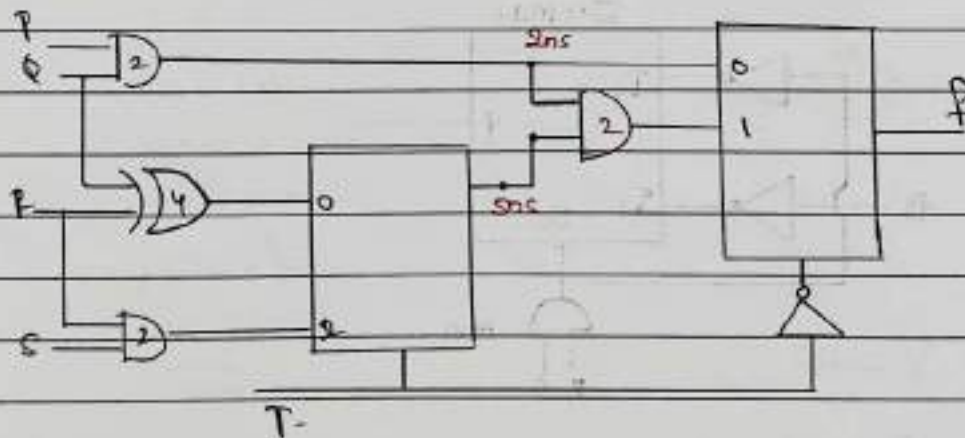
Case 2:  $T=1$ 

$$T = T_{\text{AND}} + T_{\text{MUX}} + T_{\text{AND}} + T_{\text{MUX}}$$

$$= 2 + 1 + 2 + 1 = \underline{6\text{ns}}$$

$$\therefore \text{max} = \underline{6\text{ns}}$$

Q7.

Case (1):  $T=0$ 

$$T = T_{\text{AND}} + T_{\text{MUX}} + T_{\text{AND}} + T_{\text{MUX}}$$

$$= 1 + 2 + 1 + 1 = \underline{5\text{ns}}$$

Case (2):  $T=1$ 

$$T = T_{\text{AND}} + T_{\text{MUX}}$$

$$T = 2 + 1 = \underline{3\text{ns}}$$

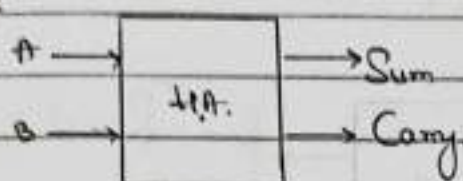
## Half Adder

$$\begin{array}{r}
 0011 \\
 +1010 \\
 \hline
 0001010
 \end{array}$$

Carry.

\* Two bit adder are known as half adder.

Step 1:



Step 2:

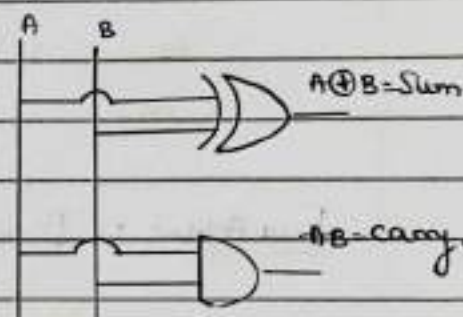
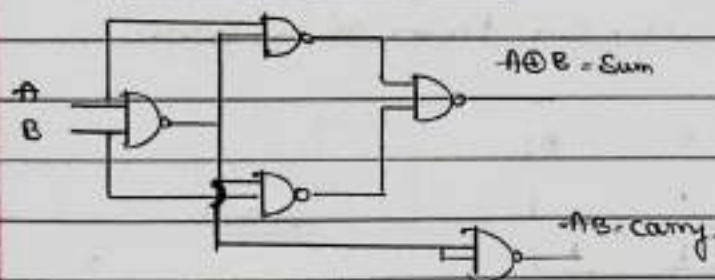
| A | B | Sum | Carry |
|---|---|-----|-------|
| 0 | 0 | 0   | 0     |
| 0 | 1 | 1   | 0     |
| 1 | 0 | 1   | 0     |
| 1 | 1 | 0   | 1     |

Step 3:  $Sum = \bar{A}B + A\bar{B} = A \oplus B$   
 $Carry = AB$

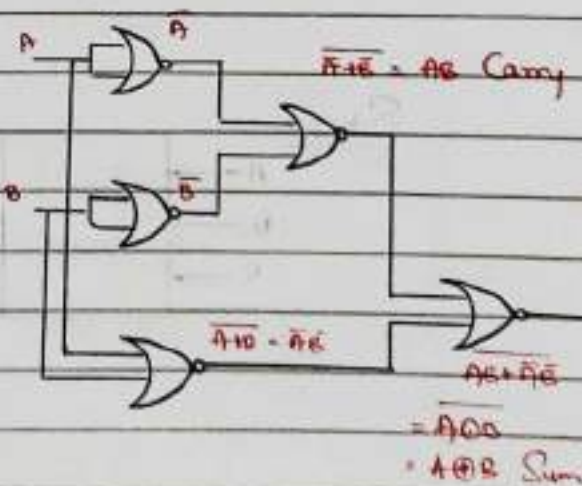
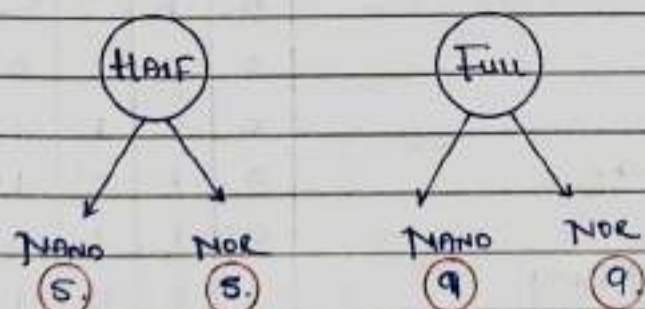
Steps:

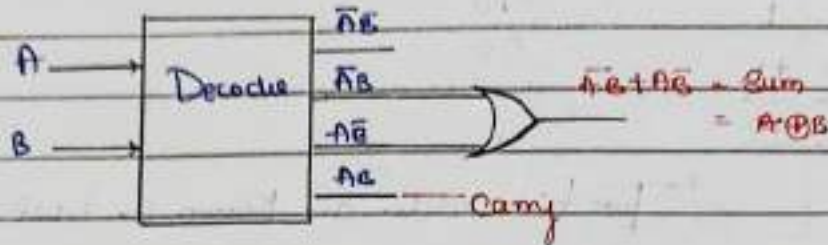
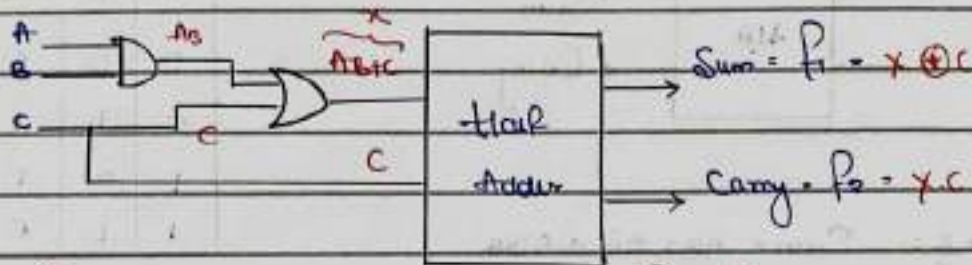
Step 1: Minimization.

NAND Gate Required = 5



By NOR Gate



Half Adder By Using Decoder.Q1. What is the value of  $f_1$  and  $f_2$ ?

$$\begin{aligned}
 f_1 &= (AB+C) \oplus C \\
 &= \overline{AB+C} \cdot C + (AB+C) \cdot \overline{C} \\
 &= \overline{AB} \cdot \overline{C} \cdot C + AB\overline{C} + C\overline{C} \\
 &= \underline{\underline{AB\overline{C}}}
 \end{aligned}$$

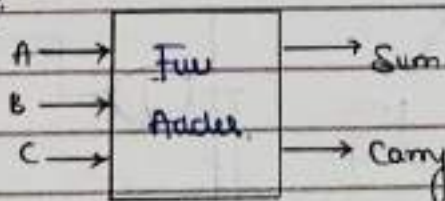
$$\begin{aligned}
 f_2 &= (AB+C) \cdot C \\
 &= ABC + CC = ABC + C \\
 &= C[AB+1] \\
 &= \underline{\underline{C \cdot AB}}
 \end{aligned}$$

Full Adder : Three bit adder are known as Full adder.

|    |    |    |    |
|----|----|----|----|
| 0  | 0  | 0  | 0  |
| 0  | 0  | 1  | 1  |
| +0 | +1 | +1 | +1 |
| 00 | 01 | 10 | 11 |

Step 2:

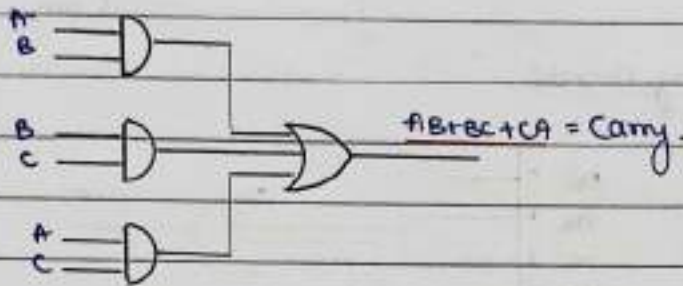
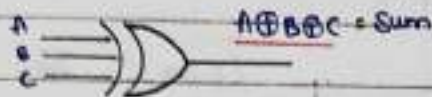
Step 1:



| A | B | C | Sum | Carry |
|---|---|---|-----|-------|
| 0 | 0 | 0 | 0   | 0     |
| 0 | 0 | 1 | 1   | 0     |
| 0 | 1 | 0 | 1   | 0     |
| 0 | 1 | 1 | 0   | 1     |
| 1 | 0 | 0 | 1   | 0     |
| 1 | 0 | 1 | 0   | 1     |
| 1 | 1 | 0 | 0   | 1     |
| 1 | 1 | 1 | 1   | 1     |

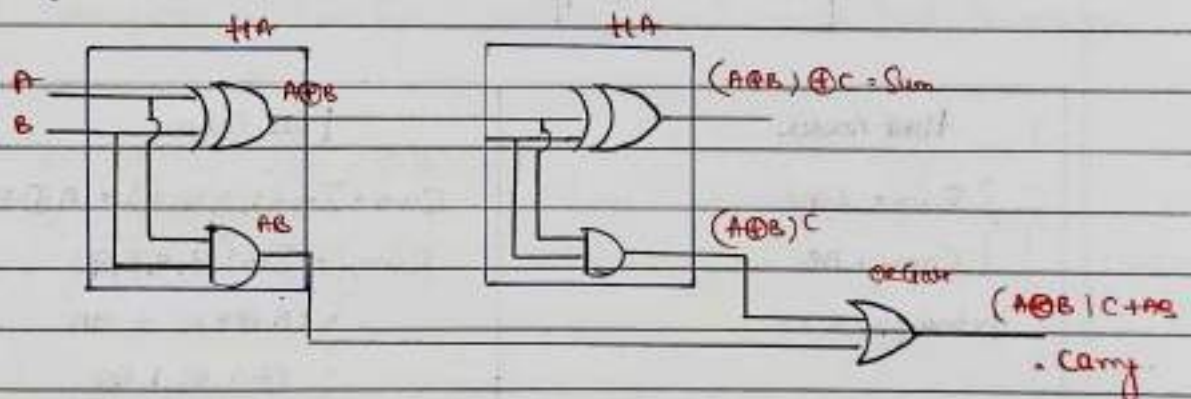
Step 3:  $\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$   
 (Sum) =  $\sum m(1, 2, 4, 7)$   
 =  $A \oplus B \oplus C$

Steps)

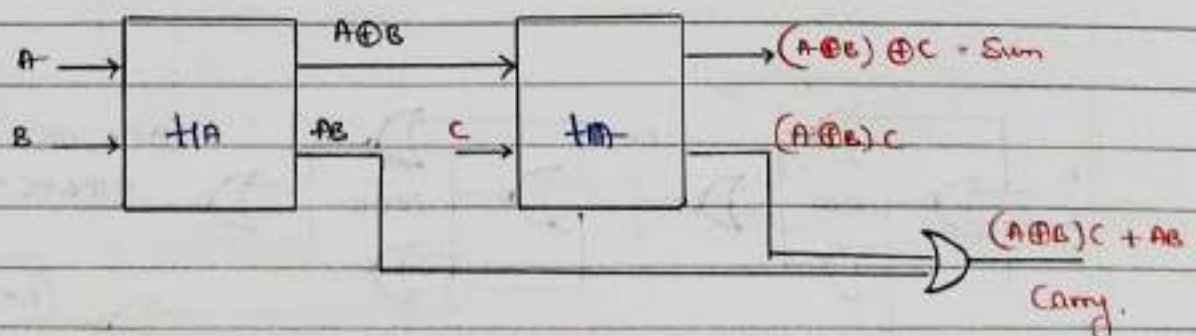


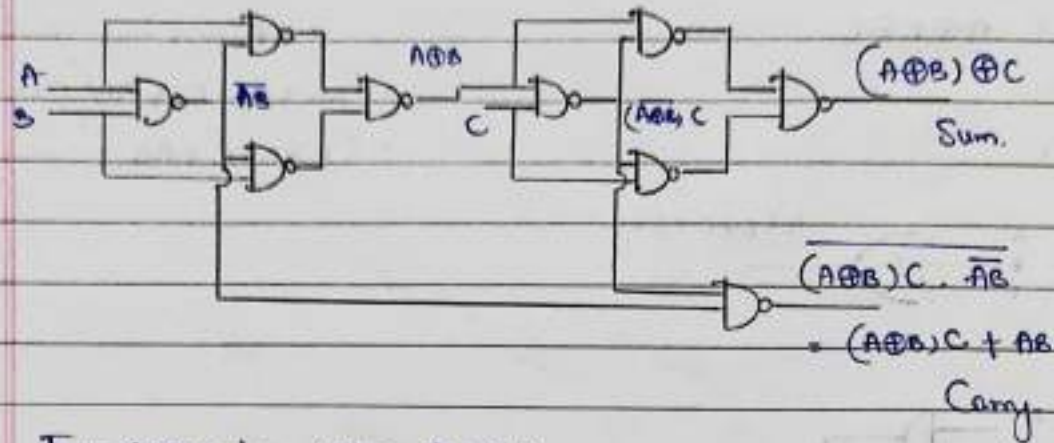
$$\text{Sum} = (A \oplus B) \oplus C$$

$$\text{Carry} = (A \oplus B)C + AB$$

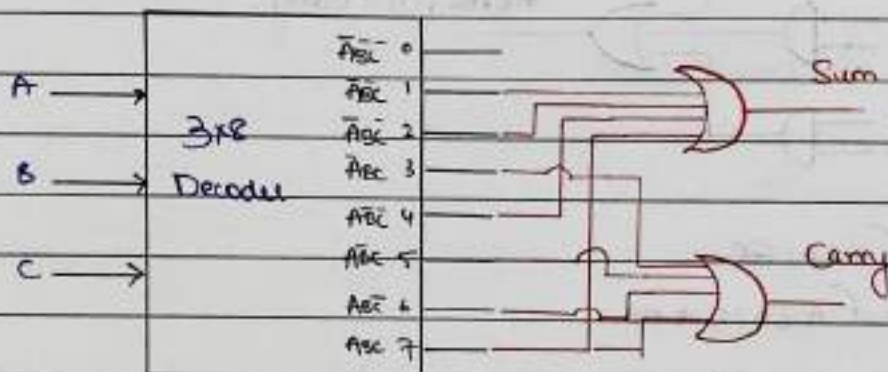


$$1 \text{ Full Adder} = 2 \text{ H.A.} + 1 \text{ OR Gate}$$

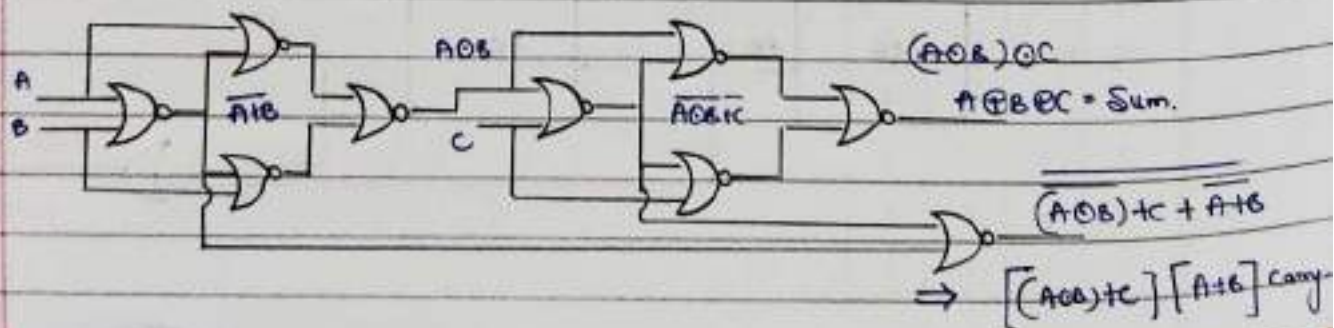


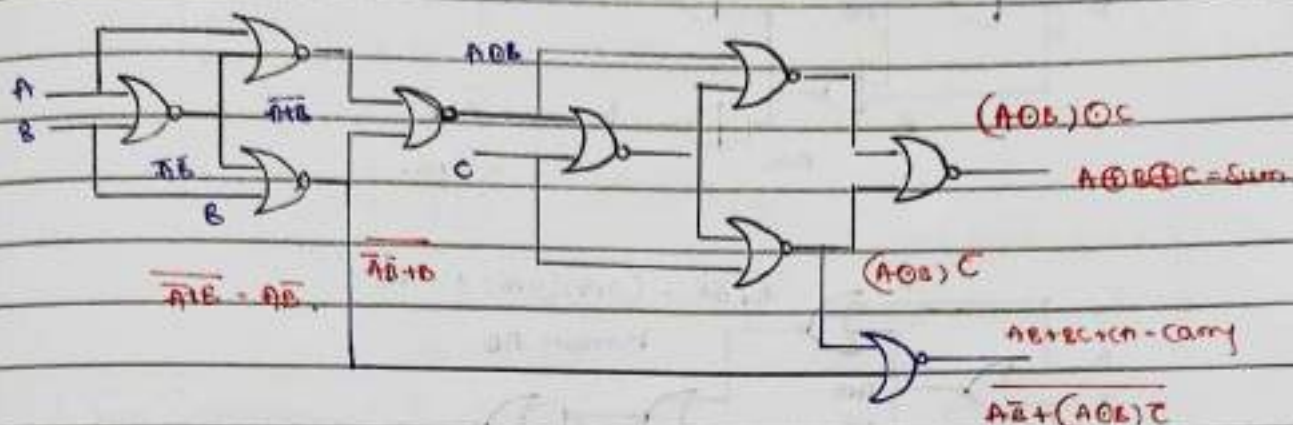
Full Adder by using NAND Gate.

HW  
Full Adder by using NOR Gate.

Full Adder by using decoder.

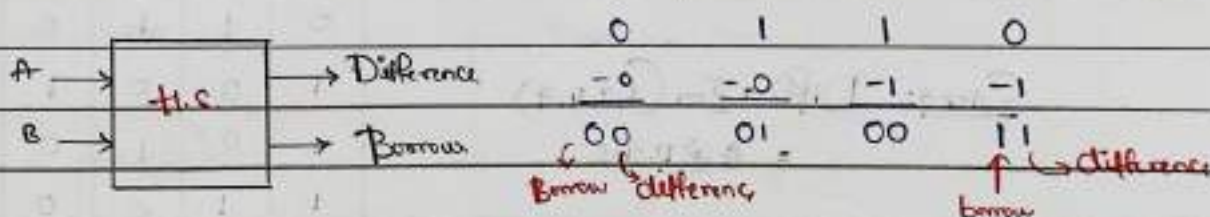
| Half Adder                                                               | Full Adder                                                                                                                                                               |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\begin{cases} \text{Sum} = A \oplus B \\ \text{Carry} = AB \end{cases}$ | $\begin{aligned} \text{Sum} &= \sum m(1, 2, 4, 7) = A \oplus B \oplus C \\ \text{Carry} &= \sum m(3, 5, 6, 7) \\ &= (A \oplus B)C + AB \\ &= AB + BC + AC \end{aligned}$ |
| NAND/NOR = 5                                                             | NAND/NOR = 9                                                                                                                                                             |
|                                                                          | 1 Full = 2 Half + 1 OR                                                                                                                                                   |



Full Subtractor

$$\begin{aligned}
 &= \overline{A}B + (A \odot B)C \\
 &= \overline{A}B + (\overline{A}B + AB)C \\
 &= (\overline{A} + B)[(\overline{A}B) + C] \\
 &= (\overline{A} + B)[\overline{A}B + AB + C] \\
 &= \overline{A} + B + \overline{A}C + AB + BC \\
 &\Rightarrow \overline{A}B + \overline{A}C + BC
 \end{aligned}$$

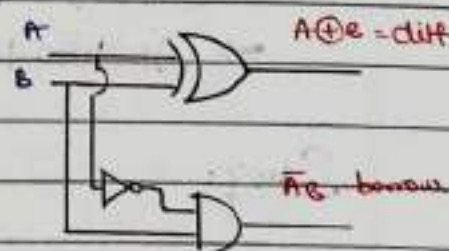
Borrow.

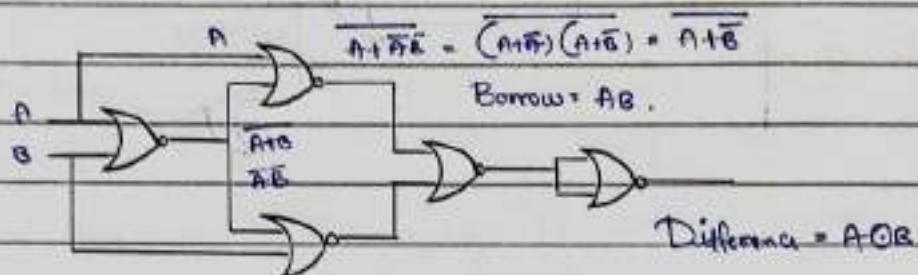
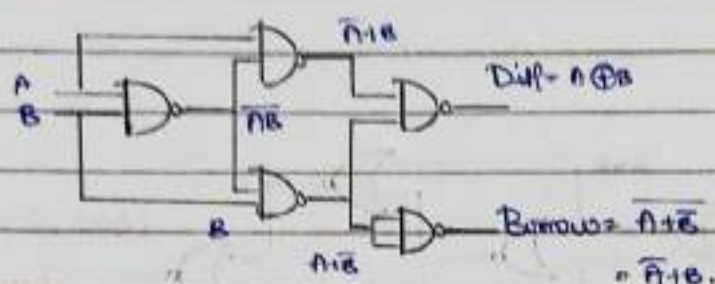
Half Subtractor : Two bit Subtractor are known as half subtractor.Step 1:Step 2:

| A | B | Diff. | Borrow |
|---|---|-------|--------|
| 0 | 0 | 0     | 0      |
| 0 | 1 | 1     | 1      |
| 1 | 0 | 1     | 0      |
| 1 | 1 | 0     | 0      |

Step 4:Step 5:

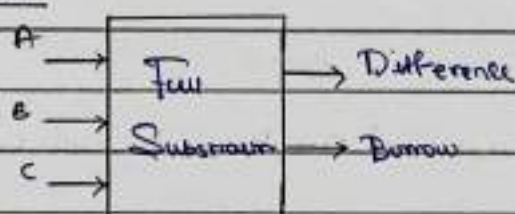
Step 3: Diff =  $A \oplus B$   
 Borrow =  $\overline{A}B$





### Full Subtractor

Step 1:



Step 2:

| Input |   |   | Output |        |
|-------|---|---|--------|--------|
| A     | B | C | Diff   | Borrow |
| 0     | 0 | 0 | 0      | 0      |
| 0     | 0 | 1 | 1      | 1      |
| 0     | 1 | 0 | 1      | 1      |
| 0     | 1 | 1 | 0      | 1      |
| 1     | 0 | 0 | 1      | 0      |
| 1     | 0 | 1 | 0      | 0      |
| 1     | 1 | 0 | 0      | 0      |
| 1     | 1 | 1 | 1      | 1      |

Step 3:  $\text{Diff} = \sum m(1, 2, 4, 7)$   
 $= A \oplus B \oplus C$

$\text{Borrow} = \sum m(1, 2, 3, 7)$

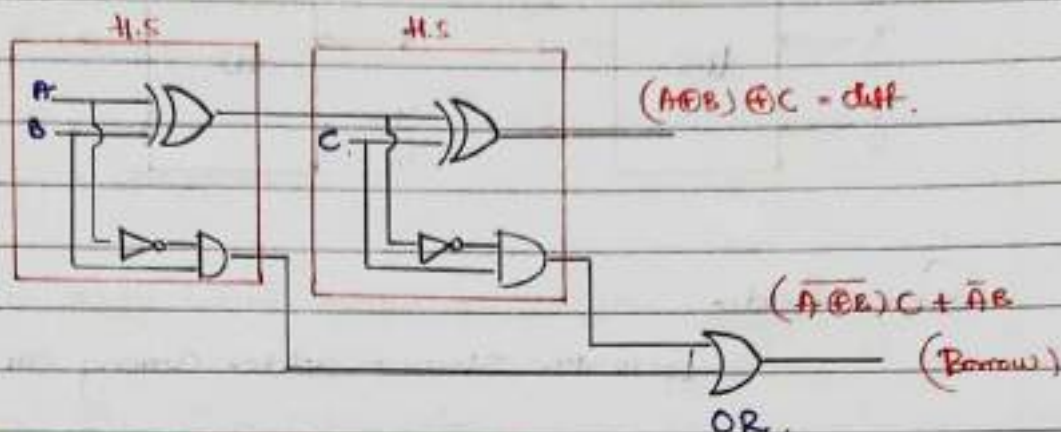
$$= \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}B C + A\bar{B}C$$

$$= (\bar{A}\bar{B} + \bar{A}B)C + \bar{A}B(\bar{C} + C)$$

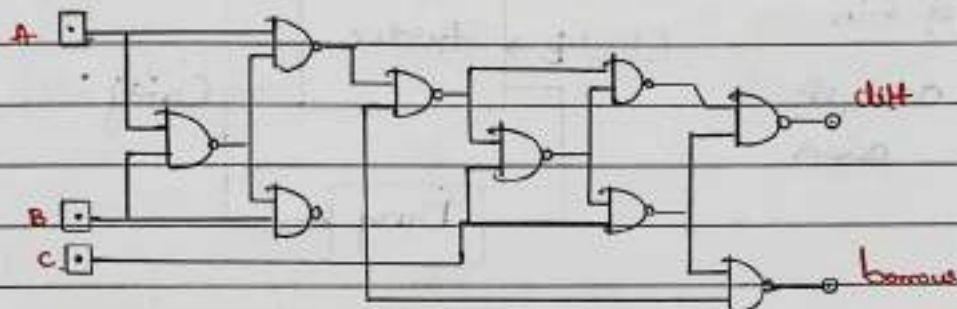
$$= (\bar{A} \oplus B)C + \bar{A}B \rightarrow \text{Semi minimized expression}$$

$$\rightarrow \bar{A}B + \bar{A}C + BC$$

Step 4:

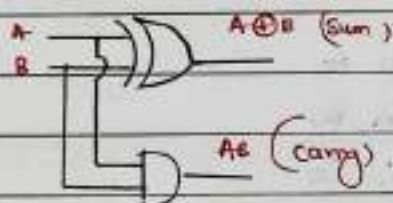
Steps:

1 Full Subtractor = 2 H.S. + Or Gate.

Half adder

$$\text{Sum} = A \oplus B$$

$$\text{Carry} = AB$$

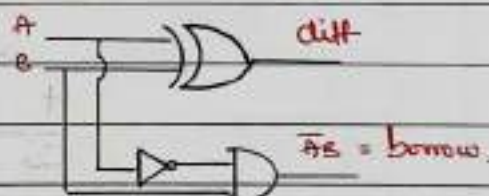


NAND } 5.  
NOR }

Half Subtractor

$$\text{Diff} = A \oplus B$$

$$\text{Borrow} = \overline{A}B$$



NAND } 5.  
NOR }

Full Adder

$$\text{Sum} = A \oplus B \oplus C$$

$$\begin{aligned} \text{Carry} &= \sum m(3, 5, 6, 7) \\ &= (A \oplus B)C + AB \\ &= AB + AC + BC \end{aligned}$$

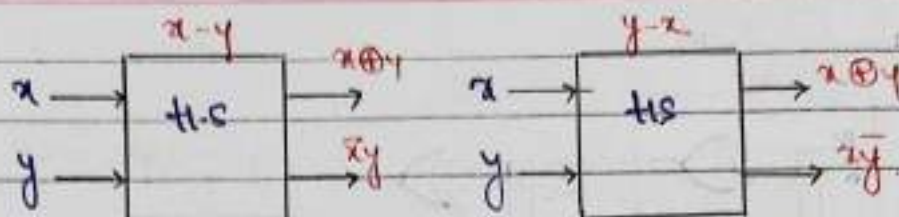
NAND/NOR = 9.

Full Subtractor

$$\text{Difference} = A \oplus B \oplus C$$

$$\begin{aligned} \text{Borrow} &= \sum m(1, 2, 3, 7) \\ &= (\overline{A \oplus B})C + \overline{A}B \\ &= \overline{A}B + \overline{A}C + BC \end{aligned}$$

NAND/NOR = 9.



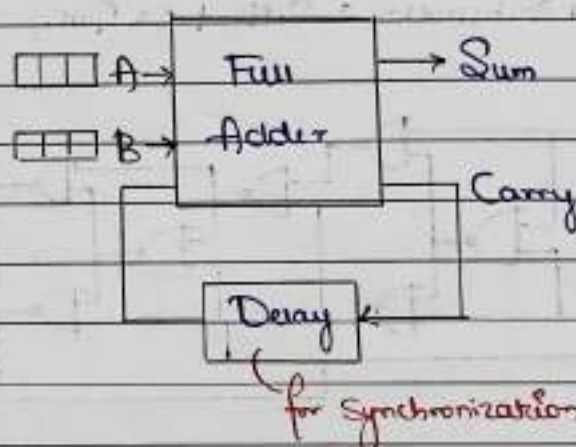
### Serial Adder "One by one addition"

It is the slowest adder among all the adders.

$$T = n \cdot T_{\text{delay}}$$

$n \rightarrow$  no. of bits

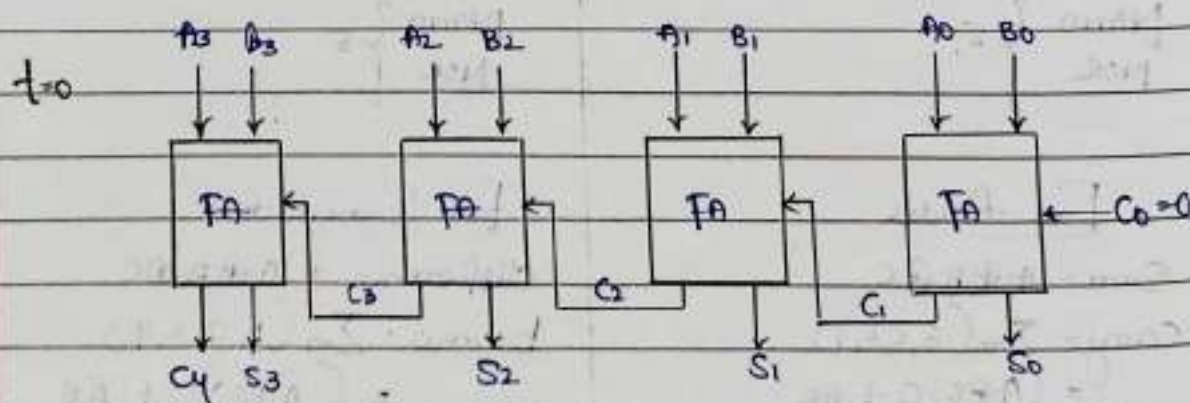
$T_{\text{delay}} \rightarrow$  delay of adder.



### Parallel Adder (Ripple Carry Adder)

Inputs are connected simultaneously

$$\begin{array}{r}
 \text{Bin} \\
 \begin{array}{cccc}
 C_3 & C_2 & C_1 & \\
 A_3 & A_2 & A_1 & A_0 \\
 + & B_3 & B_2 & B_1 & B_0 \\
 \hline
 C_4 & S_3 & S_2 & S_1 & S_0
 \end{array}
 \end{array}$$

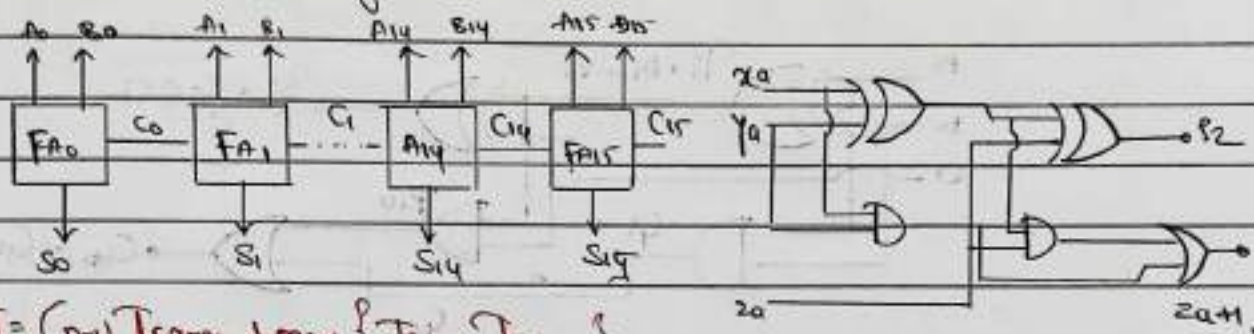


n bit Parallel adder

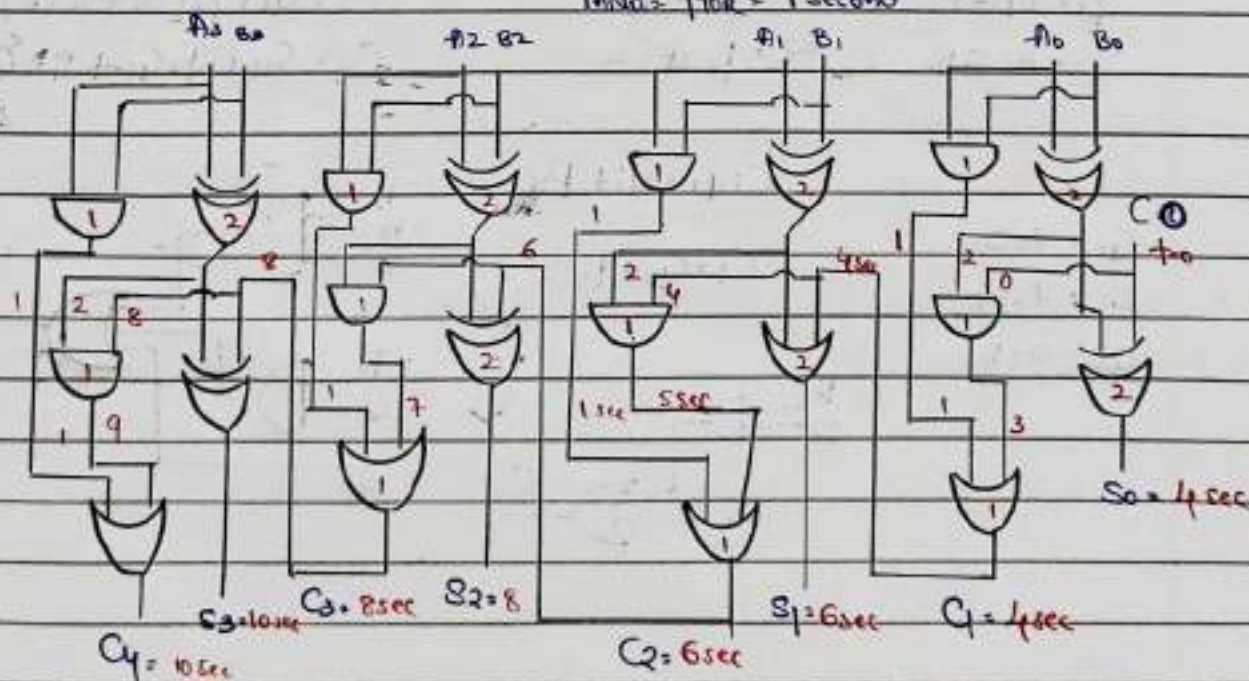
1.  $(n-1)$  FA + 1 HA
2. n-FA
3.  $(2n-1)$  HA +  $(n-1)$  OR.

$$T = (n-1) T_{\text{carry}} + \max(T_{\text{sum}}, T_{\text{carry}})$$

Q1. A 16-bit ripple carry adder is realised using 16 identical full adders (FA) as shown. The carry propagation delay of each FA is 12 ns and the sum propagation delay of each FA is 15 ns. The worst case delay (in ns) of the 16-bit adder will be \_\_\_\_\_



$$\begin{aligned}
 T &= (n-1) T_{\text{carry}} + \max\{T_{\text{sum}}, T_{\text{carry}}\} \\
 &= (16-1) \times 12 \text{ ns} + \max\{15 \text{ ns}, 12 \text{ ns}\} \\
 &= 15 \times 12 + 15 \\
 &= 195 \text{ ns}
 \end{aligned}$$

Parallel Adder (4 bit)
 $T_{\text{xor}} = 2 \text{ second}$ 
 $T_{\text{mux}} = T_{\text{xor}} = 1 \text{ second}$ 


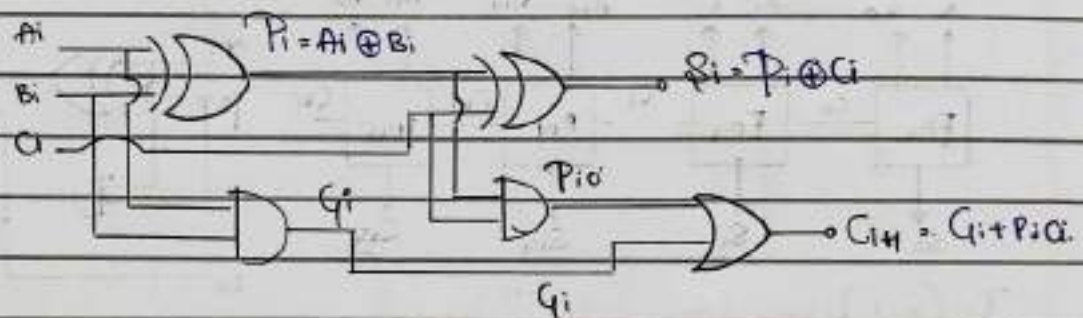
$$T = (n-1) \{ T_{AND} + T_{OR} \} + \max \{ T_{sum}, T_{carry} \}$$

$$\begin{aligned} T &= (4-1) \{ 1+1 \} + \max \{ 4, 4 \} \\ &= 3 \times 2 + 4 \\ &= 10 \text{ ns.} \end{aligned}$$

Parallel Adder, LACA

[Look Ahead Carry Adder]

→ faster adder - among all the adders



$P_i \rightarrow$  Carry propagating term  
 $G_i \rightarrow$  Carry generating term

$$P_i = A_i \oplus B_i$$

$$S_i = P_i \oplus C_i$$

$$P_0 = A_0 \oplus B_0$$

$$S_0 = P_0 \oplus C_0$$

$$P_1 = A_1 \oplus B_1$$

$$S_1 = P_1 \oplus C_1$$

$$P_2 = A_2 \oplus B_2$$

$$S_2 = P_2 \oplus C_2$$

$$C_2 = G_1 + P_1 C_1$$

$$C_2 = G_1 + P_1 [G_0 + P_0 C_0]$$

$$C_2 = G_1 + P_1 G_0 + P_1 P_0 C_0$$

$$G_i = A_i B_i$$

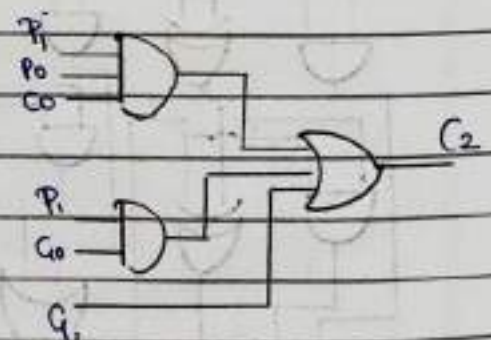
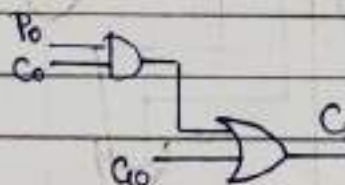
$$C_{i+1} = G_i + P_i C_i$$

$$G_0 = A_0 B_0$$

$$C_1 = G_0 + P_0 C_0$$

$$G_1 = A_1 B_1$$

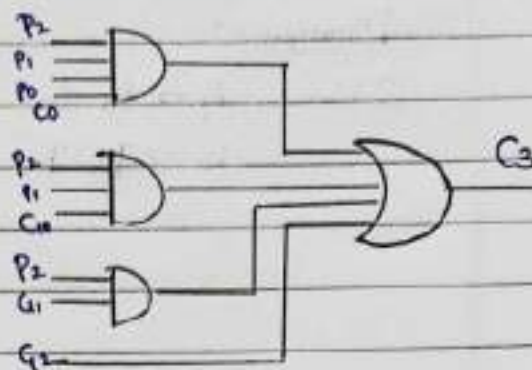
$$G_2 = A_2 B_2$$



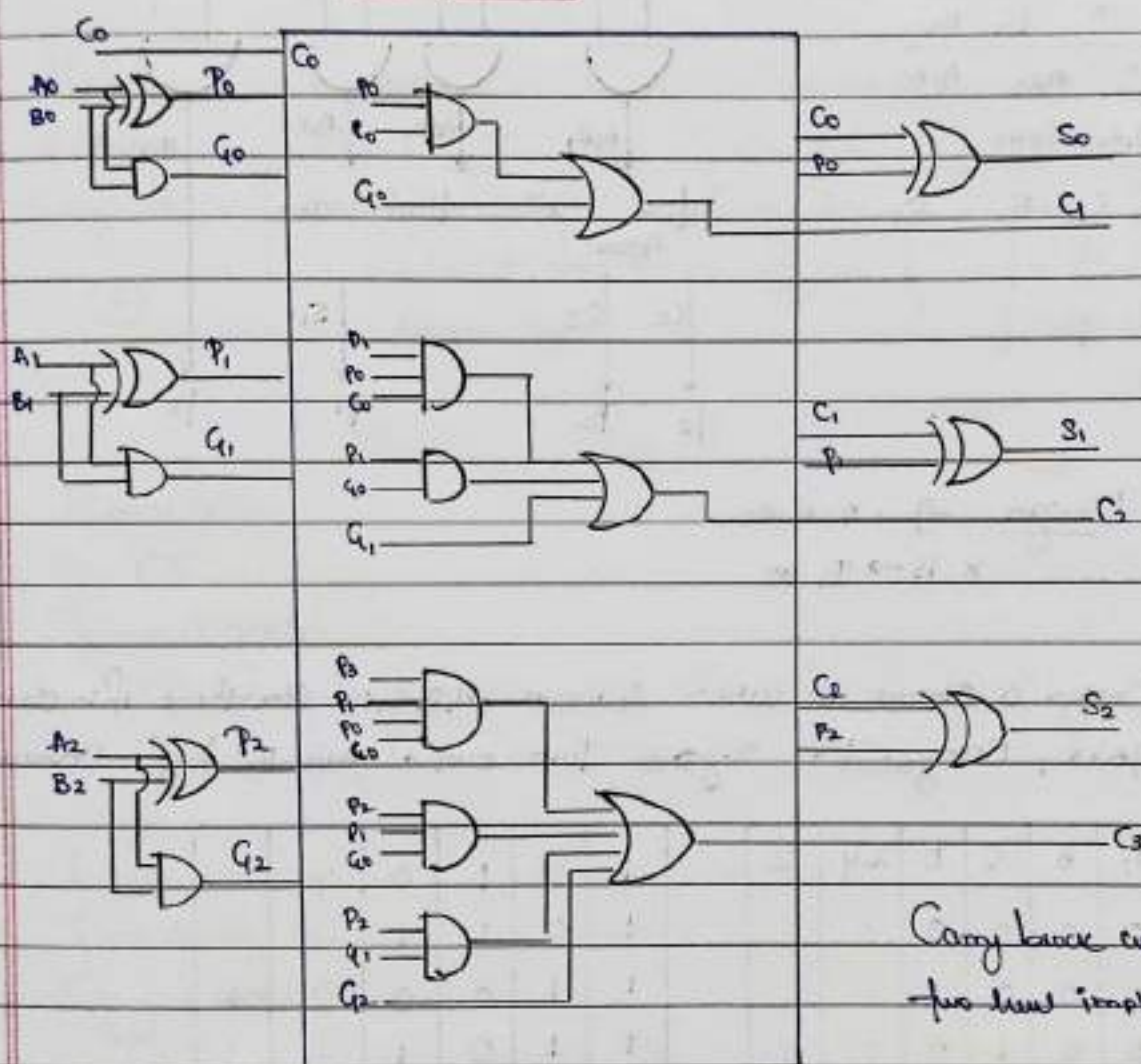
$$C_3 = G_2 + P_2 C_2$$

$$C_3 = G_2 + P_2 [G_1 + P_1 G_0 + P_1 P_0 G_0]$$

$$C_3 = G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 G_0$$



### CARRY BLOCK



Carry block can be always  
two level implementation.

'n' bit RACA

For carry block:

$$\text{No. of AND Gate} = \frac{n(n+1)}{2}$$

$$\text{No. of OR Gate} = n$$

\* Delay for Carry Block  $\Rightarrow 27$

\* For entire circuit  $\Rightarrow 47$

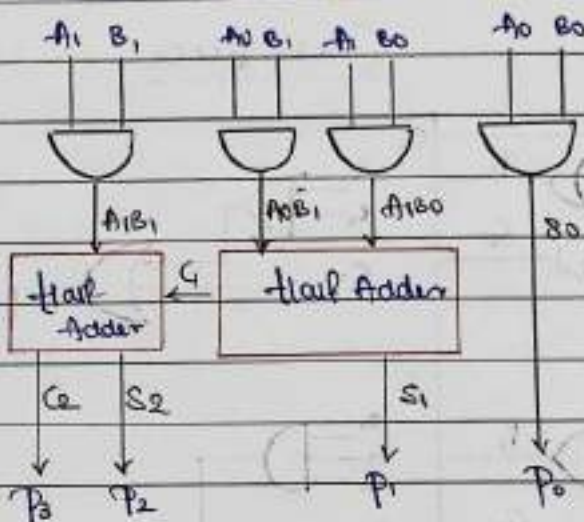
Multiplication:

2 bit,  $A \rightarrow A_1 A_0$   
 $B \rightarrow B_1 B_0$

$$\begin{array}{r} 52 \\ \times 52 \\ \hline 2704 \\ \hline \end{array}$$

$P_3 \ P_2 \ P_1 \ P_0$

$$\begin{array}{r} A_1 A_0 \\ \times B_1 B_0 \\ \hline C_1 A_0 B_0 A_0 B_0 \\ \hline C_2 S_2 S_1 S_0 \\ \hline P_3 P_2 P_1 P_0 \end{array}$$

Ans

Design:  $A \rightarrow A_2 A_1 A_0$   
 $\times B \rightarrow B_1 B_0$

Q1. Design a circuit in which 4 inputs  $A, B, C, D$  are there if in decimal  $AB = x$ ,  $CD = y$ . Then  $x \cdot y \leq 3$ . Then output will be high otherwise low.

| A | B | C | D | $x \cdot y$ | Z |
|---|---|---|---|-------------|---|
| 1 | 0 | 1 | 0 | 4           | 0 |
| 1 | 0 | 1 | 1 | 6           | 0 |
| 1 | 1 | 0 | 0 | 0           | 1 |
| 1 | 1 | 0 | 1 | 3           | 1 |
| 1 | 1 | 1 | 0 | 6           | 0 |
| 1 | 1 | 1 | 1 | 9           | 0 |
| 0 | 0 | 0 | 0 | 0           | 1 |
| 0 | 0 | 0 | 1 | 0           | 1 |
| 0 | 0 | 1 | 0 | 0           | 1 |
| 0 | 0 | 1 | 1 | 0           | 1 |
| 0 | 1 | 0 | 0 | 0           | 1 |
| 0 | 1 | 0 | 1 | 1           | 1 |
| 0 | 1 | 1 | 0 | 2           | 1 |
| 0 | 1 | 1 | 1 | 3           | 1 |
| 1 | 0 | 0 | 0 | 0           | 1 |
| 1 | 0 | 0 | 1 | 2           | 1 |

| $AB \backslash CD$ | 00 | 01 | 11 | 10 |
|--------------------|----|----|----|----|
| 00                 | 1  | 1  | 1  | 1  |
| 01                 | 1  | 1  | 1  | 1  |
| 11                 | 1  | 1  |    |    |
| 10                 | 1  | 1  |    |    |

$$\bar{A} + \bar{C}$$

$$= \bar{AC}$$

Combinational Circuit - { Static Circuit }Quick Summary!Comparator

$$(A > B) \rightarrow A_1 \bar{B}_1 + (A_1 + \bar{B}_1) A_0 \bar{B}_0 \rightarrow \text{minimised}$$

$$A_1 \bar{B}_1 + (A_1 \odot B_1) A_0 \bar{B}_0 \rightarrow \text{Semi-minimised}$$

Mux, Demux

Encoder, Decoder

Half Adder

$$S = A \oplus B$$

$$\text{Carry} = AB$$

(5)

Full Adder

$$\text{Sum} = A \oplus B \oplus C$$

$$\text{Carry} = \sum m(3, 5, 6, 7)$$

$$= (A \oplus B)C + AB$$

$$= AB + AC + BC$$

(9)

Half Subtractor

$$\text{Diff} = A \oplus B$$

$$\text{Carry} = \bar{A}B$$

(5)

Full Subtractor

$$\text{Diff} = A \oplus B \oplus C$$

$$\text{Carry} = \sum m(1, 2, 3, 7)$$

$$= \bar{A}B + (A \oplus B)C$$

$$= \bar{A}B + \bar{A}C + BC$$

(9)

Serial adder

n.T.

Parallel adder

$$T = (n-1) T_{\text{carry}} + \max \{ T_{\text{sum}}, T_{\text{carry}} \}$$

$$T = (n-1) \{ T_{\text{AND}} + T_{\text{OR}} \} + \max \{ T_{\text{sum}}, T_{\text{carry}} \}$$

LACA

$$\text{Total delay} = 4\tau$$

$$\text{Carry block} = 2\tau$$

Carry Block

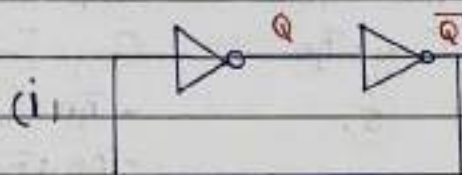
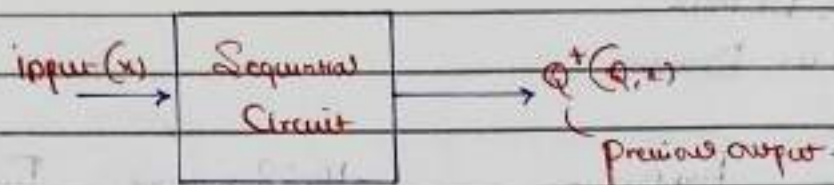
$$A_{\text{no}} = \frac{n(n+1)}{2}$$

$$OR = n$$

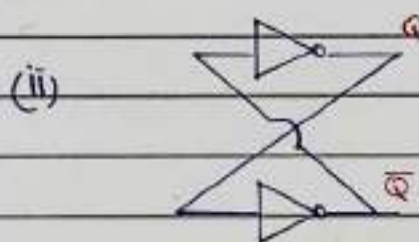
## SEQUENTIAL CIRCUIT

Sequential Circuit:

- \* A circuit with feedback and memory are called Sequential Circuit.
- \* Output of the Sequential Circuit depends on previous output as well as present state of input.

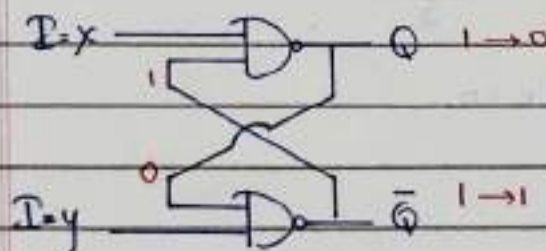


Basic Memory Element  
i = ii



Latches : Basic memory element

- \* Latches are level triggered
- \* Latches have two output which is complement of each other



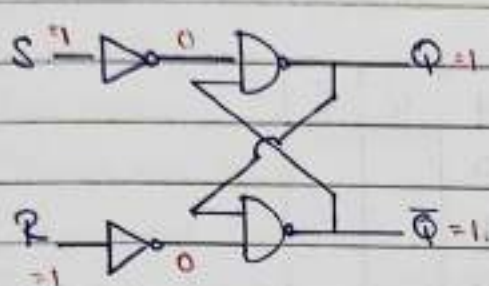
| x | y | Q | Q-bar |           |
|---|---|---|-------|-----------|
| 0 | 0 | 1 | 1     | * Invalid |
| 0 | 1 | 1 | 0     |           |
| 1 | 0 | 0 | 1     |           |
| 1 | 1 | Q | Q-bar | → Hold    |

NAND

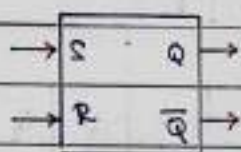
| A | B | y = A.B |
|---|---|---------|
| 0 | 0 | 1       |
| 0 | 1 | 1       |
| 1 | 0 | 1       |
| 1 | 1 | 0       |

Note: Whenever  $x \cdot y = 1$  is applied and invalid condition occurs then a NAND having lower Propagation Delay first change its Output and other remain on its previous state one called racing problem or raising problem.

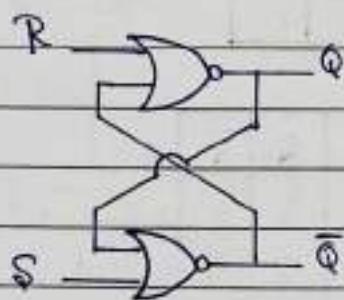
### SR: Set-Reset Latch



| S | R | Q | $\bar{Q}$ |           |
|---|---|---|-----------|-----------|
| 0 | 0 | Q | $\bar{Q}$ | Hold      |
| 0 | 1 | 0 | 1         | Reset     |
| 1 | 0 | 1 | 0         | Set       |
| 1 | 1 | X | X         | (Invalid) |



### S-R Latch by using NOR Gate:

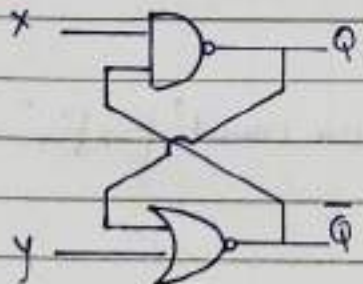


| S | R | Q | $\bar{Q}$ |         |
|---|---|---|-----------|---------|
| 0 | 0 | Q | $\bar{Q}$ | Hold    |
| 0 | 1 | 0 | 1         | Reset   |
| 1 | 0 | 1 | 0         | Set     |
| 1 | 1 | X | X         | Invalid |

| NOR |   |                      |
|-----|---|----------------------|
| A   | B | $y = \overline{A+B}$ |
| 0   | 0 | 1                    |
| 0   | 1 | 0                    |
| 1   | 0 | 0                    |
| 1   | 1 | 0                    |

Hlw

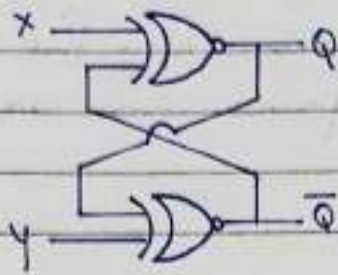
①



| x | y | Q | $\bar{Q}$ |
|---|---|---|-----------|
| 0 | 0 |   |           |
| 0 | 1 |   |           |
| 1 | 0 |   |           |
| 1 | 1 |   |           |

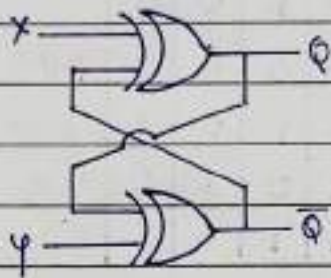
HW

②



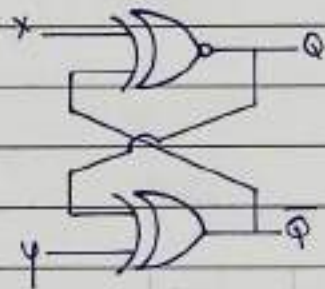
| x | y | Q | $\bar{Q}$ |
|---|---|---|-----------|
| 0 | 0 |   |           |
| 0 | 1 |   |           |
| 1 | 0 |   |           |
| 1 | 1 |   |           |

③



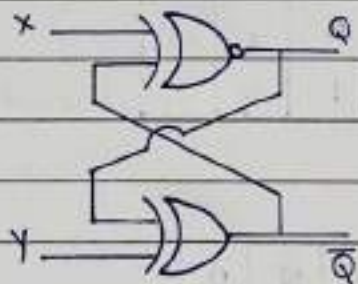
| x | y | Q | $\bar{Q}$ |
|---|---|---|-----------|
| 0 | 0 |   |           |
| 0 | 1 |   |           |
| 1 | 0 |   |           |
| 1 | 1 |   |           |

④



| x | y | Q | $\bar{Q}$ |
|---|---|---|-----------|
| 0 | 0 | 1 | 0         |
| 0 | 1 | 1 | 0         |
| 1 | 0 | 0 | 1         |
| 1 | 1 | 1 | 0         |

⑤



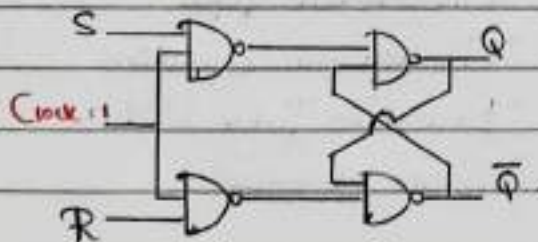
| x | y | Q         | $\bar{Q}$ |           |
|---|---|-----------|-----------|-----------|
| 0 | 0 | Q         | $\bar{Q}$ | → Invalid |
| 0 | 1 | Q         | $\bar{Q}$ | → Hold    |
| 1 | 0 | $\bar{Q}$ | Q         | → Toggle  |
| 1 | 1 | $\bar{Q}$ | $\bar{Q}$ | → Invalid |

→ Latches with Control Phenomena are called Flip Flop.



1. Circuit diagram
2. Truth table
3. Characteristic table
4. Characteristic equation
5. Excitation table
6. State diagram

Circuit diagram:



| S | R | $Q_{n+1}$ | $\bar{Q}_{n+1}$ |            |
|---|---|-----------|-----------------|------------|
| 0 | 0 | $Q_n$     | $\bar{Q}_n$     | Hold/Pass. |
| 0 | 1 | 0         | 1               | → Reset.   |
| 1 | 0 | 1         | 0               | → Set.     |
| 1 | 1 | X         | X               | → Invalid  |

| $Q$ | $S$ | $R$ | $Q_n$ | $Q_{n+1}$ |
|-----|-----|-----|-------|-----------|
| 0   | 0   | 0   | 0     | 0         |
| ①   | 0   | 0   | 1     | 1         |
| 2   | 0   | 1   | 0     | 0         |
| 3   | 0   | 1   | 1     | 0         |
| ④   | 1   | 0   | 0     | 1         |
| ⑤   | 1   | 0   | 1     | 1         |
| 6   | 1   | 1   | 0     | x         |
| 7   | 1   | 1   | 1     | x         |

$$Q_{n+1} = (S, R, Q_0) = \sum_m (1, 4, 5) + \sum_0 (6, 7)$$

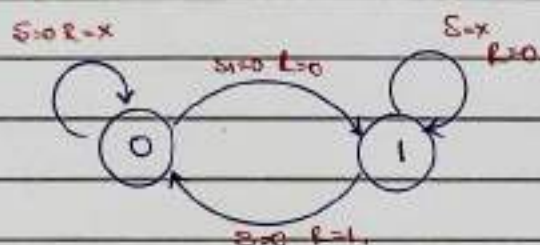
| $S$ | $\overline{R} \overline{Q}$ | $\overline{R} Q$ | $R \overline{Q}$ | $R Q$ |
|-----|-----------------------------|------------------|------------------|-------|
| 00  |                             | 1                |                  |       |
| 01  | 1                           | 1                | x                | x     |

$$Q_{n+1} = 3 + \overline{R}Q_n$$

Excitation Table:

| $Q_n$ | $Q_{n+1}$ | S | R |
|-------|-----------|---|---|
| 0     | 0         | 0 | x |
| 0     | 1         | 1 | 0 |
| 1     | 0         | 0 | 1 |
| 1     | 1         | x | 0 |

| S | R | $Q_n$ | $Q_{n+1}$ |
|---|---|-------|-----------|
| 0 | 0 | 0     | 0         |
| 0 | 0 | 1     | 1         |
| 0 | 1 | 0     | 0         |
| 0 | 1 | 1     | 0         |
| 1 | 0 | 0     | 1         |
| 1 | 0 | 1     | 1         |
| 1 | 1 | 0     | x         |
| 1 | 1 | 1     | x         |

State Diagram:

| $Q_n$ | $Q_{n+1}$ | S | R |
|-------|-----------|---|---|
| 0     | 0         | 0 | x |
| 0     | 1         | 1 | 0 |
| 1     | 0         | 0 | 1 |
| 1     | 1         | x | 0 |

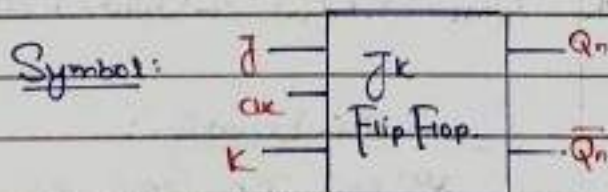
HW

Q. Truth Table:

| C | J | $Q_{n+1}$        |
|---|---|------------------|
| 0 | 0 | $\overline{Q_n}$ |
| 0 | 1 | $Q_n$            |
| 1 | 0 | 1                |
| 1 | 1 | 0                |

Write:

- \* Characteristic Table
- \* Characteristic equation
- \* Excitation Table
- \* State diagram

Jk Flip Flop

Truth Table:

| J | K | $Q_{n+1}$        |
|---|---|------------------|
| 0 | 0 | $Q_n$            |
| 0 | 1 | 0                |
| 1 | 0 | 1                |
| 1 | 1 | $\overline{Q_n}$ |

Characteristic Table:

| J | K | Q <sub>n</sub> | Q <sub>n+1</sub> |
|---|---|----------------|------------------|
| 0 | 0 | 0              | 0                |
| 0 | 0 | 1              | 1                |
| 0 | 1 | 0              | 0                |
| 0 | 1 | 1              | 0                |
| 1 | 0 | 0              | 1                |
| 1 | 0 | 1              | 1                |
| 1 | 1 | 0              | 1                |
| 1 | 1 | 1              | 0                |

$$Q_{n+1}(J, K, Q_n) = \sum m(1, 4, 5, 6)$$

Characteristic Equation:

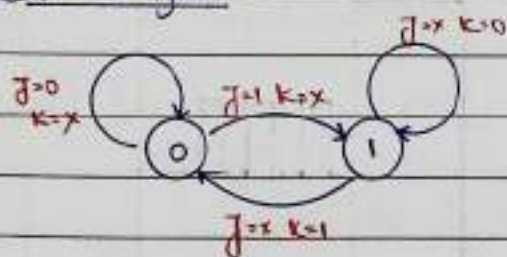
| J \ K       | $\bar{Q}_n$<br>00 | $\bar{Q}_n$<br>01 | $Q_n$<br>11 | $Q_n$<br>10 |
|-------------|-------------------|-------------------|-------------|-------------|
| $\bar{J}$ 0 |                   | 1                 |             |             |
| J 1         | 1                 | 1                 |             | 1           |

$$Q_{n+1} = \underline{\underline{\bar{J}\bar{Q}_n + JQ_n}}$$

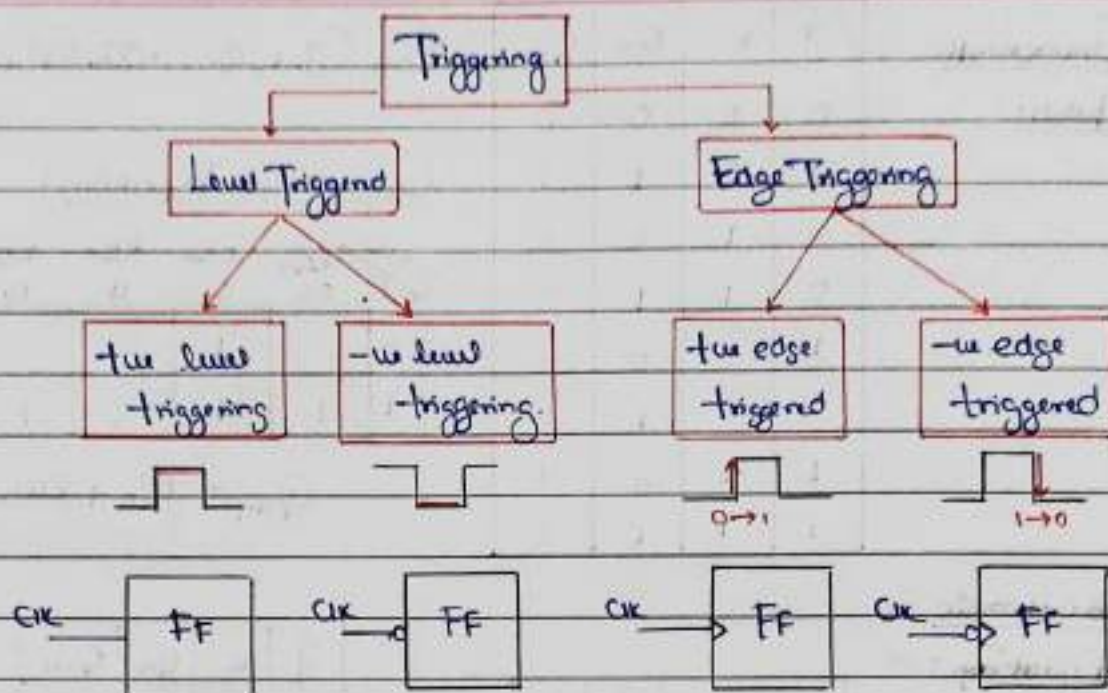
Excitation Table:Table:

| Q <sub>n</sub> | Q <sub>n+1</sub> | J | K |
|----------------|------------------|---|---|
| 0              | 0                | 0 | x |
| 0              | 1                | 1 | x |
| 1              | 0                | x | 1 |
| 1              | 1                | x | 0 |

| J | K | Q <sub>n</sub> | Q <sub>n+1</sub> |
|---|---|----------------|------------------|
| 0 | 0 | 0              | 0                |
| 0 | 0 | 1              | 1                |
| 0 | 1 | 0              | 0                |
| 0 | 1 | 1              | 0                |
| 1 | 0 | 0              | 1                |
| 1 | 0 | 1              | 1                |
| 1 | 1 | 0              | 1                |
| 1 | 1 | 1              | 0                |

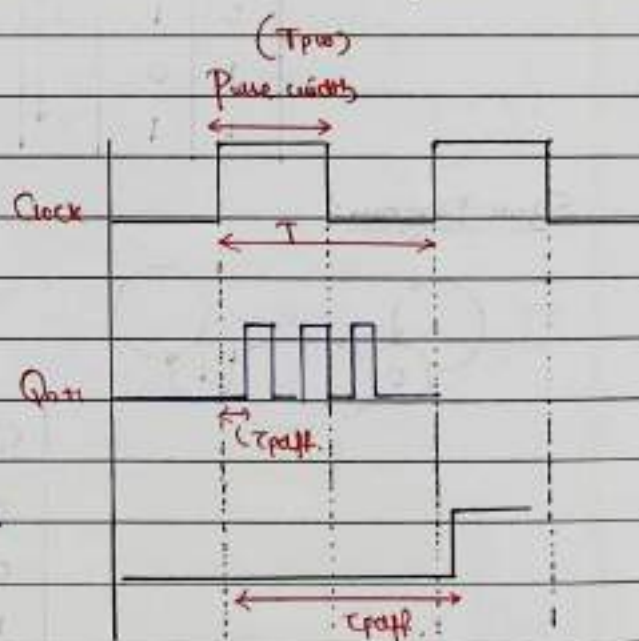
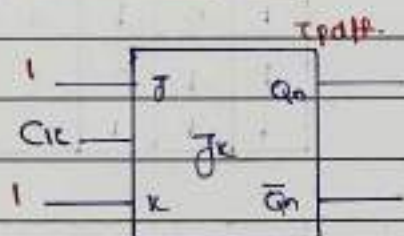
State Diagram:

| Q <sub>n</sub> | Q <sub>n+1</sub> | J | K |
|----------------|------------------|---|---|
| 0              | 0                | 0 | x |
| 0              | 1                | 1 | x |
| 1              | 0                | x | 1 |
| 1              | 1                | x | 0 |



Note: Level triggered JK flip-flop suffers from the problem of "RACE AROUND".

Race Around Problem:

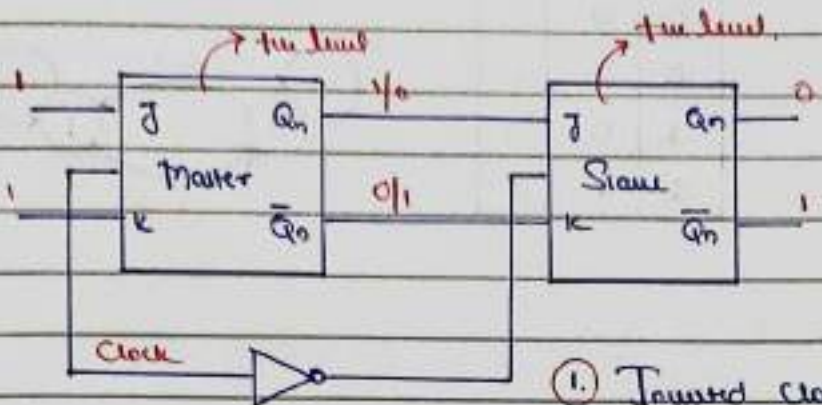


To avoid the Race Around Problem.

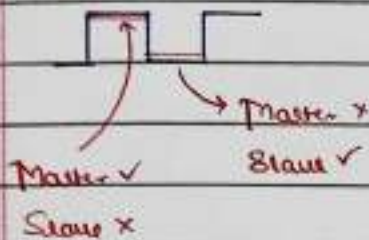
1.  $T_{pw} < T_{pd} < T_{clk}$
2. By Master Slave FF.

→ When J-K=1 is applied to the level sensitive of JK FF, then within the duration of pulse output of the FF toggle more than one times are called "Race Around Problem".

## Master Slave FF.



(1) Inverted clock is applied to the slave or compared to master.



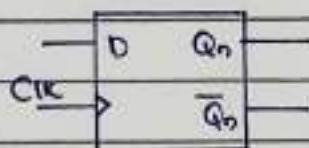
(2) Master-Slave flip-flop is used to store single bit because output is taken only from slave.

(3) By the operation it seems that master is level triggered whereas slave is negative edge triggered.

## D Flip Flop.

- \* It is known as Delay FF or Transparent FF
- \* In the D FF whenever the input is applied it will directly come to the output along with the clock.

Symbol:



Characteristic Table:

| D | Qn | Qn+1 |
|---|----|------|
| 0 | 0  | 0    |
| 0 | 1  | 0    |
| 1 | 0  | 1    |
| 1 | 1  | 1    |

Truth Table:

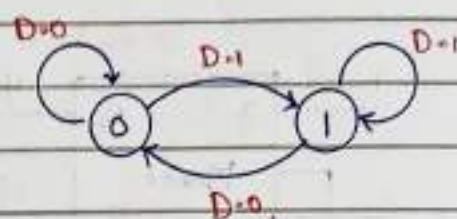
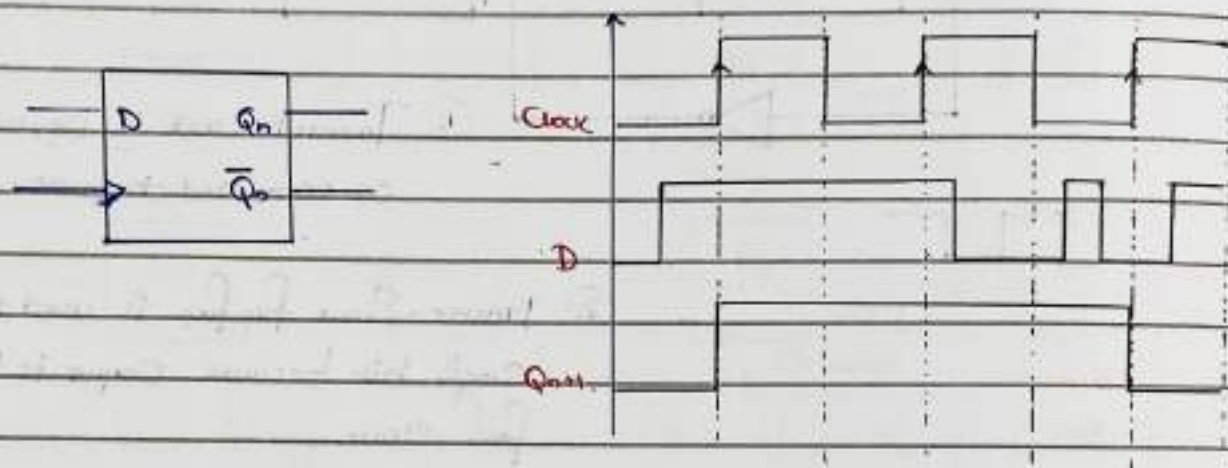
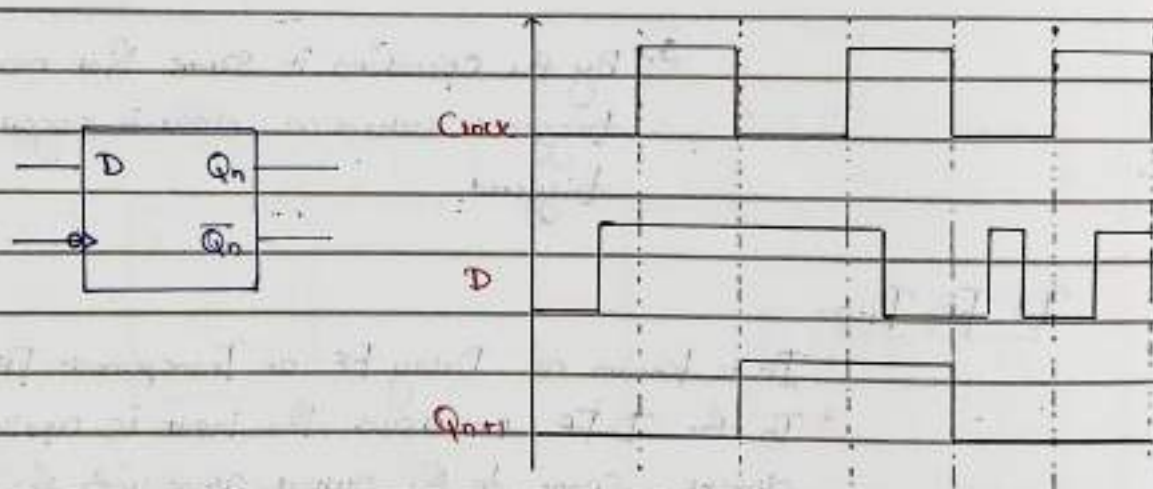
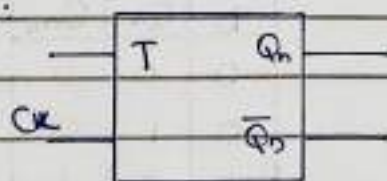
| D | Qn+1 |
|---|------|
| 0 | 0    |
| 1 | 1    |

Characteristic Equation:

$$Q_{n+1} = D$$

Excitation Table:

| $Q_n$ | $Q_{n+1}$ | $D$ |
|-------|-----------|-----|
| 0     | 0         | 0   |
| 0     | 1         | 1   |
| 1     | 0         | 0   |
| 1     | 1         | 1   |

State Diagram:Q<sub>1</sub>Q<sub>2</sub>T Flip Flop (Toggle Flip Flop)Symbol:Characteristic Table:

| $T$ | $Q_n$ | $Q_{n+1}$ |
|-----|-------|-----------|
| 0   | 0     | 0         |
| 0   | 1     | 1         |
| 1   | 0     | 0         |
| 1   | 1     | 1         |

Truth Table:

|  |  |
|--|--|
|  |  |
|  |  |

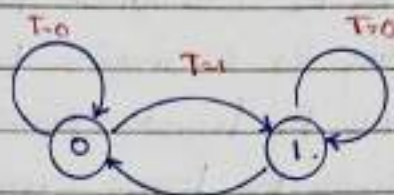
Characteristic Equations:

$$Q_{n+1} = \bar{T}Q_n + T\bar{Q}_n$$

$$Q_{n+1} = T \oplus Q_n$$

Excitation Table:

| $Q_n$ | $Q_{n+1}$ | $T$ |
|-------|-----------|-----|
| 0     | 0         | 0   |
| 0     | 1         | 1   |
| 1     | 0         | 1   |
| 1     | 1         | 0   |

State Diagram:

Note: J-K FF is also known as universal flip flop.

|          | J | K | $Q_{n+1}$   |
|----------|---|---|-------------|
| SR<br>FF | 0 | 0 | $Q_n$       |
|          | 0 | 1 | 0           |
|          | 1 | 0 | 1           |
|          | 1 | 1 | $\bar{Q}_n$ |

T-FF

SR FF

$$\rightarrow Q_{n+1} = S + \bar{R}Q_n$$

J-K FF

$$\rightarrow Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

D-FF

$$\rightarrow Q_{n+1} = D$$

T-FF

$$\rightarrow Q_{n+1} = T \oplus Q_n$$

| S | R | $Q_{n+1}$ |
|---|---|-----------|
| 0 | 0 | $Q_n$     |
| 0 | 1 | 0         |
| 1 | 0 | 1         |
| 1 | 1 | x         |

| J | K | $Q_{n+1}$   |
|---|---|-------------|
| 0 | 0 | $Q_n$       |
| 0 | 1 | 0           |
| 1 | 0 | 1           |
| 1 | 1 | $\bar{Q}_n$ |

| D | $Q_{n+1}$ |
|---|-----------|
| 0 | 0         |
| 1 | 1         |

| T | $Q_{n+1}$   |
|---|-------------|
| 0 | $Q_n$       |
| 1 | $\bar{Q}_n$ |

Excitation Table

| $Q_n$ | $Q_{n+1}$ | S | R | J | K | D | T |
|-------|-----------|---|---|---|---|---|---|
| 0     | 0         | 0 | x | 0 | x | 0 | 0 |
| 0     | 1         | 1 | 0 | 1 | x | 1 | 1 |
| 1     | 0         | 0 | 1 | x | 1 | 0 | 1 |
| 1     | 1         | x | 0 | x | 0 | 1 | 0 |

Designing of Flip Flop:

- Step 1: Write the characteristic table of desired FF  
 Step 2: Write the excitation table of available FF  
 Step 3: Write the logical expression  
 Step 4: Minimize the logical expression  
 Step 5: Hardware implementation.

Q3. Design JK flip flop using SR Flip Flop.  
 (desired) (available)

| Step 1 | J | K | Q <sub>n</sub> | Q <sub>n+1</sub> | S | R |
|--------|---|---|----------------|------------------|---|---|
| Step 2 | 0 | 0 | 0              | 0                | 0 | x |
|        | 0 | 0 | 1              | 1                | x | 0 |
|        | 0 | 1 | 0              | 0                | 0 | x |
|        | 0 | 1 | 1              | 0                | 0 | 1 |
|        | 1 | 0 | 0              | 1                | 1 | 0 |
|        | 1 | 0 | 1              | 1                | x | 0 |
|        | 1 | 1 | 0              | 1                | 1 | 0 |
|        | 1 | 1 | 1              | 0                | 0 | 1 |

$$\text{Step 3: } S(J, K, Q_n) = \sum m(4, 5) + \sum d(1, 5)$$

$$R(J, K, Q_n) = \sum m(3, 7) + \sum d(6, 2)$$

Step 4:

| J \ Q <sub>n</sub> | 0 | 1 | 1 | 0 |
|--------------------|---|---|---|---|
| 0                  | 0 | x |   |   |
| 1                  | 1 | x |   | 1 |

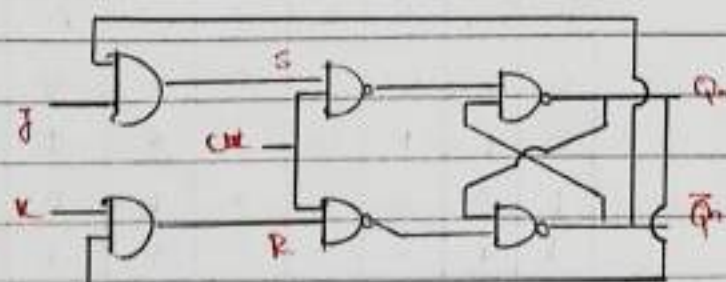
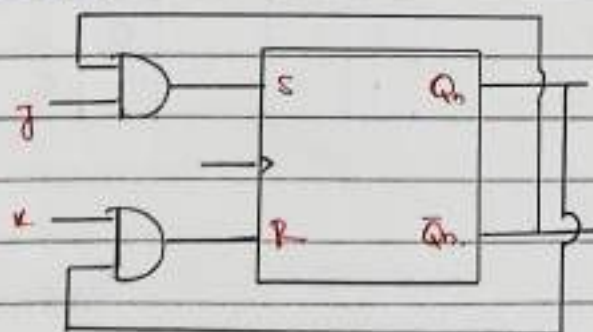
$$S = J\bar{Q}_n$$

Step 4:

| J \ Q <sub>n</sub> | 0 | 1 | 1 | 0 |
|--------------------|---|---|---|---|
| 0                  | x |   | 1 | x |
| 1                  |   |   | 1 |   |

$$R = KQ_n$$

Step 5:



Q4. Design a D FF by using SR FF.

Step 1:

Step 2:

| D | Qn | Qnm | S | R |
|---|----|-----|---|---|
| 0 | 0  | 0   | 0 | x |
| 0 | 1  | 0   | 0 | 1 |
| 1 | 0  | 1   | 1 | 0 |
| 1 | 1  | 1   | x | 0 |

Step 3:

Step 4:

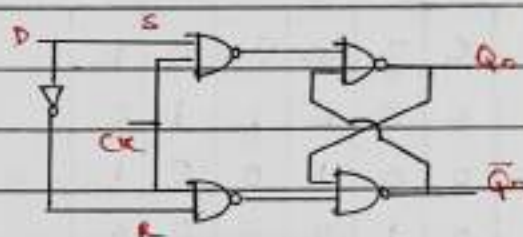
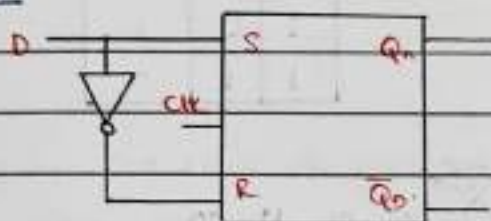
| D \ Qn | 0 | 1 |
|--------|---|---|
| 0      |   |   |
| 1      | 1 | x |

$$S = D$$

| D \ Qn | 0 | 1 |
|--------|---|---|
| 0      | x | 1 |
| 1      |   |   |

$$R = \bar{D}$$

Steps:



Q5. Design a T-FF using SR Flip Flop.

Step 1:

Step 2:

| T | Qn | Qnm | S | R |
|---|----|-----|---|---|
| 0 | 0  | 0   | 0 | x |
| 0 | 1  | 1   | x | 0 |
| 1 | 0  | 1   | 1 | 0 |
| 1 | 1  | 0   | 0 | 1 |

Step 3:

Step 4:

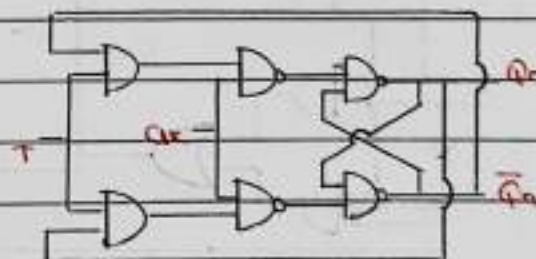
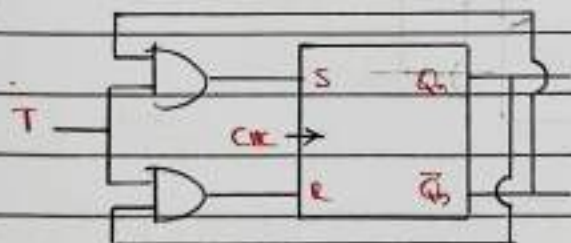
| T \ Qn | 0 | 1 |
|--------|---|---|
| 0      |   | x |
| 1      | 1 |   |

$$S = T\bar{Qn}$$

| T \ Qn | 0 | 1 |
|--------|---|---|
| 0      | x |   |
| 1      |   | 1 |

$$R = TQn$$

Steps:



Q6. Design TFF using D-FF.

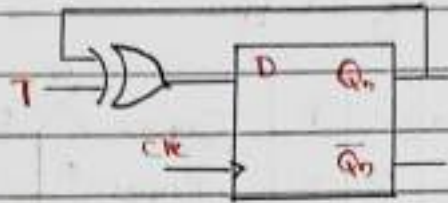
Step 1:

Step 2:

| T | Qn | Qnm | D |
|---|----|-----|---|
| 0 | 0  | 0   | 0 |
| 0 | 1  | 1   | 1 |
| 1 | 0  | 1   | 1 |
| 1 | 1  | 0   | 0 |

Step 3 & Step 4:

$$D = T \oplus Qn$$

Steps:

Q7. Design a Cg FF by using SR FF

Truth Table:

| C | J | Qn | Qn+1 | S | R |
|---|---|----|------|---|---|
| 0 | 0 | 0  | 1    | 1 | 0 |
| 0 | 0 | 1  | 0    | 0 | 1 |
| 0 | 1 | 0  | 0    | 0 | x |
| 0 | 1 | 1  | 1    | x | 0 |
| 1 | 0 | 0  | 1    | 1 | 0 |
| 1 | 0 | 1  | 1    | x | 0 |
| 1 | 1 | 0  | 0    | 0 | x |
| 1 | 1 | 1  | 0    | 0 | 1 |

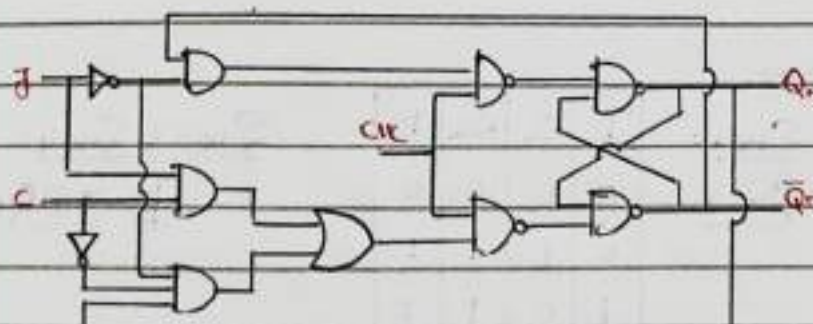
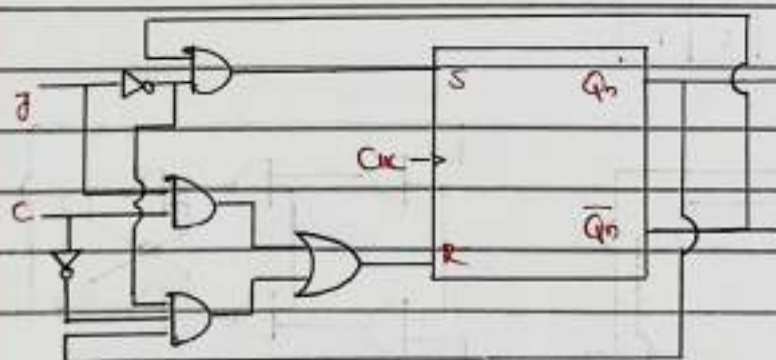
| C | J | Qn+1        |
|---|---|-------------|
| 0 | 0 | $\bar{Q}_n$ |
| 0 | 1 | $Q_n$       |
| 1 | 0 | 1           |
| 1 | 1 | 0           |

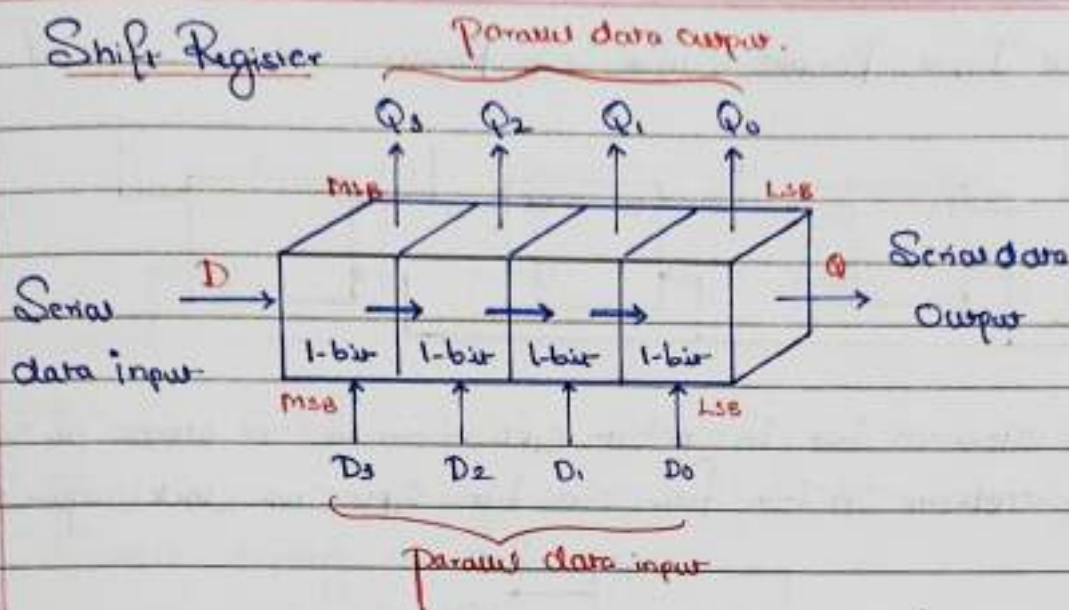
| C \ J Qn | 00 | 01 | 11 | 10 |
|----------|----|----|----|----|
| 0        | 1  |    | x  |    |
| 1        | 1  | x  |    |    |

$$S = \bar{J} \bar{Q}_n$$

| C \ J Qn | 00 | 01 | 11 | 10 |
|----------|----|----|----|----|
| 0        |    | 1  |    | x  |
| 1        |    |    | 1  | x  |

$$R = \bar{Q} \bar{J} Q_n + C J$$

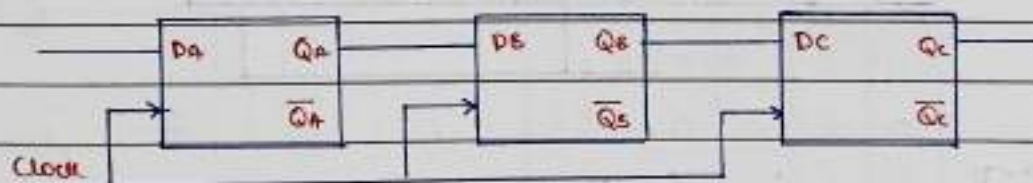




- \* Registers are used to store group of bits
- \* To store 'n' bits minimum 'n' flip flops are required.
- \* Generally D Flip Flops are used to Design registers.

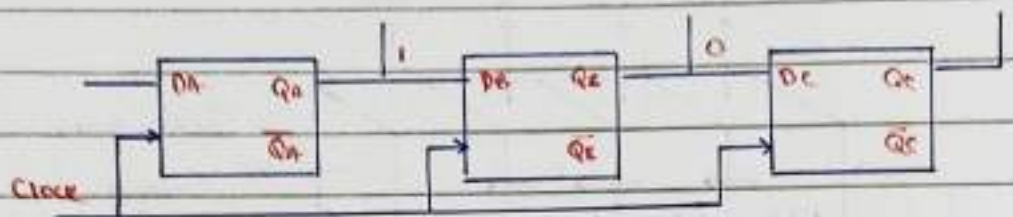
1. Serial input Serial output Shift register (SISO)
2. Serial input Parallel output Shift register (SIPO)
3. Parallel input Serial output Shift register (PISO)
4. Parallel input Parallel Output Shift register (PIPO)

### Serial Input Serial Output Shift Register

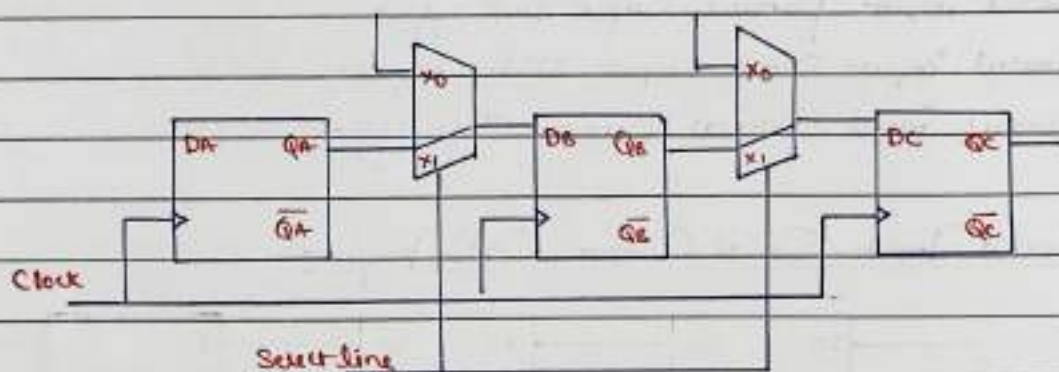


| Clock | Input | QA | QB | QC |
|-------|-------|----|----|----|
| 0     | 101   | 0  | 0  | 0  |
| 1     |       | 1  | 0  | 0  |
| 2     |       | 0  | 1  | 0  |
| 3     |       | 1  | 0  | 1  |

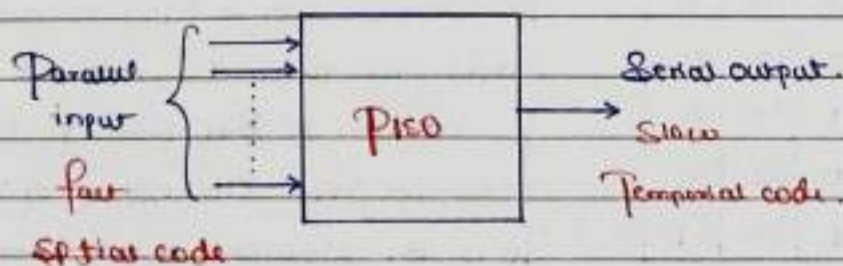
- \* To store 'n' bits in SISO minimum 'n' clocks are required
- \* To retrieve 'n' bits from 'n' bit SISO minimum 'n' clocks are required
- \* It is the slowest register among all the Shift registers.

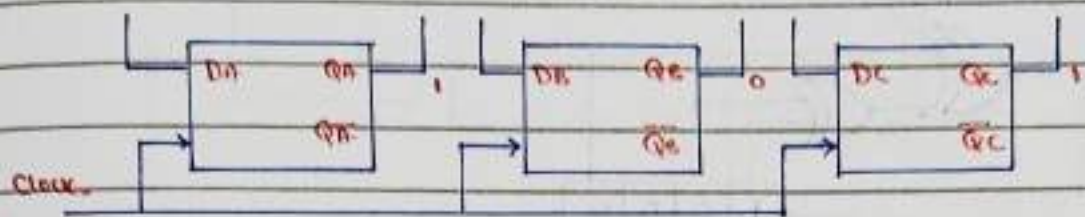
Serial Input Parallel Output Shift Register

- \* To store 'n' bit in 'n' bit SIPO minimum 'n' clocks are required.
- \* To retrieve 'n' bit from 'n' bit SIPO no clock required.

Parallel Input Serial Output Shift Register

- \* To store 'n' bit in PISO only one clock is required.
- \* To retrieve 'n' bit from 'n' bit PISO minimum (n-1) clocks are required.

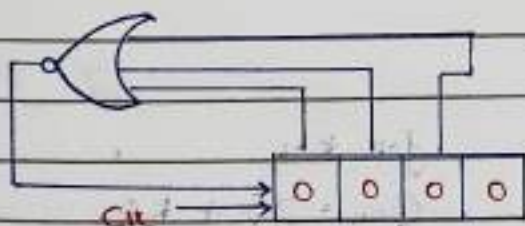


Parallel Input-Parallel Output Shift Register.

- \* To store 'n' bit in 'n' bit PISO only one clock is required.
- \* There is no clock requirement in PISO to retrieve 'n' bits.
- \* Fastest among all shift registers.

|      | Store | Retrieve | Total         |
|------|-------|----------|---------------|
| SISO | n     | n-1      | 2n-1 → Slower |
| SIPO | n     | 0        | n             |
| PISO | 1     | n-1      | n             |
| PIPO | 1     | 0        | 1 → Faster    |

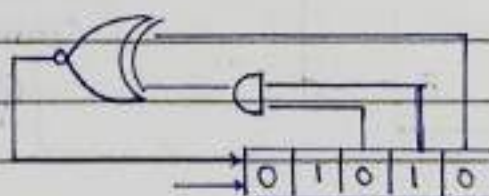
Q1 Write the state of the register given below? Assume all flip flops are reset initially



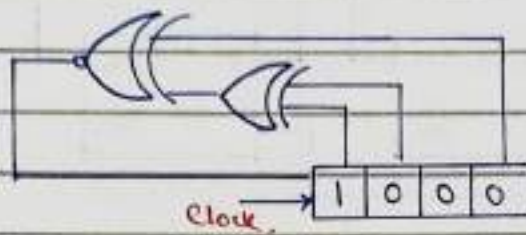
| Clock | Q3 | Q2 | Q1 | Q0 |
|-------|----|----|----|----|
| 0     | 0  | 0  | 0  | 0  |
| 1     | 1  | 0  | 0  | 0  |
| 2     | 0  | 1  | 0  | 0  |
| 3     | 0  | 0  | 1  | 0  |
| 4     | 0  | 0  | 0  | 1  |
| 5     | 1  | 0  | 0  | 0  |
| 6     | 0  | 1  | 0  | 0  |
| 7     | 0  | 0  | 1  | 0  |
| 8     | 0  | 0  | 0  | 1  |

HW

Q2 Output after 13<sup>th</sup> clock will be?

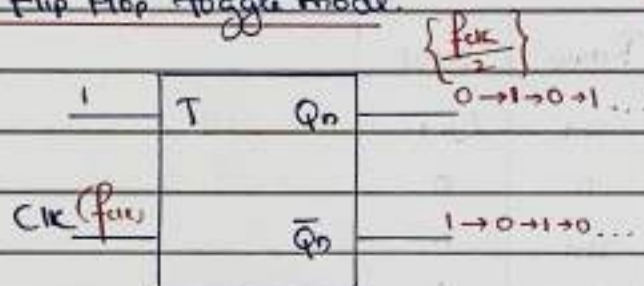


Qs. After 9<sup>th</sup> clock the output will be?



### Toggle Mode of flip flops

#### 1. T Flip Flop toggle mode



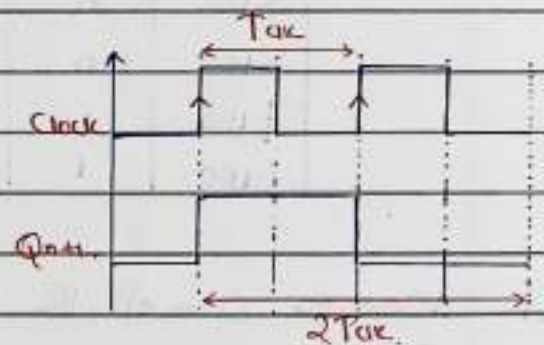
$$Q_{n+1} = T \oplus Q_n$$

$$Q_{n+1} = 1 \oplus Q_n = \bar{Q}_n \Rightarrow Q_{n+1} = \bar{Q}_n$$

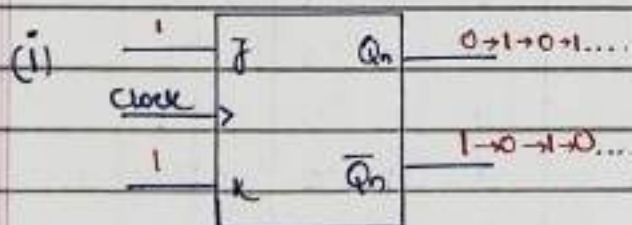
Toggle Mode

$$f_{clk} = \frac{1}{T_{clk}}$$

$$f_{Q_{n+1}} = \frac{1}{2T_{clk}} = \frac{f_{clk}}{2}$$



#### 2. JK Flip Flop toggle mode

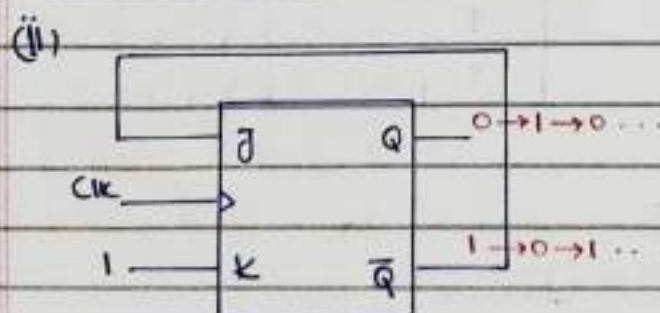


$$J=1 \quad K=1$$

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

$$= 1\bar{Q}_n + 0Q_n$$

$$Q_{n+1} = \bar{Q}_n$$

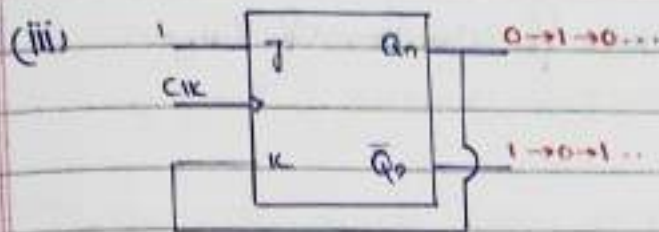


$$J = \bar{Q}_n \quad K=1$$

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

$$Q_{n+1} = \bar{Q}_n \cdot \bar{Q}_n + 0 \cdot Q_n$$

$$Q_{n+1} = \bar{Q}_n$$



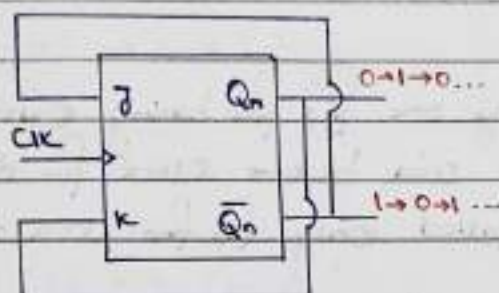
$$J=1 \quad K=Q_n$$

$$Q_{n+1} = J\bar{Q}_n + KQ_n$$

$$= 1 \cdot \bar{Q}_n + \bar{Q}_n \cdot Q_n$$

$$Q_{n+1} = \bar{Q}_n$$

(iv)



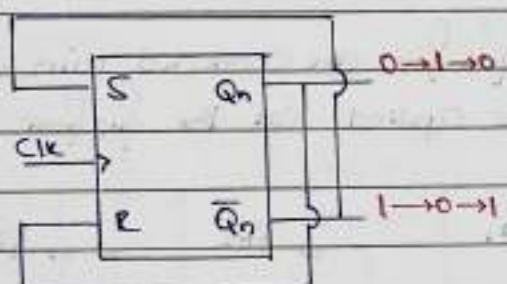
$$J=\bar{Q}_n \quad K=Q_n$$

$$Q_{n+1} = J\bar{Q}_n + KQ_n$$

$$= \bar{Q}_n \cdot \bar{Q}_n + \bar{Q}_n \cdot Q_n$$

$$Q_{n+1} = \bar{Q}_n$$

### 3 Toggle Mode of SR Flip Flop



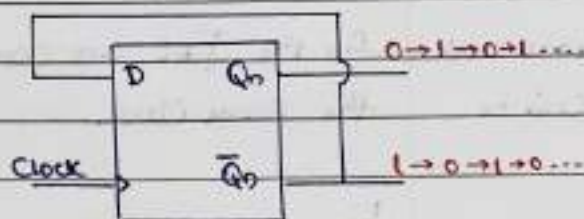
$$S=\bar{Q}_n \quad R=Q_n$$

$$Q_{n+1} = S + \bar{R}Q_n$$

$$Q_{n+1} = \bar{Q}_n + \bar{Q}_n \cdot Q_n$$

$$Q_{n+1} = \bar{Q}_n$$

### 4 D-Flip Flop Toggle Mode



$$D=\bar{Q}_n$$

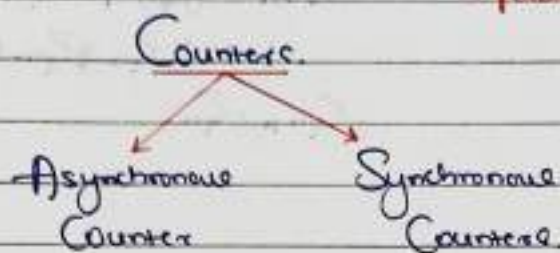
$$Q_{n+1} = D$$

$$Q_{n+1} = \bar{Q}_n$$

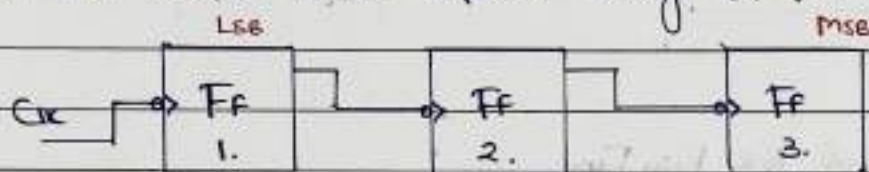
### Counters:

- \* Counters are used to count number of clock
- \* Counter are used as frequency divider circuit.
- \* Counter are also used in A/D (Analog to Digital Converter)
- \* Counters are also known as pulse stretcher circuit.

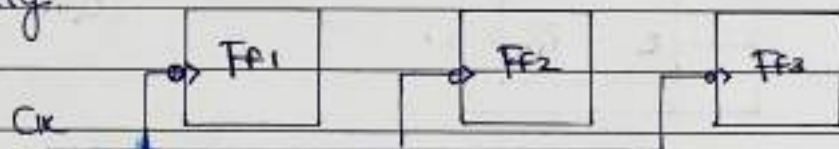
- \* Counters also used in RADAR for detection of Range  
↳ Radio detection & Ranging.



Asynchronous Counter: Only one flip flop having external clock and output of that flip flop will become clock for the next flip flop. So when clock applied only flip flop will work at that time.



Synchronous Counter: All flip flops are connected with the same clock. Hence when the clock is applied all the flip flops work simultaneously.

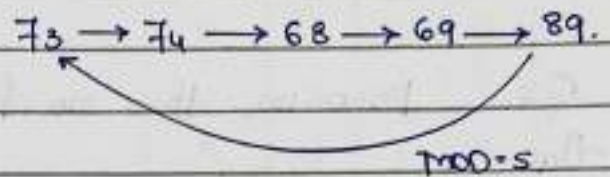
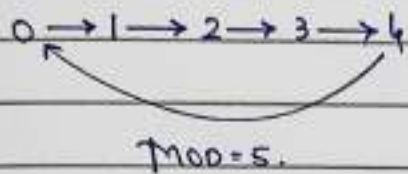


|   | Asynchronous Counter                                                               | Synchronous Counter.                           |
|---|------------------------------------------------------------------------------------|------------------------------------------------|
| ① | Only one FF having external clock and output of that FF will be clock for next FF. | All the FFs are connected with the same clock. |
| ② | Slower.                                                                            | Faster                                         |
| ③ | Only increasing and decreasing counting possible.                                  | All types of counting are possible.            |
| ④ | There is transition error.                                                         | No transition error                            |
| ⑤ | Eg: Ripple counter                                                                 | Eg: Ring counter, Johnson Counter.             |

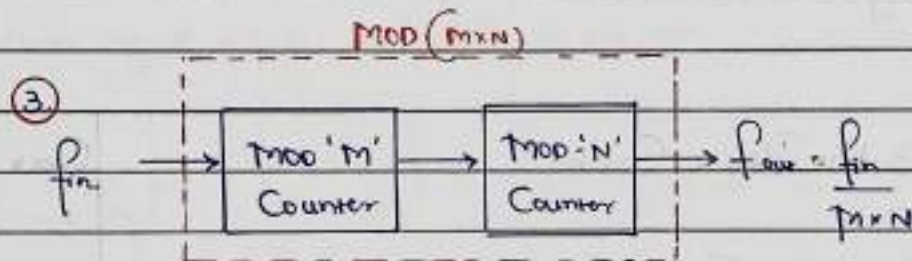
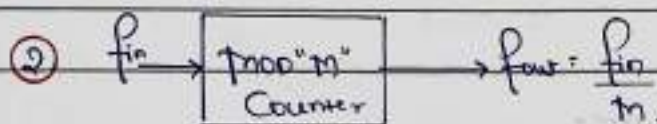
Maximum no. of States =  $2^n$

$\hookrightarrow n$ : no. of FFs.

Module of Counter: Total no. of States used by the counter are called mod of the counter.



Note: ①  $\text{MOD}(m) \leq 2^n$



Bcd Counter:

0  $\rightarrow$  0000  
1  $\rightarrow$  0001  
2  $\rightarrow$  0010  
3  $\rightarrow$  0011  
4  $\rightarrow$  0100  
5  $\rightarrow$  0101  
6  $\rightarrow$  0110  
7  $\rightarrow$  0111  
8  $\rightarrow$  1000  
9  $\rightarrow$  1001

$\text{MOD} \rightarrow 10$

$$2^n \leq 10$$

$$n \geq \log_2 10$$

$$n \geq \log_2 10$$

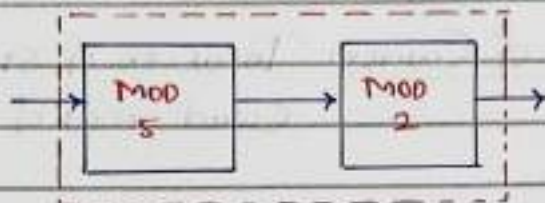
$$n \geq 3.32 \dots$$

$$\boxed{n = 4}$$

Q1. If mod-5 counter is cascaded with mod-2 counter, then it will become?

Ans. MOD-10 counter.

$$M_{\text{MOD}}(5 \times 2) = M_{\text{MOD}} = 10$$



Q2. MOD-100, then no. of FF?

Ans

$$m \leq 2^n$$

$$n \geq \log_2 m$$

$$\geq \log_2 100$$

$$\geq 6.64$$

$$n \geq 7$$

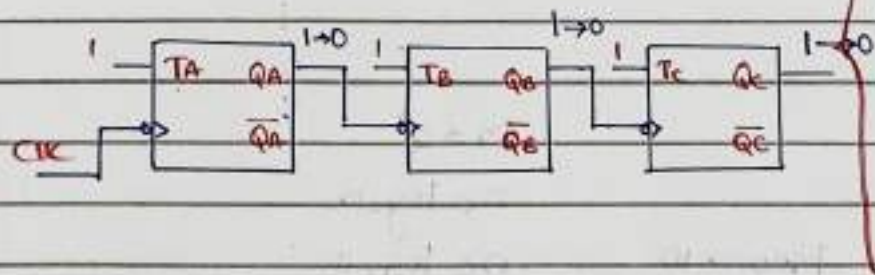
### Asynchronous Counter

→ All the FFs are used in toggle mode.



### 3-bit Ripple Counter

#### MOD-8 UP Ripple Counter:



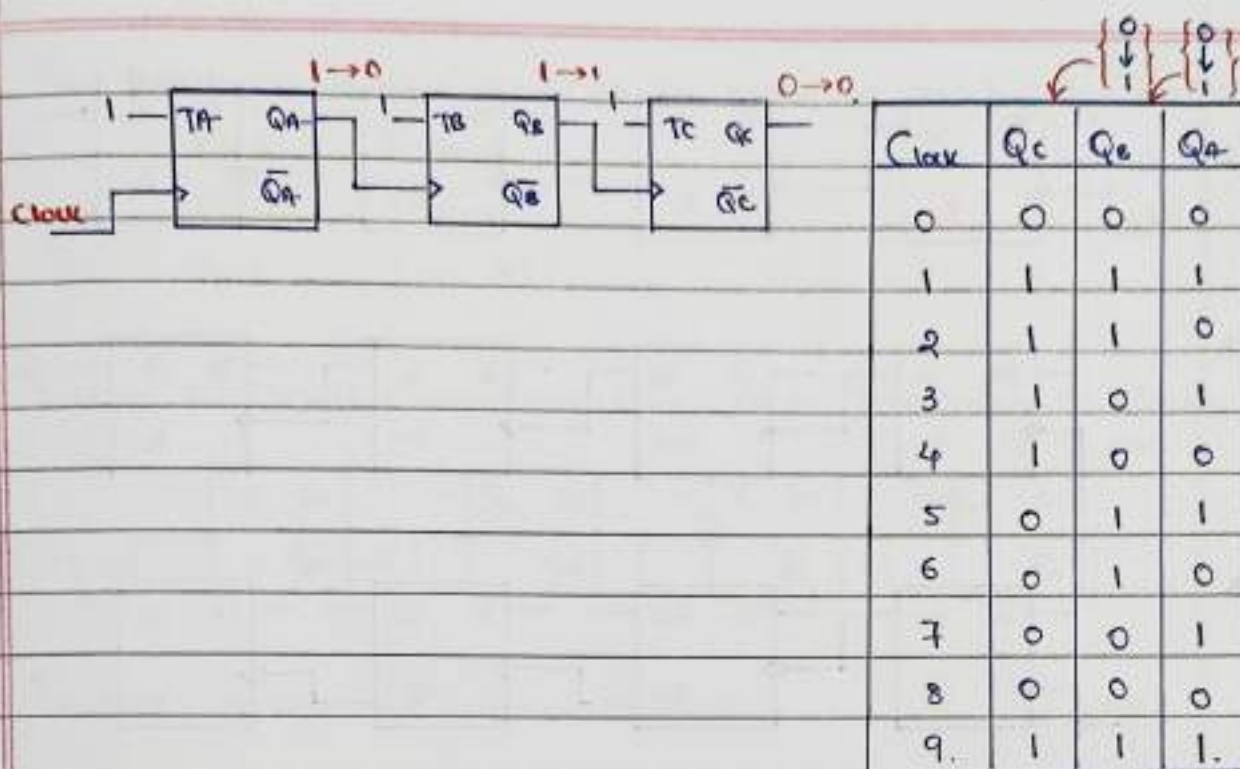
→ QA will toggle when the edge of the external clock appears.

→ QB will toggle when QA goes from 1 to 0.

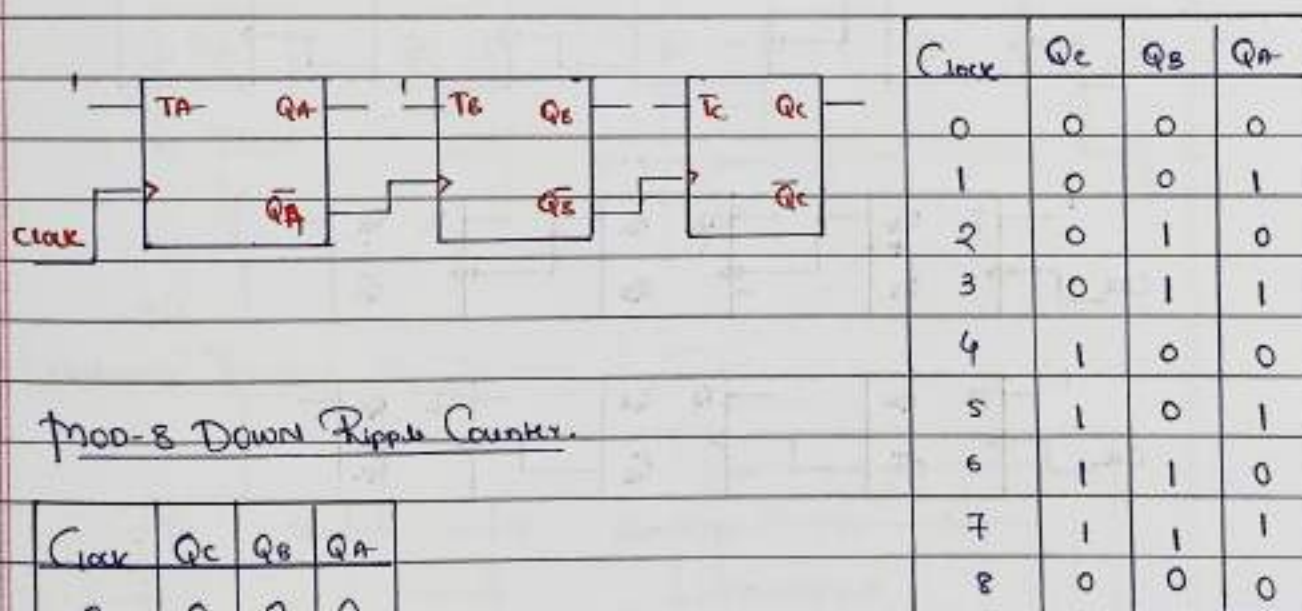
| Clock | Qc | Qb | Qa |
|-------|----|----|----|
| 0     | 0  | 0  | 0  |
| 1     | 0  | 0  | 1  |
| 2     | 0  | 1  | 0  |
| 3     | 0  | 1  | 1  |
| 4     | 1  | 0  | 0  |
| 5     | 1  | 0  | 1  |
| 6     | 1  | 1  | 0  |
| 7     | 1  | 1  | 1  |
| 8     | 0  | 0  | 0  |
| 9     | 0  | 0  | 1  |
| 10    | 0  | 1  | 0  |

#### MOD-8 Down Ripple Counter:

111 → 110 → 101 → 100 → 011 → 010 → 001 → 000

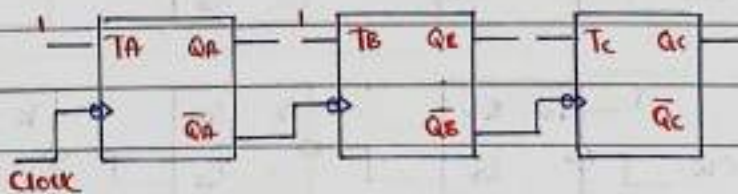


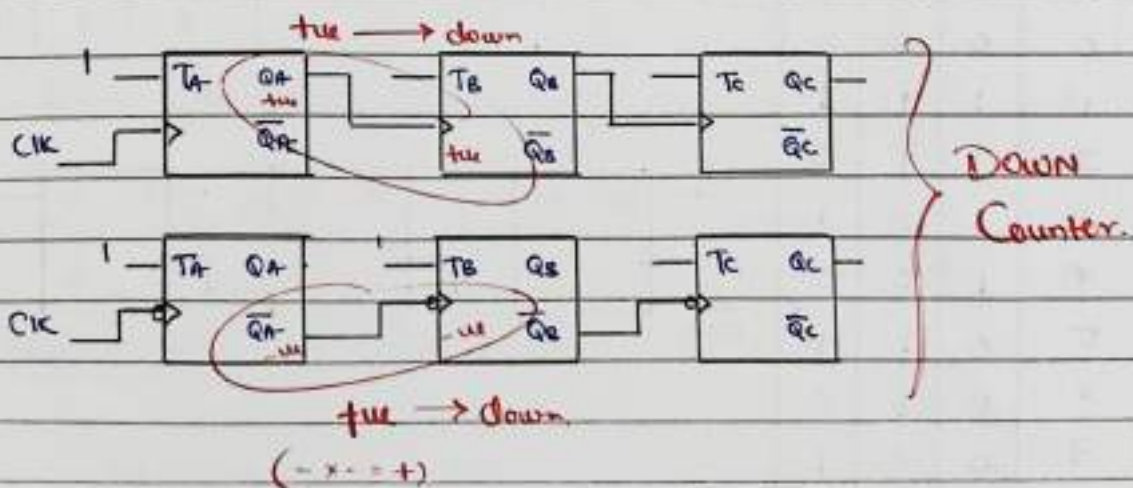
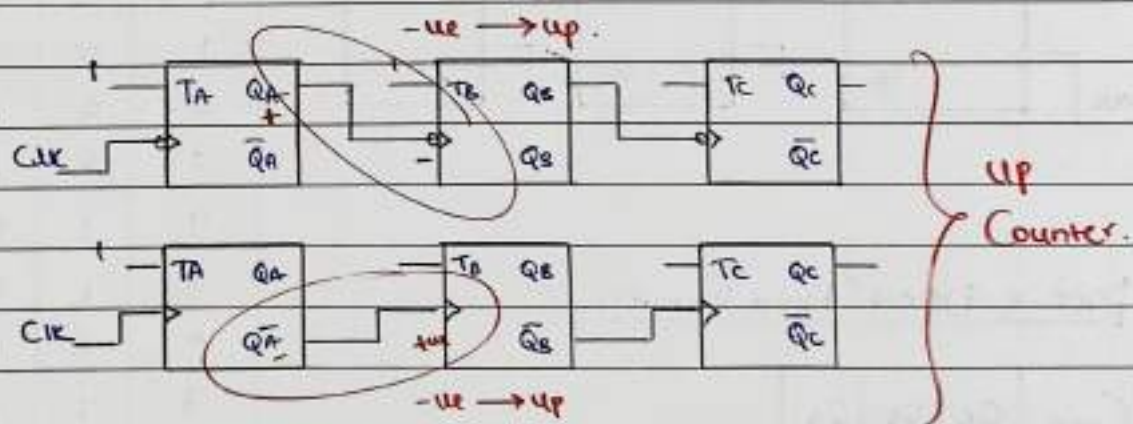
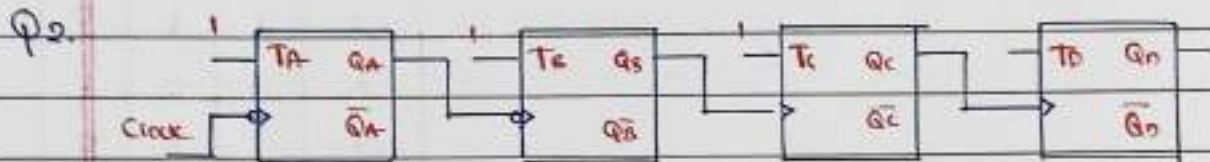
Mod-8 Up Ripple Counter



Mod-8 Down Ripple Counter

| Clock | QC | QB | QA |
|-------|----|----|----|
| 0     | 0  | 0  | 0  |
| 1     | 1  | 1  | 1  |
| 2     | 1  | 1  | 0  |
| 3     | 1  | 0  | 1  |
| 4     | 1  | 0  | 0  |
| 5     | 0  | 1  | 1  |
| 6     | 0  | 1  | 0  |
| 7     | 0  | 0  | 1  |
| 8     | 0  | 0  | 0  |



HW

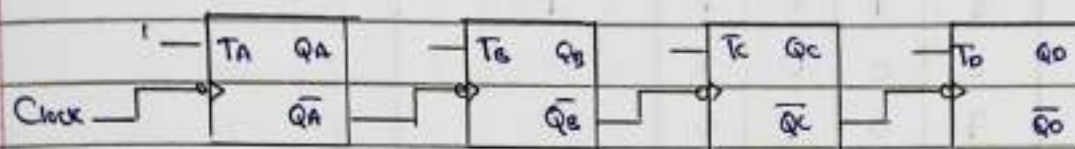
→ Full Mode Counter → when all the states are used by the counter.

$$n=3 \quad \text{Mod}=8$$

$$n=4 \quad \text{Mod}=16$$

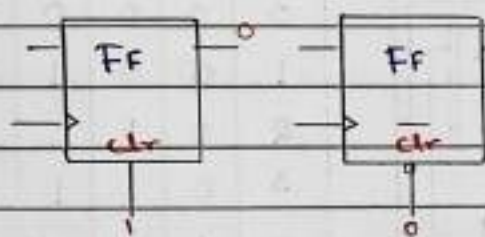
$$n=5 \quad \text{Mod}=32$$

Q1. Design a Mod 16 down counter in which  $\bar{Q}$  is taken as clock.

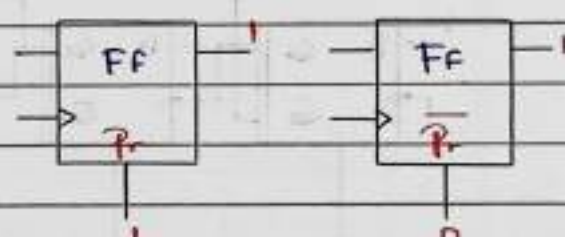


→ Feedback reduces the number of states:

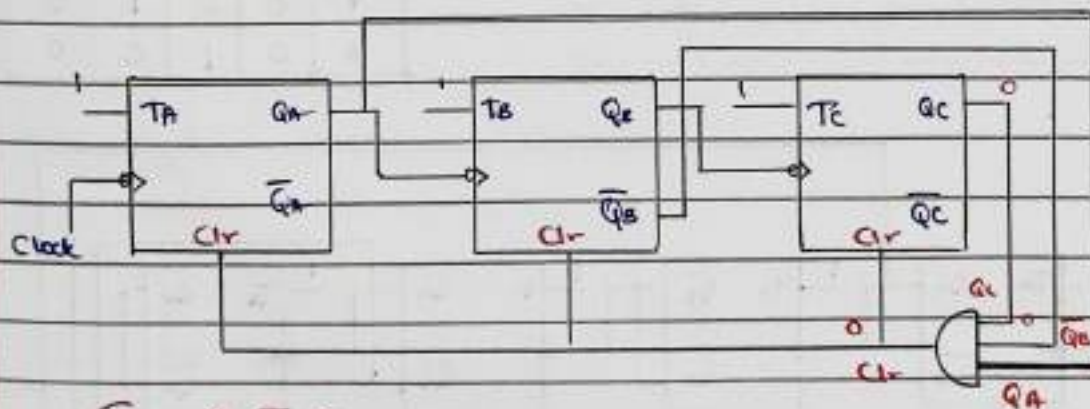
RESET - Clear



PRESET



Feedback Reduces the Number of States

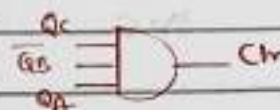


$$\text{Clr} = \bar{Q}_C \bar{Q}_B Q_A$$

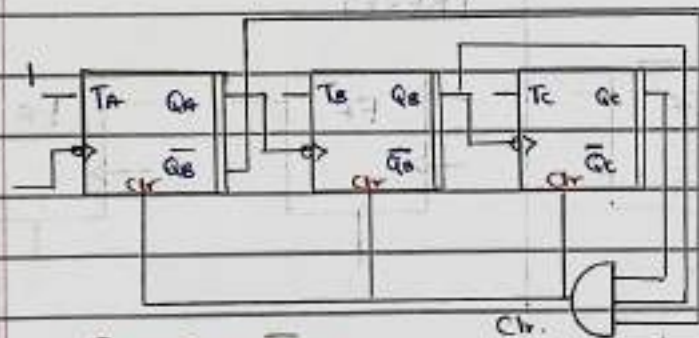
$$= 101$$

$$= 5$$

| Clock | Q <sub>c</sub> | Q <sub>b</sub> | Q <sub>a</sub> | CLR = Q <sub>c</sub> Q <sub>b</sub> Q <sub>a</sub> |
|-------|----------------|----------------|----------------|----------------------------------------------------|
| 0     | 0              | 0              | 0              | 0                                                  |
| 1     | 0              | 0              | 1              | 0                                                  |
| 2     | 0              | 1              | 0              | 0                                                  |
| 3     | 0              | 1              | 1              | 0                                                  |
| 4     | 1              | 0              | 0              | 0                                                  |
| 5     | ↑ <sup>0</sup> | ↑ <sup>0</sup> | ↑ <sup>0</sup> | ↑ <sup>0</sup>                                     |
| 6     | 0              | 0              | 1              | 0                                                  |
| 7     | 0              | 1              | 0              | 0                                                  |
| 8     | 0              | 1              | 1              | 0                                                  |

Mod-6 UP Ripple Counter

$$Clr = Q_c Q_b \bar{Q}_a$$



$$Clr = Q_c Q_b \bar{Q}_a$$

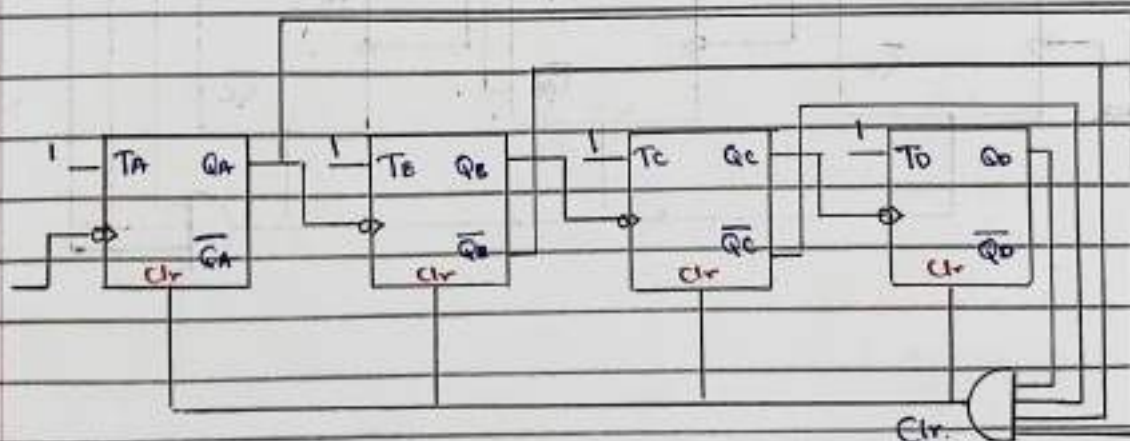
$$Q_c Q_b \bar{Q}_a$$

$$110 \Rightarrow 6$$

000 → 001 → 010 → 011 → 100 → 101



| Clock | Q <sub>c</sub> | Q <sub>b</sub> | Q <sub>a</sub> | Clr            |
|-------|----------------|----------------|----------------|----------------|
| 0     | 0              | 0              | 0              | 0              |
| 1     | 0              | 0              | 1              | 0              |
| 2     | 0              | 1              | 0              | 0              |
| 3     | 0              | 1              | 1              | 0              |
| 4     | 1              | 0              | 0              | 0              |
| 5     | 1              | 0              | 1              | 0              |
| 6     | ↑ <sup>0</sup> | ↑ <sup>0</sup> | ↑ <sup>0</sup> | ↑ <sup>0</sup> |
| 7     | 0              | 0              | 1              | 0              |
| 8     | 0              | 1              | 0              | 0              |

Q<sub>2</sub>Mod-9 UP Ripple Counter.

$$Clr = Q_d \bar{Q}_c \bar{Q}_b \bar{Q}_a$$

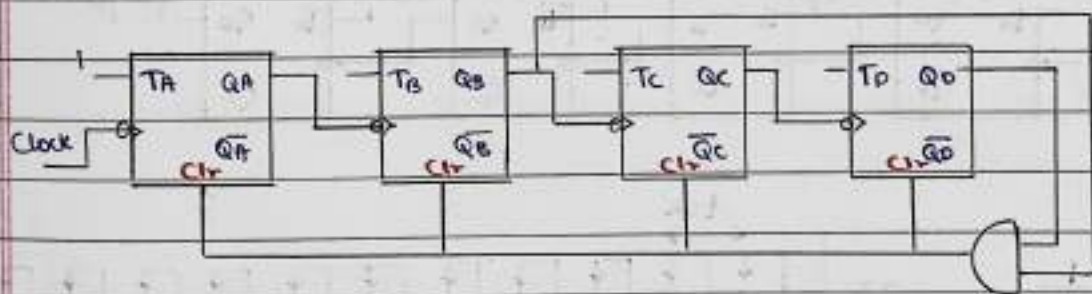
$$1001 \rightarrow 9$$

| Clock | Q <sub>0</sub>            | Q <sub>1</sub>            | Q <sub>2</sub>            | Q <sub>3</sub>            | Clr                       |
|-------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| 0     | 0                         | 0                         | 0                         | 0                         | 0                         |
| 1     | 0                         | 0                         | 0                         | 1                         | 0                         |
| 2     | 0                         | 0                         | 1                         | 0                         | 0                         |
| 3     | 0                         | 0                         | 1                         | 1                         | 0                         |
| 4     | 0                         | 1                         | 0                         | 0                         | 0                         |
| 5     | 0                         | 1                         | 0                         | 1                         | 0                         |
| 6     | 0                         | 1                         | 1                         | 0                         | 0                         |
| 7     | 0                         | 1                         | 1                         | 1                         | 0                         |
| 8     | 1                         | 0                         | 0                         | 0                         | 0                         |
| 9     | <del>1</del> <sup>0</sup> | <del>0</del> <sup>0</sup> | <del>0</del> <sup>0</sup> | <del>1</del> <sup>0</sup> | <del>0</del> <sup>0</sup> |
| 10    | 0                         | 0                         | 0                         | 1                         |                           |
| 11    | 0                         | 0                         | 1                         | 0                         |                           |

$$Clr = Q_0 \bar{Q}_1 \bar{Q}_2 \bar{Q}_3$$

$$1001 \Rightarrow 9$$

Q3 Design a BCD Ripple Carry Counter.

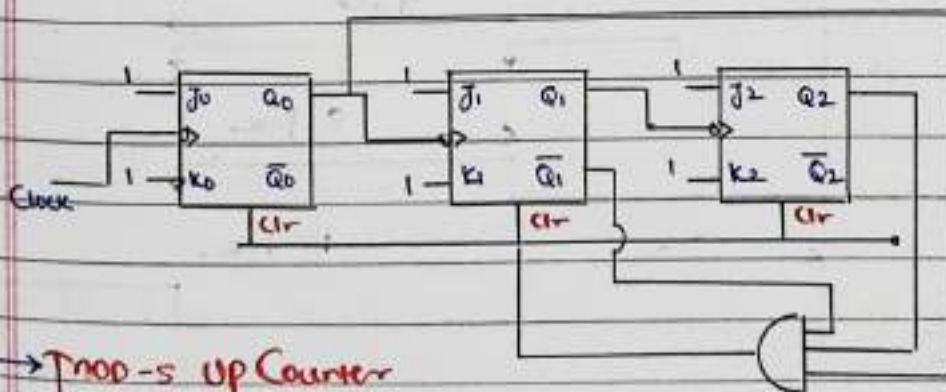


$$Clr = Q_0 \bar{Q}_1 \bar{Q}_2 \bar{Q}_3$$

$$1010$$

$$\Rightarrow 10$$

Q4 Which type of counter is shown below?

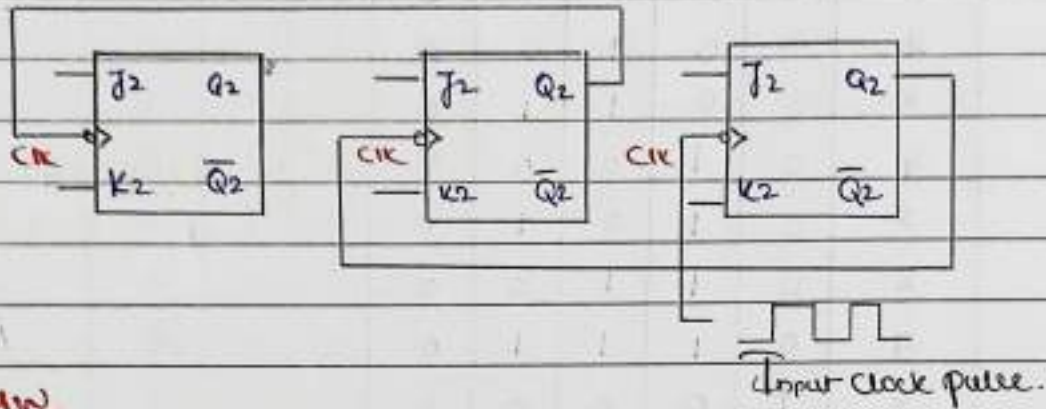


$$Clr = Q_2 \bar{Q}_1 \bar{Q}_0$$

$$101 \Rightarrow 5$$

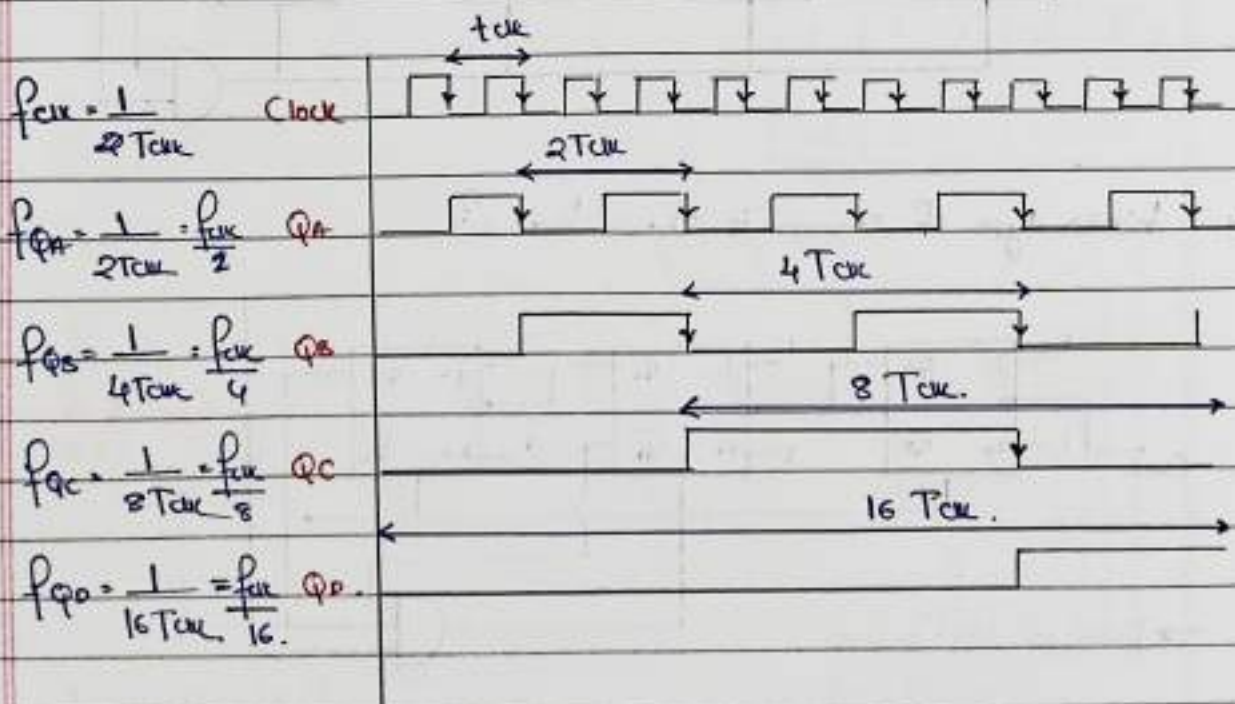
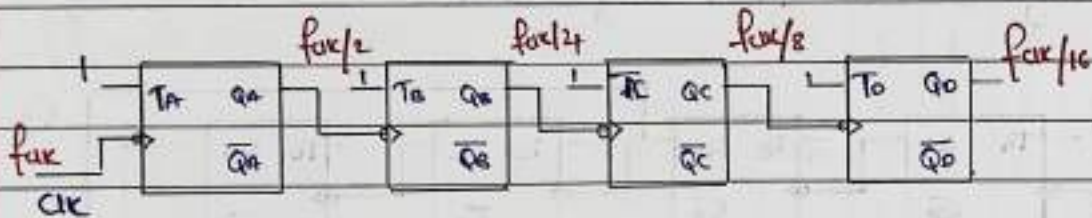
→ MOD-5 Up Counter

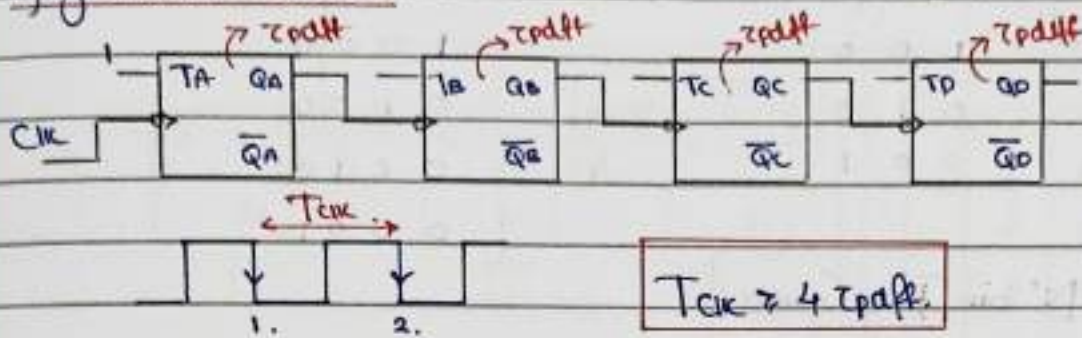
HW  
Q5. Consider the following clock. If Counter starts at 000, what will be count after 13 clock pulses?



HW  
Q6. Design MOD-13, MOD-21, MOD-27, MOD-14 (Answer) Ripple Counter.

Asynchronous [Ripple Counter]



Asynchronous Counter.N-bit Asynchronous counter

$$T_{clk} \geq n \cdot t_{pdff}$$

$$(f_{clk})_{max} = \frac{1}{n \cdot t_{pdff}}$$

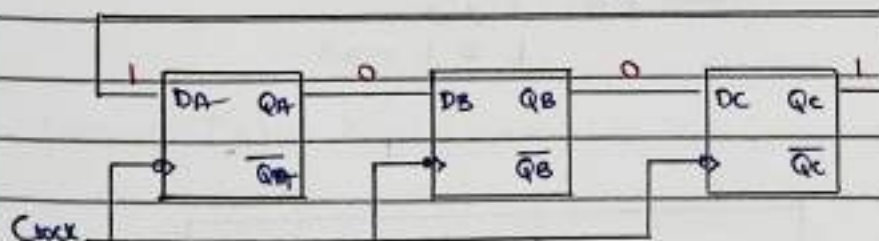
$$\frac{1}{T_{clk}} \leq \frac{1}{n \cdot t_{pdff}}$$

$$f_{clk} \leq \frac{1}{n \cdot t_{pdff}}$$

Synchronous Counter : All the FlipFlops are Synchronized with Same Clock  
eg: Ring Counter, Johnson Counter.

Ring Counter

→ It is also a SISO shift register in the form of Ring.



| Clock | QA | QB | QC |
|-------|----|----|----|
| 0     | 0  | 0  | 0  |
| 1     | 1  | 0  | 0  |
| 2     | 0  | 1  | 0  |
| 3     | 0  | 0  | 1  |
| 4     | 1  | 0  | 0  |

→ Ring Counter is not a self starting counter, to start the ring counter we have to place '1' at first flip-flop. This '1' will rotate among all the flip-flops.

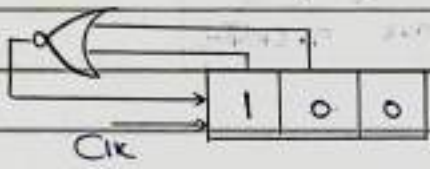
4 bit Ring Counter

$$\frac{100}{25} = 4$$

$\left\{ \begin{array}{l} 1000 \\ 0100 \\ 0010 \\ 0001 \end{array} \right.$

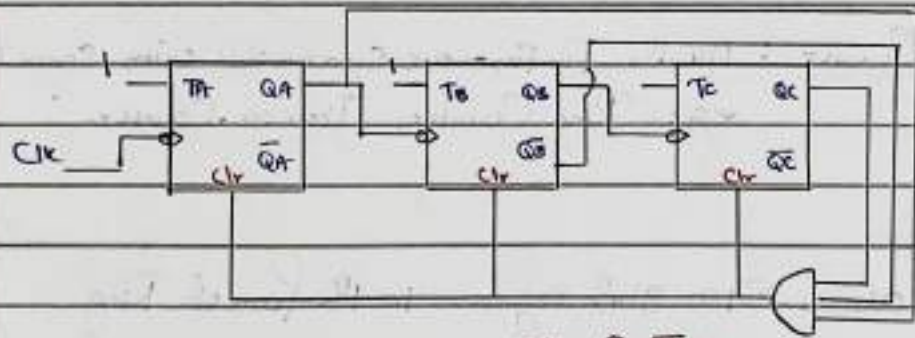
'N' bit Ring Counter  
 $\Rightarrow \text{Prod} = N$

| Clock | Q <sub>A</sub> | Q <sub>B</sub> | Q <sub>C</sub> |
|-------|----------------|----------------|----------------|
| 0     | 0              | 0              | 0              |
| 1     | 1              | 0              | 0              |
| 2     | 0              | 1              | 0              |
| 3     | 0              | 0              | 1              |
| 4     | 1              | 0              | 0              |



{ How  
Assume  
of power  
Class? }

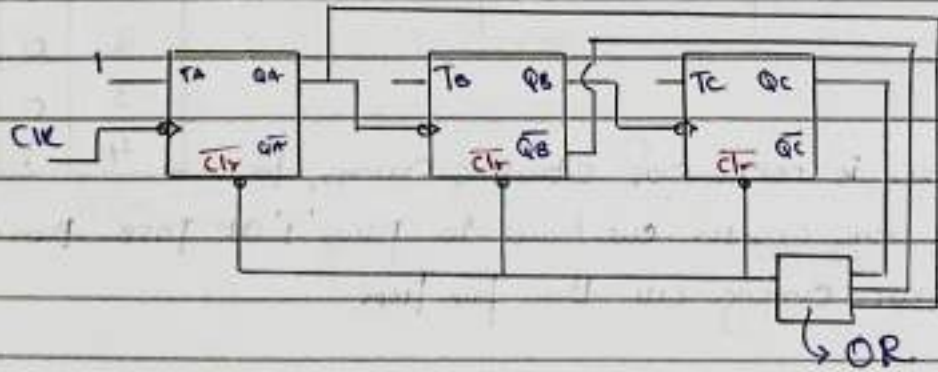
MOD-5 UP

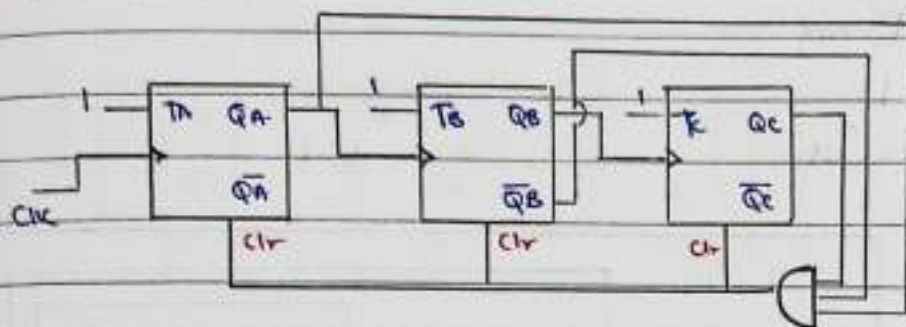


$$C1r = Qc \bar{Q}BQA$$

$$101 \Rightarrow 5$$

mod-2 up



MOD-3 Down Ripple Counter

111 → 110 → 000

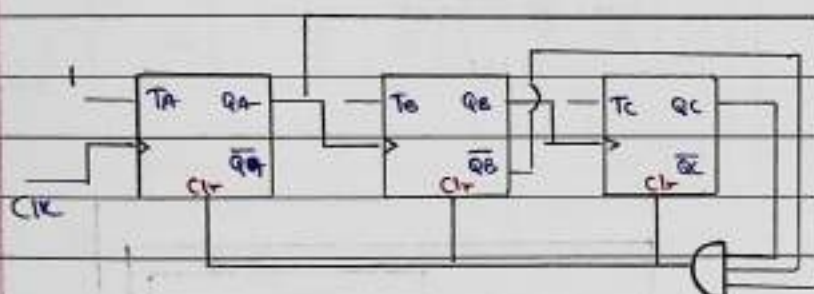
$Clr = Q_C Q_B Q_A$

Down = Full-up

$= 8 - 5$

$= 3$

| Clk | Qc           | Qb           | Qa           | Clr          |
|-----|--------------|--------------|--------------|--------------|
| 0   | 0            | 0            | 0            | 0            |
| 1   | 1            | 1            | 1            | 0            |
| 2   | 1            | 1            | 0            | 0            |
| 3   | <del>1</del> | <del>0</del> | <del>1</del> | <del>0</del> |
| 4   | 1            | 1            | 1            | 0            |
| 5   | 1            | 1            | 0            | 0            |
| 6   | <del>1</del> | <del>0</del> | <del>1</del> | <del>0</del> |

MOD-4 Down Ripple Counter

111 → 110 → 101 → 000

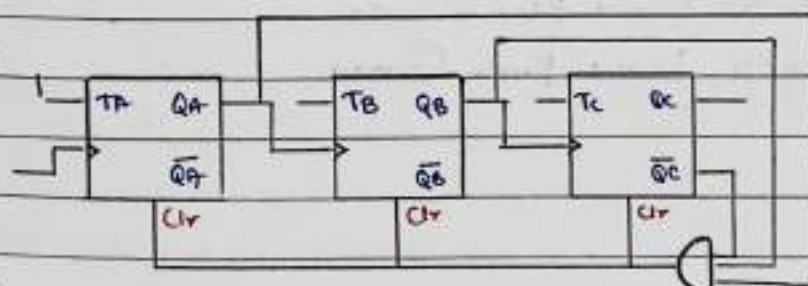
$Clr = Q_C Q_B Q_A$

100

Down = Full-up

$= 8 - 4 \rightarrow 4$

| Clock | Qc           | Qb           | Qa           | Clr          |
|-------|--------------|--------------|--------------|--------------|
| 0     | 0            | 0            | 0            | 0            |
| 1     | 1            | 1            | 1            | 0            |
| 2     | 1            | 1            | 0            | 0            |
| 3     | 1            | 0            | 1            | 0            |
| 4     | <del>1</del> | <del>0</del> | <del>0</del> | <del>0</del> |

MOD-5 Down Ripple Counter

111 → 110 → 101 → 100 → 000

Down = Full-up

$(5) = 8 - 3$

$\bar{Q}_C Q_B Q_A$

011 → 3

| Clk | Qc           | Qb           | Qa           | Clr |
|-----|--------------|--------------|--------------|-----|
| 0   | 0            | 0            | 0            | 0   |
| 1   | 1            | 1            | 1            | 0   |
| 2   | 1            | 1            | 0            | 0   |
| 3   | 1            | 0            | 1            | 0   |
| 4   | 1            | 0            | 0            | 0   |
| 5   | <del>0</del> | <del>1</del> | <del>1</del> | 1   |

Q1. Design a Mod-13 Down Ripple Counter.

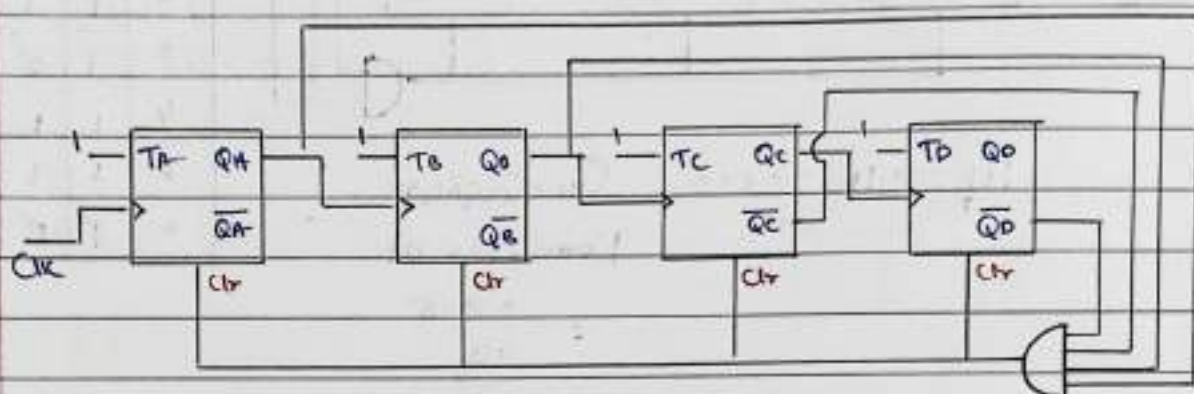
Full-Up = Down

$$16 - \text{UP} = 13$$

$$16 - 13 = 3 \text{ (UP)}$$

$$\text{Clr} = \overline{Q_D} \overline{Q_C} Q_B Q_A$$

$$0 \ 0 \ 1 \ 1$$



Q2. Design a Mod-9 Down Pipeline Counter.

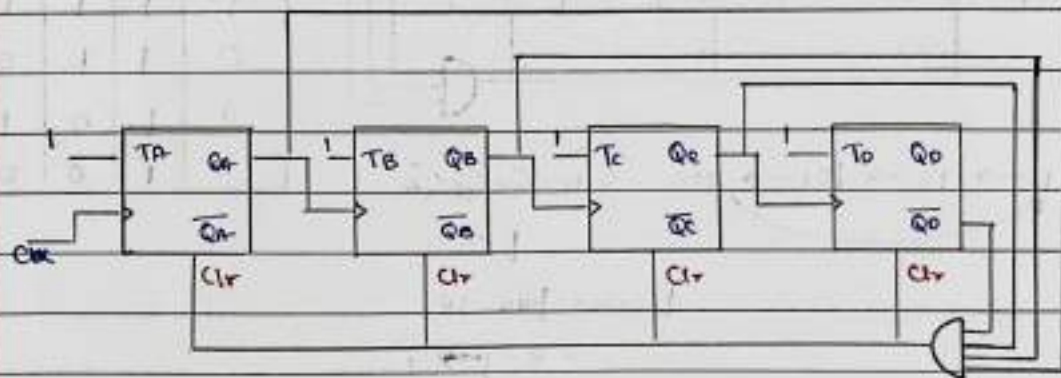
Full-Up = Down

$$16 - \text{UP} = 9$$

$$\text{UP} = 16 - 9 = 7$$

$$\text{Clr} = \overline{Q_D} \overline{Q_C} Q_B Q_A$$

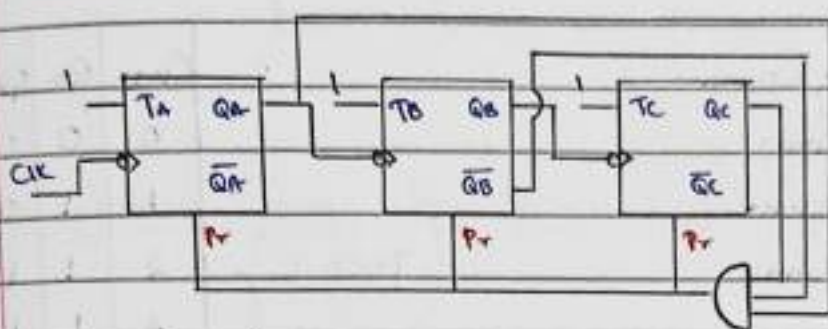
$$0 \ 1 \ 1 \ 1$$



Q3. Design a Mod-19 Down Ripple Counter.

Q4. Design a Mod-23 Down Ripple Counter.

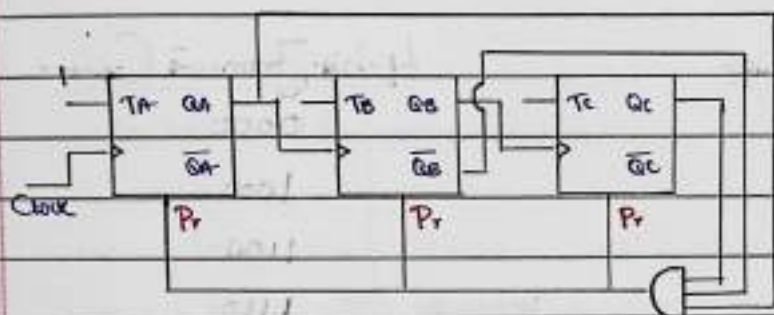
Q5. Design a Mod-11 Down Ripple Counter.

Prob-6

000 → 001 → 010 → 011 → 100 → 111

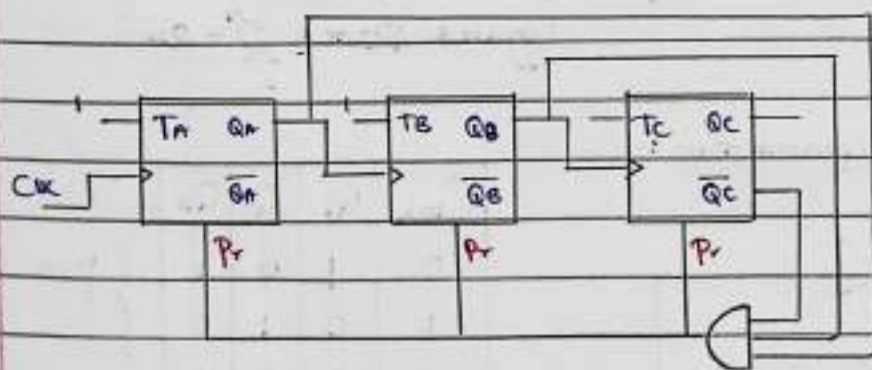
 $Pr = Q_C \bar{Q}_B \bar{Q}_A$ 

| Clock | Qc        | Qb        | Qa        | Pr        |
|-------|-----------|-----------|-----------|-----------|
| 0     | 0         | 0         | 0         | 0         |
| 1     | 0         | 0         | 1         | 0         |
| 2     | 0         | 1         | 0         | 0         |
| 3     | 0         | 1         | 1         | 0         |
| 4     | 1         | 0         | 0         | 0         |
| 5     | $\bar{x}$ | $\bar{x}$ | $\bar{x}$ | $\bar{x}$ |
| 6     | 0         | 0         | 0         | 0         |
| 7     | 0         | 0         | 1         | 0         |

Prob-2

000 → 111 → 110

| Clock | Qc        | Qb        | Qa        | Pr        |
|-------|-----------|-----------|-----------|-----------|
| 0     | 0         | 0         | 0         | 0         |
| 1     | 1         | 1         | 1         | 0         |
| 2     | 1         | 1         | 0         | 0         |
| 3     | $\bar{x}$ | $\bar{x}$ | $\bar{x}$ | $\bar{x}$ |
| 4     | 1         | 1         | 0         | 0         |
| 5     | $\bar{x}$ | $\bar{x}$ | $\bar{x}$ | $\bar{x}$ |
| 6     | 1         | 1         | 0         | 0         |
| 7     | $\bar{x}$ | $\bar{x}$ | $\bar{x}$ | $\bar{x}$ |

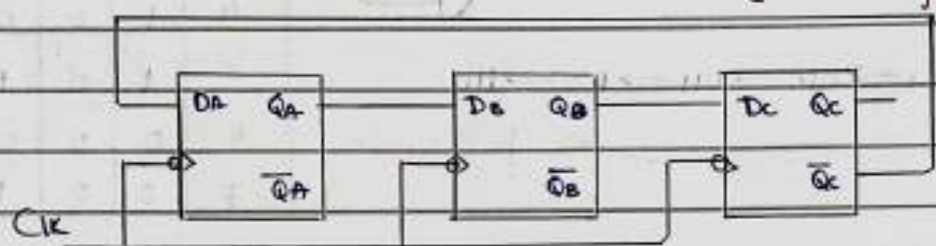
Prob-4

000 → 111 → 110 → 101 → 100

| Clock | Qc        | Qb        | Qa        | Pr        |
|-------|-----------|-----------|-----------|-----------|
| 0     | 0         | 0         | 0         | 0         |
| 1     | 1         | 1         | 1         | 0         |
| 2     | 1         | 1         | 0         | 0         |
| 3     | 1         | 0         | 1         | 0         |
| 4     | 1         | 0         | 0         | 0         |
| 5     | $\bar{x}$ | $\bar{x}$ | $\bar{x}$ | $\bar{x}$ |
| 6     | 1         | 1         | 0         | 0         |
| 7     | 1         | 0         | 1         | 0         |

Johnson Counter

- Twisted Ring Counter
- Creeping Counter
- Probies Counter
- Walking Counter



| Clock | QA | QB | QC |
|-------|----|----|----|
| 0     | 0  | 0  | 0  |
| 1     | 1  | 0  | 0  |
| 2     | 1  | 1  | 0  |
| 3     | 1  | 1  | 1  |
| 4     | 0  | 1  | 1  |
| 5     | 0  | 0  | 1  |
| 6     | 0  | 0  | 0  |
| 7     | 1  | 0  | 0  |

{ mod-6 }

3 bit Johnson Counter

mod 6

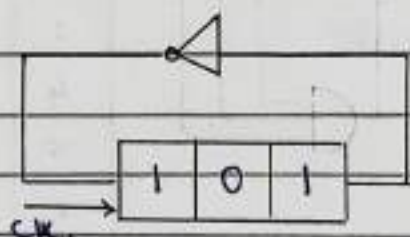
- 000
- 100
- 110
- 111
- 011
- 001

4-bit Johnson Counter

mod-8

- 0000
- 1000
- 1100
- 1110
- 1111
- 0111
- 0011
- 0001

→ N bit Johnson Counter  $\Rightarrow \text{mod (used state)} = 2N$   
 Unused State =  $2^N - 2N$

Symbolic Representation:

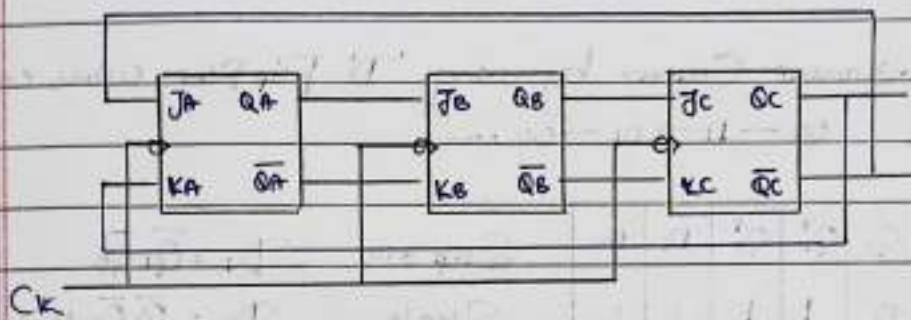
Lockout Problem!

| Clock | Q1 | Q2 | Q3 |
|-------|----|----|----|
| 0     | 1  | 0  | 1  |
| 1     | 0  | 1  | 0  |
| 2     | 1  | 0  | 1  |
| 3     | 0  | 1  | 0  |
| 4     | 1  | 0  | 1  |

{ mod-2 }

Lockout Problem: Whenever Johnson Counter enters its unused state, it will trap or lock into unused state, are called Lockout Problem.

### Johnson Counter by Jk Flip Flop



### Synchronous Counter Design:

- Step 1: Write the Previous and Present State
- Step 2: Write the Excitation table of flip flop
- Step 3: Write the logical expression
- Step 4: Minimise the logical expression
- Step 5: Hardware implementation

Q1. Design a Synchronous Counter using 'T' Flip Flop which counts the sequence  $\Rightarrow 0-3-1-2-0 \dots$   $\{ \infty - 11 - 01 - 10 \rightarrow \infty \dots \}$   
 $\hookrightarrow$  2 FFs are required.

Step 1:  
& Step 2

| $Q_1$ | $Q_0$ | $Q_1^+$ | $Q_0^+$ | $T_1$ | $T_0$  |
|-------|-------|---------|---------|-------|--------|
| 0     | 0     | 1       | 1       | 1     | 1      |
| 0     | 1     | 1       | 0       | 1     | $\phi$ |
| 1     | 0     | 0       | 0       | 1     | 0      |
| 1     | 1     | 0       | 1       | 1     | 0      |

Step 3

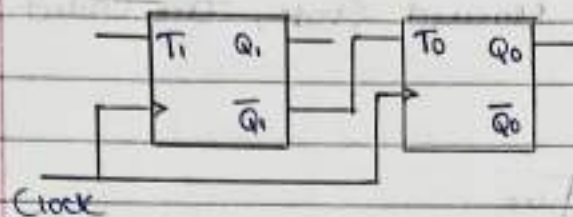
& Step 4:

$$T_1 = 1$$

$$T_0 = \bar{Q}_1 \bar{Q}_0 + \bar{Q}_1 Q_0$$

$$T_0 = \bar{Q}_1 [\bar{Q}_0 + Q_0]$$

$$T_0 = \bar{Q}_1$$

Steps:Justification:

| Clock | Q <sub>1</sub> | Q <sub>0</sub> |
|-------|----------------|----------------|
| 0     | 0              | 0              |
| 1     | 1              | 1              |
| 2     | 0              | 1              |
| 3     | 1              | 0              |

Q2. Design a Synchronous Counter by using 'D' Flip Flops which count the Sequence 00-11-01-00....

Method 1Step 1 & Step 2

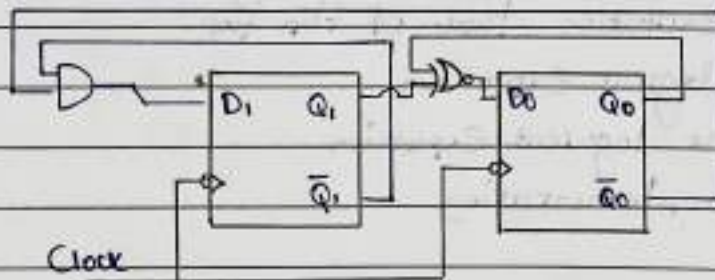
| Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1</sub> <sup>+</sup> | Q <sub>0</sub> <sup>+</sup> | D <sub>1</sub> | D <sub>0</sub> |
|----------------|----------------|-----------------------------|-----------------------------|----------------|----------------|
| 0              | 0              | 1                           | 1                           | 1              | 1              |
| 1              | 1              | 0                           | 1                           | 0              | 1              |
| 0              | 1              | 0                           | 0                           | 0              | 0              |

Step 3 & Step 4

$$D_1 = \bar{Q}_1 \bar{Q}_0$$

$$D_0 = \bar{Q}_1 \bar{Q}_0 + Q_1 Q_0$$

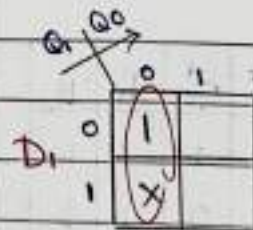
$$D_0 = Q_1 \odot Q_0$$

Step 5:

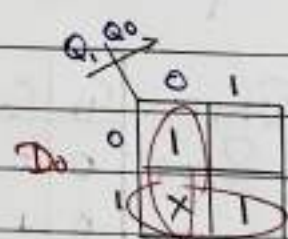
| Clock | Q <sub>1</sub> | Q <sub>0</sub> |
|-------|----------------|----------------|
| 0     | 0              | 0              |
| 1     | 1              | 1              |
| 2     | 0              | 1              |
| 3     | 0              | 0              |

Method 2

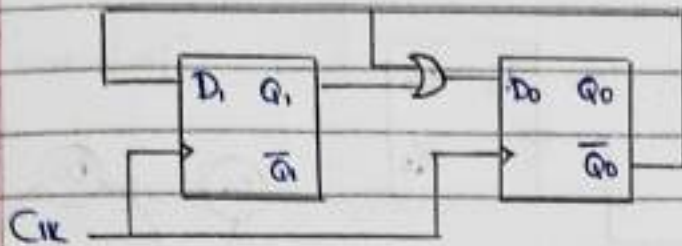
| Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1</sub> <sup>+</sup> | Q <sub>0</sub> <sup>+</sup> | D <sub>1</sub> | D <sub>0</sub> |
|----------------|----------------|-----------------------------|-----------------------------|----------------|----------------|
| 0              | 0              | 1                           | 1                           | 1              | 1              |
| 0              | 1              | 0                           | 0                           | 0              | 0              |
| 1              | 0              | x                           | x                           | x              | x              |
| 1              | 1              | 0                           | 1                           | 0              | 1              |



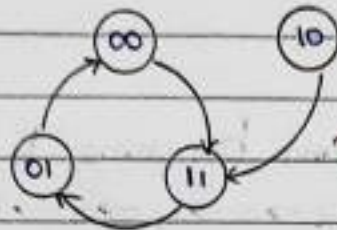
$$D_1 = \bar{Q}_1 \bar{Q}_0$$



$$D_0 = \bar{Q}_1 \bar{Q}_0 + Q_1 Q_0$$



| Clock | Q <sub>1</sub> | Q <sub>0</sub> |
|-------|----------------|----------------|
| 0     | 0              | 0              |
| 1     | 1              | 1              |
| 2     | 0              | 1              |
| 3     | 0              | 0              |
| 4     | 1              | 1              |
| 5     | 0              | 1              |



No lockout!

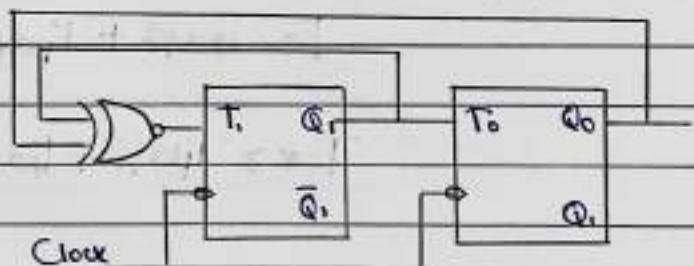
HW

Q3. Design a synchronous counter by using P-FF which count:  
000 → 001 → 010 → 011 → 100 → 101 → 110 → 111 → 000 ...

Q4. Design a synchronous counter by using T-flip flop which count  
the sequence 0 → 2 → 3 → 0  
00 → 10 → 11 → 00      2 bin → 2 flip flops are required.

Method 1

| Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1</sub> <sup>+</sup> | Q <sub>0</sub> <sup>+</sup> | T <sub>1</sub> | T <sub>0</sub> |
|----------------|----------------|-----------------------------|-----------------------------|----------------|----------------|
| 0              | 0              | 1                           | 0                           | 1              | 0              |
| 1              | 0              | 1                           | 1                           | 0              | 1              |
| 1              | 1              | 0                           | 0                           | 1              | 1              |

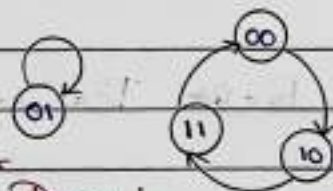


$$T_1 = \underline{Q_1 + Q_0}$$

$$T_0 = \underline{Q_1}$$

Lockout

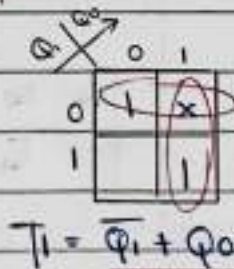
Problem!



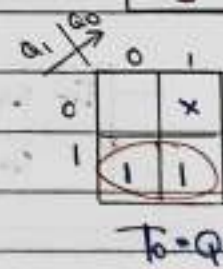
| Clock | Q <sub>1</sub> | Q <sub>0</sub> |
|-------|----------------|----------------|
| 0     | 0              | 0              |
| 1     | 1              | 0              |
| 2     | 1              | 1              |
| 3     | 0              | 0              |

Method 2

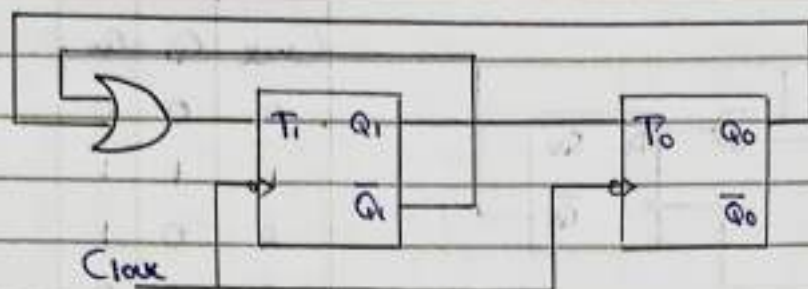
| Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1</sub> <sup>+</sup> | Q <sub>0</sub> <sup>+</sup> | T <sub>1</sub> | T <sub>0</sub> |
|----------------|----------------|-----------------------------|-----------------------------|----------------|----------------|
| 0              | 0              | 1                           | 0                           | 1              | 0              |
| 0              | 1              | x                           | x                           | x              | x              |
| 1              | 0              | 1                           | 1                           | 0              | 1              |
| 1              | 1              | 0                           | 0                           | 1              | 1              |



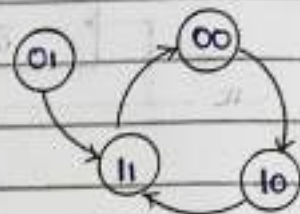
$$T_1 = \underline{Q_1 + Q_0}$$



$$T_0 = \underline{Q_1}$$



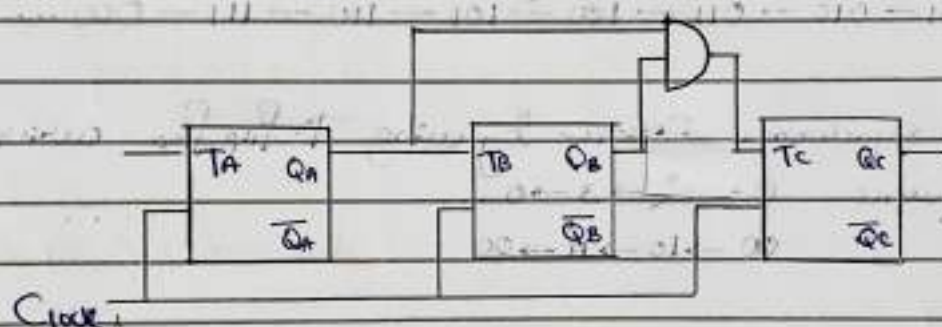
No Lockout Problem!



Q1. Design a synchronous counter by using T Flip Flop which count the sequence  $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 0$ .

Three bit  $\rightarrow$  Three Flip Flops Required

$\rightarrow T_A = 1, T_B = Q_A, T_C = Q_A Q_B$



$$T = T_{pdff} + T_{pdand}$$

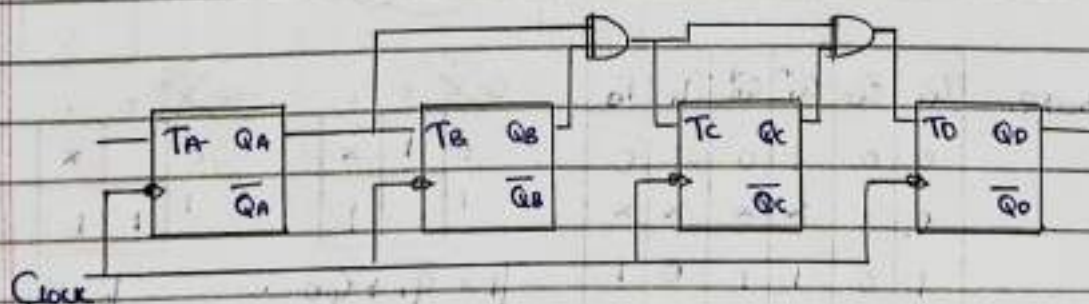
$$\frac{1}{T_{clk}} \leq \frac{1}{T_{pdff} + T_{pdand}}$$

$$T_{clk} \geq T_{pdff} + T_{pdand}$$

$$f_{clk} \leq \frac{1}{T_{pdff} + T_{pdand}}$$

4 bit

$T_A = 1, T_B = Q_A, T_C = Q_A Q_B, T_D = Q_A Q_B Q_C$



$$T_{clk} \geq T_{pdf} + 2 \cdot T_{pd\&no}$$

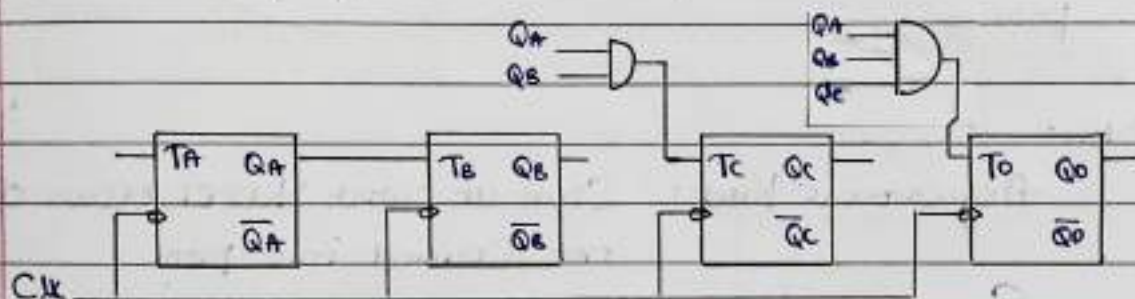
$$f_{clk} \leq \frac{1}{T_{pdf} + 2 \cdot T_{pd\&no}}$$

### Serial Carry Synchronous Counter

$$T_{clk} \geq T_{pdf} + (N-2) \cdot T_{pd\&no}$$

$$f_{clk} \leq \frac{1}{T_{pdf} + (N-2) \cdot T_{pd\&no}}$$

### Parallel Carry Synchronous Counter

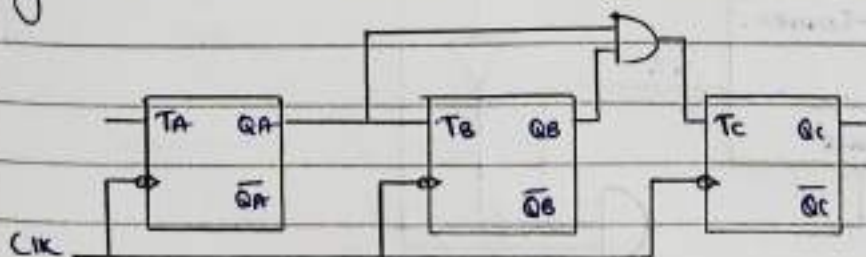


$$T_{clk} \geq T_{pdf} + T_{pd\&no}$$

$$f_{clk} \leq \frac{1}{T_{pdf} + T_{pd\&no}}$$

Q1. Minimum no. of flip flops required to construct BCD counter is  
 Ans. 0000 } 4 bits  $\rightarrow$  Four Flip Flops.  
 1001

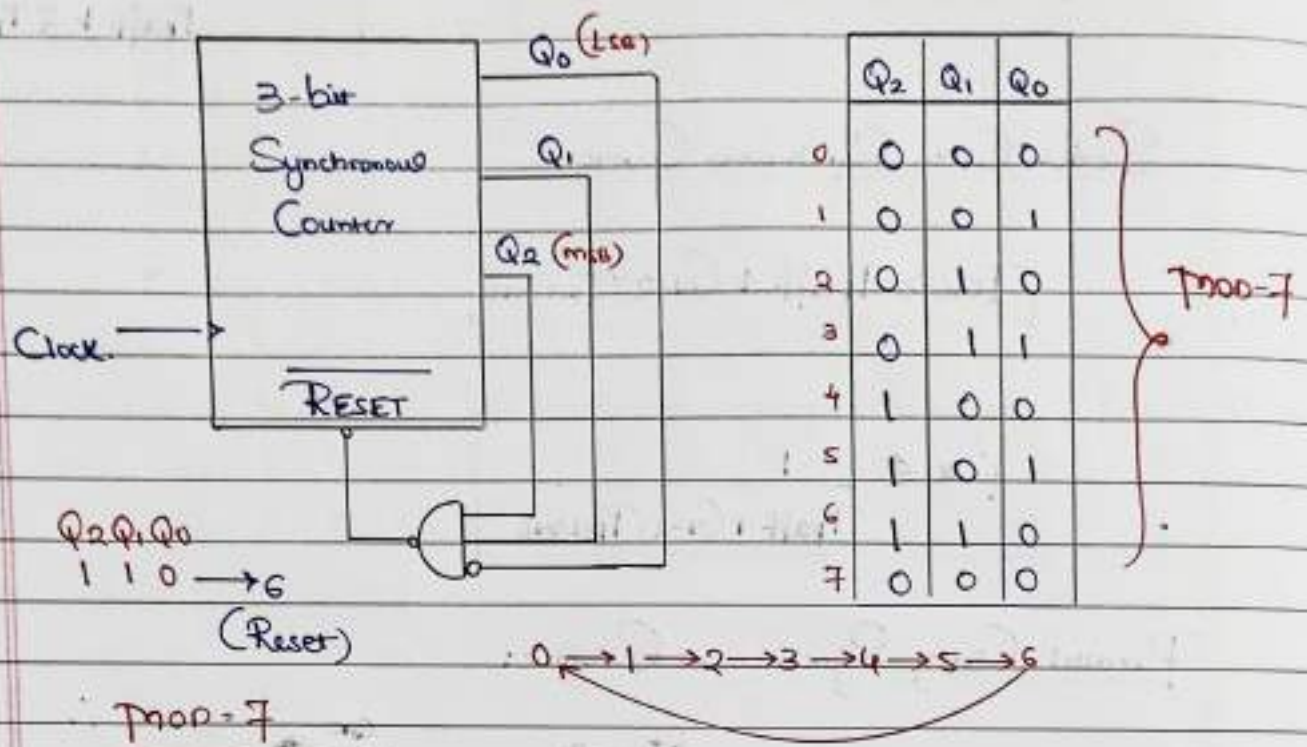
Q2. Minimum no. of Flip Flops required to design counter whose given below  $0 \rightarrow 0 \rightarrow 1 \rightarrow 1 \rightarrow 2 \rightarrow 2 \rightarrow 3 \rightarrow 3 \rightarrow 0 \rightarrow 0 \dots$



3-bit Synchronous up Counter

$\rightarrow$  3 Flip Flops Required

Q3. What is the mod of the counter?



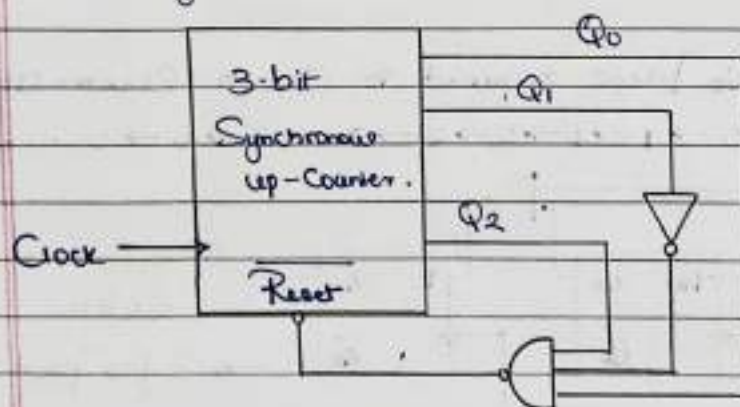
Note:

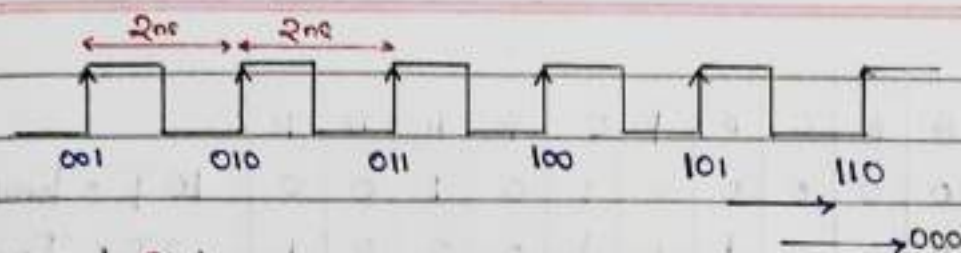
Asynchronous Reset: State at which RESET occurs does not Counted into mod.

Synchronous Reset: State at which RESET occurs is Counted into mod.

Q4. Consider the below Circuit:

The delay of NAND, NOR, gate is 3ns, 1ns respectively and that of the Counter is assumed to be zero. If the clock frequency is 500 kHz, then the Counter behaves as:





$$Cir = Q_2 \bar{Q}_1 \bar{Q}_0$$

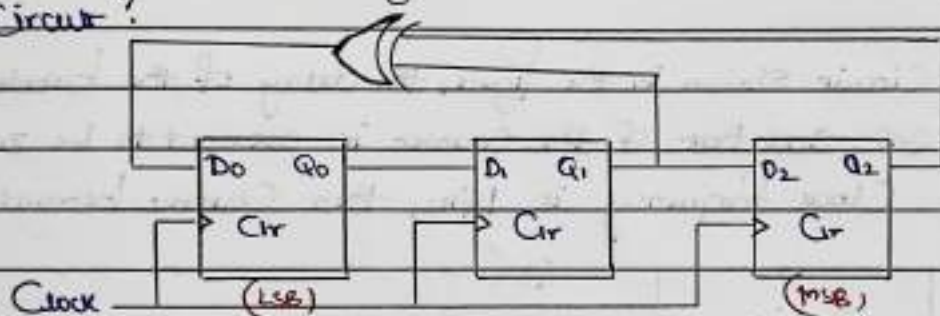
$$f_{clk} = 500 \times 10^6 \text{ Hz}$$

$$T_{clk} = \frac{1}{500 \times 10^6} \text{ Sec} = \frac{1000 \times 10^{-9}}{500}$$

$$\therefore T_{mod} = 7$$

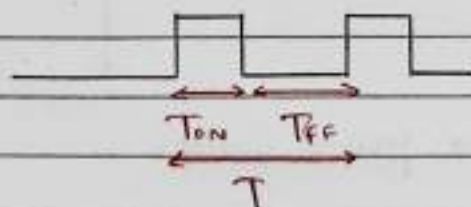
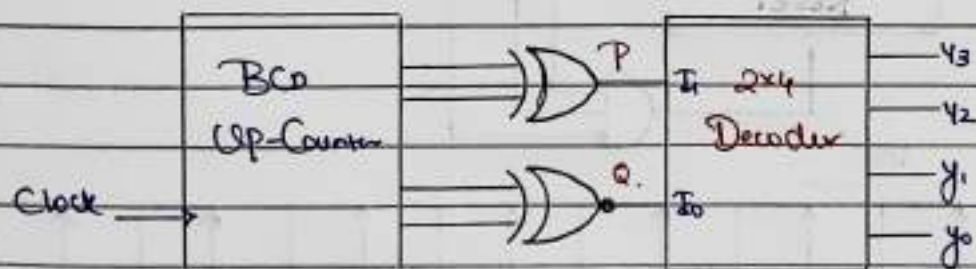
$$= 2ns$$

Q5. Consider the Circuit below with initial state  $Q_0=1$ ,  $Q_1=Q_2=0$ . The state of the circuit is given by the value of  $4Q_2+2Q_1+Q_0$ . Which of the following is the correct state sequence of the circuit?



Sequence:  $1 \rightarrow 2 \rightarrow 5 \rightarrow 3 \rightarrow 7 \rightarrow 6 \rightarrow 4$

Q6. Consider the circuit given below. The duty cycle of  $Y_2$  is



$$\text{Duty Cycle} = \frac{T_{on}}{T_{on} + T_{ff}} \times 100$$

| A | B | C | D | P | Q | Y <sub>0</sub> | Y <sub>1</sub> | Y <sub>2</sub> | Y <sub>3</sub> |
|---|---|---|---|---|---|----------------|----------------|----------------|----------------|
| 0 | 0 | 0 | 0 | 0 | 1 | 0              | 1              | 0              | 0              |
| 0 | 0 | 0 | 1 | 1 | 1 | 0              | 0              | 0              | 1              |
| 0 | 0 | 1 | 0 | 1 | 1 | 0              | 0              | 0              | 1              |
| 0 | 0 | 1 | 1 | 0 | 1 | 0              | 1              | 0              | 0              |
| 0 | 1 | 0 | 0 | 0 | 0 | 1              | 0              | 0              | 0              |
| 0 | 1 | 0 | 1 | 1 | 0 | 0              | 0              | 1              | 0              |
| 0 | 1 | 1 | 0 | 1 | 0 | 0              | 0              | 1              | 0              |
| 0 | 1 | 1 | 1 | 0 | 0 | 1              | 0              | 0              | 0              |
| 1 | 0 | 0 | 0 | 0 | 0 | 1              | 0              | 0              | 0              |
| 1 | 0 | 0 | 1 | 1 | 0 | 0              | 0              | 1              | 0              |

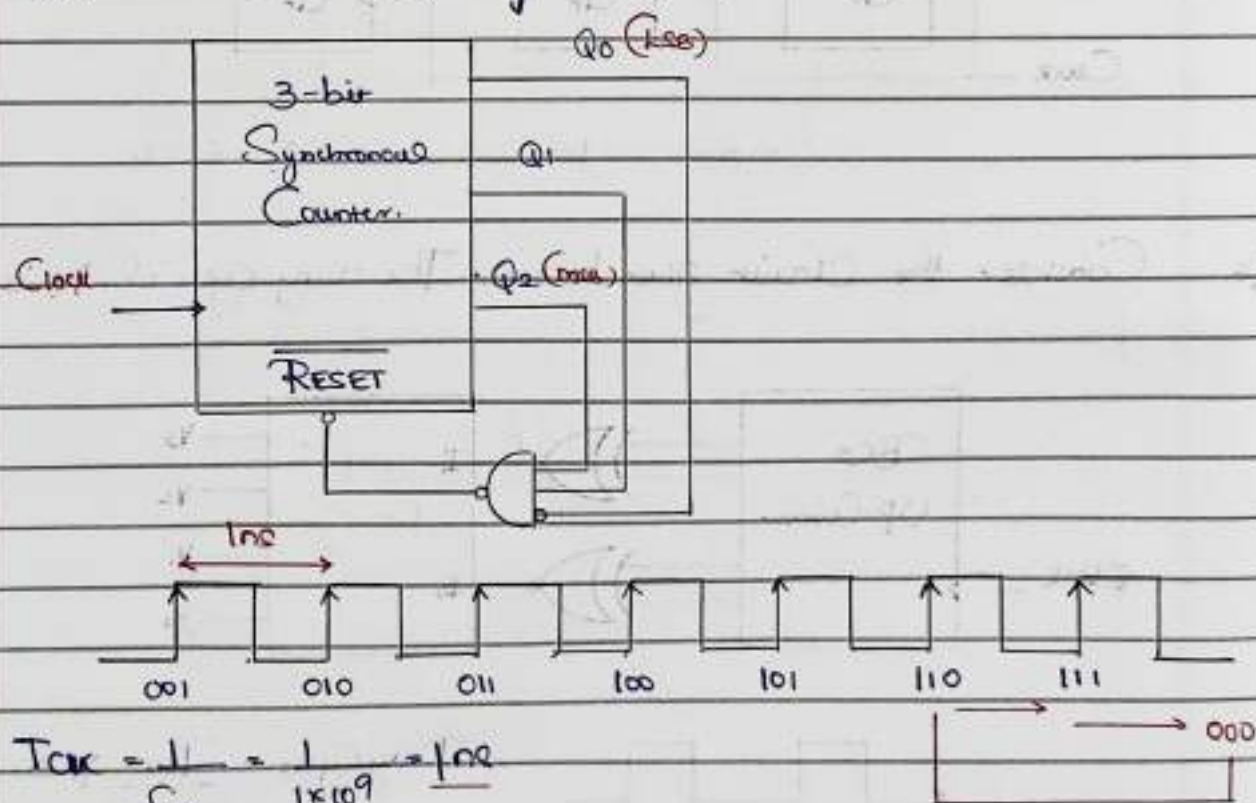
10T = Total time

3T = T<sub>on</sub>

$$D = \frac{3T}{10T} \times 100\%$$

$$= \underline{30\%}$$

Q7. For the Circuit Shown in the figure, the delay of the bubbled NAND gate is 2ns and that of the Counter is assumed to be zero. If the Clock frequency is 1GHz, then Counter behaves as a:

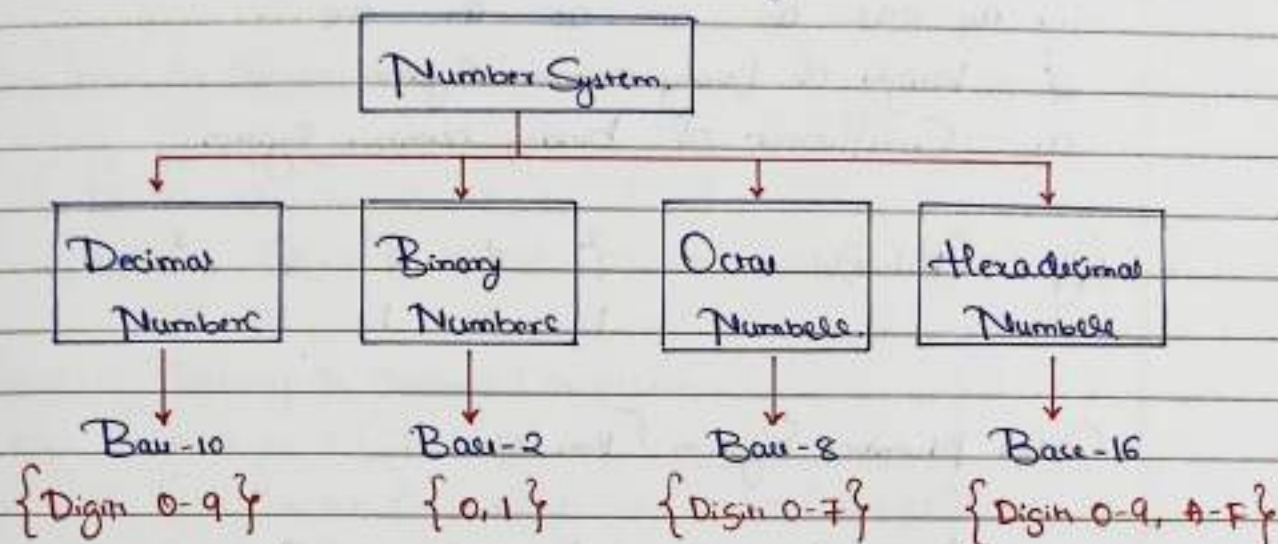


⇒ Mod-8 Counter

# NUMBER SYSTEM

## Base (Radix)

Total number of digits used in the system.



## Decimal Number System:

$$\begin{array}{cccccccc}
 10^4 & 10^3 & 10^2 & 10^1 & 10^0 & 10^{-1} & 10^{-2} & 10^{-3} \dots \\
 a_4 & a_3 & a_2 & a_1 & a_0 & a_{-1} & a_{-2} & a_{-3} \dots
 \end{array}$$

$a_i \rightarrow$  Co-efficient of decimal number system.

$10^i \rightarrow$  Weight of decimal number system.

eg:  $(501.23)_{10}$

$$\begin{array}{ccccc}
 10^2 & 10^1 & 10^0 & 10^{-1} & 10^{-2} \\
 5 & 0 & 1 & 2 & 3
 \end{array}$$

| Base | Digit |     |     |     |   |   |   |   |   |   |   |   |   |   |   |   |
|------|-------|-----|-----|-----|---|---|---|---|---|---|---|---|---|---|---|---|
| 2    | 0     | 1   |     |     |   |   |   |   |   |   |   |   |   |   |   |   |
| 3    | 0     | 1   | 2   |     |   |   |   |   |   |   |   |   |   |   |   |   |
| 4    | 0     | 1   | 2   | 3   |   |   |   |   |   |   |   |   |   |   |   |   |
| ...  | ...   | ... | ... | ... |   |   |   |   |   |   |   |   |   |   |   |   |
| 14   | 0     | 1   | 2   | 3   | 4 | 5 | 6 | 7 | 8 | 9 | A | B | C | D |   |   |
| 15   | 0     | 1   | 2   | 3   | 4 | 5 | 6 | 7 | 8 | 9 | A | B | C | D | E |   |
| 16   | 0     | 1   | 2   | 3   | 4 | 5 | 6 | 7 | 8 | 9 | A | B | C | D | E | F |

Binary Number System (Base = 2)

$$\dots 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \quad 2^{-1} \quad 2^{-2} \dots$$

$$\dots a_4 \quad a_3 \quad a_2 \quad a_1 \quad a_0 \quad a_{-1} \quad a_{-2} \dots$$

$2^i \rightarrow$  Weight of binary number System.

$a_i \rightarrow$  Co-efficient of binary number System.

eg:  $(101.11)_2$

|       |       |       |          |          |
|-------|-------|-------|----------|----------|
| $2^2$ | $2^1$ | $2^0$ | $2^{-1}$ | $2^{-2}$ |
| 1     | 0     | 1     | 1        | 1        |

Octal Number System (Base = 8)

$$\dots 8^3 \quad 8^2 \quad 8^1 \quad 8^0 \quad 8^{-1} \quad 8^{-2} \dots$$

$$\dots a_3 \quad a_2 \quad a_1 \quad a_0 \quad a_{-1} \quad a_{-2} \dots$$

$8^i \rightarrow$  Weight of Octal number System

$a_i \rightarrow$  Co-efficient of Octal number System  $\{0-7\}$ .

eg:  $(720.64)_8$

|       |       |       |          |          |
|-------|-------|-------|----------|----------|
| $8^2$ | $8^1$ | $8^0$ | $8^{-1}$ | $8^{-2}$ |
| 7     | 2     | 0     | 6        | 4        |

Hexadecimal Number System (Base = 16)

$$\dots 16^3 \quad 16^2 \quad 16^1 \quad 16^0 \quad 16^{-1} \quad 16^{-2} \dots$$

$$\dots a_3 \quad a_2 \quad a_1 \quad a_0 \quad a_{-1} \quad a_{-2} \dots$$

$16^i \rightarrow$  Weight of Hexadecimal number System

$a_i \rightarrow$  Co-efficient of Hexadecimal number System  $\{0-9, A-F\}$ .

eg:  $(A2C.F)_{16}$

|        |        |        |           |
|--------|--------|--------|-----------|
| $16^2$ | $16^1$ | $16^0$ | $16^{-1}$ |
| A      | 2      | C      | F         |

In base Conversion there are two key points:

- I. Any base to decimal conversion.
- II. Decimal to any other base conversion.

Any base to decimal conversion:

$$(a_3 \overset{r^3}{a_2} \overset{r^2}{a_1} \overset{r^1}{a_0} . \overset{r^{-1}}{a_{-1}} \overset{r^{-2}}{a_{-2}})_r = ( )_{10}$$

$$(a_3 \times r^3 + a_2 \times r^2 + a_1 \times r^1 + a_0 \times r^0 + a_{-1} \times r^{-1} + a_{-2} \times r^{-2})_{10}$$

Case (1) : Binary to decimal conversion.

eg:  $(1011.11)_2 = ( ? )_{10}$

$$= (1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2})$$

$$= (8 + 0 + 2 + 1 + 0.5 + 0.25)$$

$$= (11.75)_{10}$$

Case (2) : Octal to decimal conversion

eg:  $(721.4)_8 = ( ? )_{10}$

$$= (7 \times 8^2 + 2 \times 8^1 + 1 \times 8^0 + 4 \times 8^{-1})$$

$$= (448 + 16 + 1 + 0.5)$$

$$= (465.5)_{10}$$

Case (3) : Hexadecimal to decimal conversion

eg:  $(A8B.C)_{16} = ( ? )_{10}$

$$= (A \times 16^2 + 8 \times 16^1 + B \times 16^0 + C \times 16^{-1})$$

$$= (10 \times 256 + 8 \times 16 + 11 \times 1 + 12 \times 16^{-1})$$

$$= (2663.75)_{10}$$

Case (4) : Base 5 to decimal conversion

eg:  $(432.22)_5 = ( ? )_{10}$

$$= (4 \times 5^2 + 3 \times 5^1 + 2 \times 5^0 + 2 \times 5^{-1} + 2 \times 5^{-2})$$

$$= (100 + 15 + 2 + 0.4 + 0.08)$$

$$= (117.48)_{10}$$

Q1.  $(62c) = (?)_{10}$

By minimum base Convert into decimal:

$(62c)$

↳ 13, 14, 15, minimum = 13.

$$= (6 \times 13^2 + 2 \times 13^1 + c \times 13^0)$$

$$= (1052)_{10}$$

Decimal to any other base Conversion:

| $x$ | $a_3$ | $a_2$ | $a_1$ | $a_0$ | Remainder | $a_3$ | $a_2$ | $a_1$ | $a_0$ | $a_{-1}$ | $a_{-2}$ |
|-----|-------|-------|-------|-------|-----------|-------|-------|-------|-------|----------|----------|
|     |       |       |       |       | $b_0$     |       |       |       |       |          |          |
|     |       |       |       |       | $b_1$     |       |       |       |       |          |          |
|     |       |       |       |       | $b_2$     |       |       |       |       |          |          |
|     |       |       |       |       | $b_3$     |       |       |       |       |          |          |

Before decimal      After decimal

$$\begin{aligned}
 0.a_1a_2a_3 \times x &= x_0x_1x_2 \rightarrow x_0 \\
 0.x_1x_2x_3 \times x &= x_1x_2x_3 \rightarrow x_1 \\
 0.x_2x_3x_4 \times x &= x_2x_3x_4 \rightarrow x_2
 \end{aligned}$$

$$\begin{aligned}
 & (a_3a_2a_1a_0.a_{-1}a_{-2}a_{-3})_{10} \\
 &= (b_3b_2b_1b_0.x_0x_1x_2)_x
 \end{aligned}$$

Case (1): Decimal to Binary Base Conversion:

eg:  $(19.75)_{10} = (?)_2$

After decimal

|   |    |   |
|---|----|---|
| 2 | 19 | 1 |
| 2 | 9  | 1 |
| 2 | 4  | 0 |
| 2 | 2  | 0 |

1      1

$$\begin{aligned}
 0.75 \times 2 &= 1.5 \rightarrow 1 \\
 0.5 \times 2 &= 1.0 \rightarrow 1
 \end{aligned}$$

$$(19.75)_{10} = (10011.11)_2$$

Case (2): Decimal to Octal base Conversion:

eg:  $(210.23)_{10} = (?)_8$

|   |     |   |
|---|-----|---|
| 8 | 210 | 2 |
| 8 | 26  | 2 |
|   | 3   | 3 |

$$\begin{array}{l}
 0.23 \times 8 = 1.84 \rightarrow 1 \\
 0.84 \times 8 = 6.72 \rightarrow 6 \\
 0.72 \times 8 = 5.76 \rightarrow 5
 \end{array}$$

$$(210.23)_{10} = (32A.165)_8$$

Case (3) : Decimal to hexadecimal base conversion.

eg:  $(1288.55)_{10} = (?)_{16}$

$$\begin{array}{r|l}
 16 & 1288 \\
 \hline
 16 & 76 \\
 & 4
 \end{array}
 \begin{array}{l}
 12(C) \uparrow \\
 12(C) \uparrow \\
 4
 \end{array}$$

$$\begin{array}{l}
 0.55 \times 16 = 8.8 \rightarrow 8 \\
 0.8 \times 16 = 12.8 \rightarrow 12(C) \downarrow
 \end{array}$$

$$(1288.55)_{10} = (4CC.8C)_{16}$$

Some Special Case

Case (1) : Binary to Octal base conversion

eg:  $(10110111)_2 = (?)_8$

Octal = base 8  $\rightarrow 8 = 2^3$  ( $\therefore$  every 3 bits/digits of binary represent 1 octal).

$$\begin{array}{ccc}
 010 & 110 & 111 \\
 2 & 6 & 7
 \end{array}$$

Hence:  $(10110111)_2 = (267)_8$

Case (2) : Binary to hexadecimal base conversion.

eg:  $(1011011)_2 = (?)_{16}$

Hexadecimal = base 16 =  $2^4$  ( $\therefore$  every 4 digits of binary  $\rightarrow$  1 digit of hexadecimal).

$$0101 \quad 1011$$

$$5 \quad 11(B)$$

Hence  $(1011011)_2 = (5B)_{16}$

Q2.  $(330123)_4 = (?)_8$

Method 1:  $(330123)_4 = (?)_{10} = (?)_8$

Method 2:  $(330123)_4 = (?)_2 = (?)_8$

$$4 = 2^2$$

$$(\overline{111} \overline{100} \overline{011} \overline{011}) = (?)_8$$

$$(7433)_8$$

$$2^3$$

BCD (Binary Coded Decimal) :

In this each digit is represented by its four-bit binary equivalent. It is also called as natural BCD or 8421-code and it is weighted code.

Excess-3 Code: This is a non weighted binary code used for decimal digits. Its code assignment is obtained from the corresponding value of BCD after the addition of 3.

BCO (Binary Coded Octal) : In this each digit of the octal number is represented by its three-bit binary equivalent.

BCH (Binary Coded Hexadecimal) : In this each digit of the hexadecimal number is represented by its four bit binary equivalent.

| Decimal | BCD 8421 | Excess-3 | Octal | BCD | Hexadecimal | BCH  |
|---------|----------|----------|-------|-----|-------------|------|
| 0       | 0000     | 0011     | 0     | 000 | 0           | 0000 |
| 1       | 0001     | 0100     | 1     | 001 | 1           | 0001 |
| 2       | 0010     | 0101     | 2     | 010 | 2           | 0010 |
| 3       | 0011     | 0110     | 3     | 011 | 3           | 0011 |
| 4       | 0100     | 0111     | 4     | 100 | 4           | 0100 |
| 5       | 0101     | 1000     | 5     | 101 | 5           | 0101 |
| 6       | 0110     | 1001     | 6     | 110 | 6           | 0110 |
| 7       | 0111     | 1010     | 7     | 111 | 7           | 0111 |
| 8       | 1000     | 1011     |       |     | 8           | 1000 |
| 9       | 1001     | 1100     |       |     | 9           | 1001 |
|         |          |          |       |     | A           | 1010 |
|         |          |          |       |     | B           | 1011 |
|         |          |          |       |     | C           | 1100 |
|         |          |          |       |     | D           | 1101 |
|         |          |          |       |     | E           | 1110 |
|         |          |          |       |     | F           | 1111 |

Q1. Find the base which satisfy the equation  $\left[\left(\frac{39}{3}\right) = 13\right]_x$

Ans  $\frac{(39)}{(3)} = (13)_x$

$$\frac{3x+9}{3} = x+3$$

$3x+9 = 3x+9 \rightarrow$  This equation satisfies for all values of  $x$   
 $\therefore$  Any base  $> 9$ .

Q2. Find the base value which can satisfy the following two equations simultaneously

(i)  $2+3=5$

(i)  $2_x + (3)_x = (5)_x$

(ii)  $2 \times 4 = 10$

(ii)  $2 \times 4 = 10$

$2 \times 2^0 + 3 \times 2^0 = (5 \times 2^0)$

$(2)_x \times (4)_x = (10)_x$

Ans

$2+3=5$

$(2 \times 2)_x = (4 \times 2)_x = (1 \times 2^1 + 0 \times 2^0)$

$x > 5$

$2 \times 4 = 2+0$

$x > 6$

$x = 8$

$\therefore x = 8$  will satisfy both.

Q3. Find the number of solutions of  $x$  and  $y$  to satisfy  $(43)_8 = (20)_y$

Ans  $y > x$

$(43)_8 = (20)_y$

$(4 \times 8^1 + 3 \times 8^0)_{10} = (x \cdot y + 0 \cdot xy^0)_{10}$

$35 = x \cdot y$

$\therefore$  No. of solutions = 2

$x \quad y$

35 1

1 35 ✓

5 7 ✓

7 5

Q4. Find the number of solutions of  $x$  &  $y$  to satisfy  $(123)_5 = (x8)_y$

Ans

$y > 8$

$(123)_5 = (x8)_y$

$y > x$

$(1 \times 5^2 + 2 \times 5^1 + 3 \times 5^0) = xy^1 + 8xy^0$

$25 + 10 + 3 = xy + 8x$

$xy = 30$

$\therefore$  No. of solutions = 3

$x \quad y$

1 30 ✓

2 15 ✓

3 10 ✓

5 6

Q5. Result of addition of  $34+43$  performed on minimum base is stored in an 8-bit register. The content will be

Ans

$(34)_5 + (43)_5$

$$(3 \times 5^1 + 4 \times 5^0) + (4 \times 5^1 + 3 \times 5^0)$$

$$= (42)_{10} \Rightarrow \underline{00101010}$$

Q6  $7 \times 512 + 3 \times 256 + 6 \times 128 + 5 \times 64 + 3 \times 16 + 3$

If the number can be written in binary then number is present can be?

Ans  $2^0 = 1$   $(4+2+1)2^9 + (2+1)2^8 + (4+2)2^7 + (4+1)2^6 + (3+1)2^5 + 2+1$   
 $2^1 = 10 = 1$   $2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$   
 $2^2 = 100 = 1$   $2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2$   
 $2^3 = 1000 = 1$   $2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^4 + 2^3$   
 $\Rightarrow \underline{7}$

### Magnitude Representation

|    | Unsigned | Signed | Complement     |                |
|----|----------|--------|----------------|----------------|
|    |          |        | 1's Complement | 2's Complement |
| +5 | 101      | 0101   | 0101           | 0101           |
| -5 | X        | 1101   | 1010           | 1011           |

$8 = 10$   
 $8-1$   $8$  (10's complement)  
 (9's complement)

eg. Find 9's Complement  
 $999$   
 $732$   
 $\underline{267}$

### Unsigned Representation

3 bit Number

Range: 0 to 7

0 to  $2^3 - 1$

(n=3)

| Binary | Decimal |
|--------|---------|
| 000    | 0       |
| 001    | 1       |
| 010    | 2       |
| 011    | 3       |
| 100    | 4       |
| 101    | 5       |
| 110    | 6       |
| 111    | 7       |

Signed Representation:4 bit number:  $\{2^{4-1}-1\}$  to  $\{2^{4-1}\}$ .

| Signed Number | Equivalent decimal | Signed Number | Equivalent decimal |
|---------------|--------------------|---------------|--------------------|
| 0000          | +0                 | 1000          | -0                 |
| 0001          | +1                 | 1001          | -1                 |
| 0010          | +2                 | 1010          | -2                 |
| 0011          | +3                 | 1011          | -3                 |
| 0100          | +4                 | 1100          | -4                 |
| 0101          | +5                 | 1101          | -5                 |
| 0110          | +6                 | 1110          | -6                 |
| 0111          | +7                 | 1111          | -7                 |

1's Complement Representation:

4 bit number: -7 to +7.

'n' bit

 $-\{2^{n-1}-1\}$  to  $+\{2^{n-1}-1\}$ 

| 1's Complement | Equivalent Decimal | 1's Complement | Equivalent decimal |
|----------------|--------------------|----------------|--------------------|
| 0000           | +0                 | 1000           | -7                 |
| 0001           | +1                 | 1001           | -6                 |
| 0010           | +2                 | 1010           | -5                 |
| 0011           | +3                 | 1011           | -4                 |
| 0100           | +4                 | 1100           | -3                 |
| 0101           | +5                 | 1101           | -2                 |
| 0110           | +6                 | 1110           | -1                 |
| 0111           | +7                 | 1111           | -0                 |

Q1. 001101  $\rightarrow$  +13Q2. 110010  $\rightarrow$  -13001101  $\rightarrow$  +13

2's Complement Representation.

4 Bit Number: -8 to +7.

'n' bit

Range:  $-\{2^{n-1}\}$  to  $+\{2^{n-1}-1\}$ 

| 2's Compl. | Equivalent dec. | 2's Compl. | Equivalent dec. |
|------------|-----------------|------------|-----------------|
| 0000       | +0              | 1000       | -8              |
| 0001       | +1              | 1001       | -7              |
| 0010       | +2              | 1010       | -6              |
| 0011       | +3              | 1011       | -5              |
| 0100       | +4              | 1100       | -4              |
| 0101       | +5              | 1101       | -3              |
| 0110       | +6              | 1110       | -2              |
| 0111       | +7              | 1111       | -1              |

Q1. Convert the number given in 1's Complement into Equivalent decimal.

1. 01101  $\rightarrow$  13

5. 11110010  $\rightarrow$  -13

2. 001101  $\rightarrow$  13

6. 11101010  $\rightarrow$  -21

3. 00001101  $\rightarrow$  13

7. 0000  $\rightarrow$  +0

4. 110010  $\rightarrow$  -13

8. 1111  $\rightarrow$  -0

Q2. Convert the number given in 2's Complement into Equivalent decimal?

1. 01101  $\rightarrow$  +13

5. 11110010  $\rightarrow$  -14

2. 001101  $\rightarrow$  +13

6. 11101010  $\rightarrow$  -22

3. 00001101  $\rightarrow$  +13

7. 0000  $\rightarrow$  0

4. 110010  $\rightarrow$  -14

8. 1111  $\rightarrow$  -1

Q3. 1's Complement of (2BFD)<sub>hex</sub> is D402

FFFF

2BFD

D402

Q4. A number is expressed in binary two's Complement as 10011. Its decimal equivalent value is -13

Q5. The greatest negative number, which can be stored in a computer that has 8-bit word length and uses 2's Complement arithmetic is.

Ans  $-\{2^{n-1}\} = -2^{8-1} = \underline{-128}$

Q6. What is the resultant of  $(C4)_{16} - (7B)_{16}$ ?

Ans

Q7. The number of bytes required to represent the decimal number +1856357 in packed BCD (Binary Coded Decimal Form) is \_\_\_\_\_

Q8. 11001, 1001, and 111001 correspond to the 2's Complement representation of which of the following sets of numbers?

Ans  $-7, -7$  and  $-7$  respectively.

Q9. P, Q, and R are the decimal integers corresponding to the 4-bit binary number 1100 considered in signed magnitude, 1's complement and 2's complement representation respectively. The 6-bit 2's complement representation of  $(P+Q+R)$  is 110101

Ans.  $P = -4$

$Q = -3, R = -4$

$P+Q+R = -11$

2's  
001011  $\rightarrow$  110101

Q10. What is the possible base of the given number system,  $\sqrt{41} = 5$ .

Ans  $(\sqrt{41})_x (5)_x \quad x > 5$

$$\sqrt{4x+1} = 5$$

$$4x+1 = 25$$

$$4x = 24 \quad \therefore x = \underline{6}$$

Q11. -13 in 2's Complement can be -?

$$+13 = 01101$$

$$-13 = 10011$$

In 2's Complement: 110011

Q12. Let us consider the following equation in a 6-bit binary number system  $X = A + BA$  is given A. Point as  $(001010)_2$  in 1's Complement binary number system, B is given as  $(111010)_2$  in signed number system. What would be X in 2's Complement number system?

Ans

$$A = 10$$

$$X = A + BA$$

$$B = -26$$

$$= 10 + (-26) \times 10$$

$$= -250$$

$$01111010 : 250$$

$$\underline{10000110 : -250}$$

Q13.  $-24$  in 2's Complement form is 11101000

$$00011000 \rightarrow +24$$

$$\underline{11101000} = -24$$